

Matrix Compositions: Exact Spectra, Hankel Products, and Uniform Asymptotics

OEIS A261781, its empty-row transition,
and refined asymptotics for A261784

Part II: finite spectra, uniform asymptotics, and two Poisson limits

Part III: Cauchy limits, arithmetic atoms, and edge laws for the Hankel zeros

Part I: OpenAI ChatGPT

Research prepared for Vladimir Reshetnikov

Part II: a research report prepared for Vladimir Reshetnikov

Part III: OpenAI ChatGPT

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Abstract

Let $T(n, k)$ count nonnegative integer matrices of total sum n , with k ordered rows, an arbitrary number of ordered columns, and no zero row or column. These are the entries of OEIS A261781. We establish a collection of exact and asymptotic results for this family. First, its minimal eventual constant-coefficient recurrence has order $k(k+1)/2$, and its full-rank shifted Hankel determinants admit an explicit positive product formula, including a polynomial refinement marking the number of columns. Second, we give an exponential relative error bound for the unrestricted count that is uniform in the number of rows. This yields a Poisson empty-row law at $n = k(\log k + s)$, an explicit first correction with a summable error of order $(\log k)^2/k^2$, and the sharp total-variation rate with its leading constant. We also prove joint Gaussian–Poisson independence for column count and empty-row count in this window. Third, for n/k in compact subsets of $(1, \infty)$, a positive Stirling-number transform gives a uniform saddle expansion to every fixed order. For A261784, this identifies the exact amplitude in the recorded leading equivalent and gives the first relative correction. Finally, fixing the number of columns produces a different explicit coverage threshold.

Attribution matters here: the generating functions, a recurrence of the stated order, and fixed-row asymptotics were already published by Munarini, Poneti, and Rinaldi. Thus the two conjecture labels still present in A261781 lead in part to a reconciliation with prior literature. The main extensions of this article are the factored determinant and the uniform distributional and asymptotic results. Full proofs, reproducible computations, a bounded source audit, and further research questions are included. These are unrefereed mathematical proofs; no proof-assistant verification or historical-priority guarantee is claimed.

Editorial note (3 October 2026, batch 85B). This report is built from two manuscripts of ProveIt’s incoming-reports intake, written independently of each other on the same subject: manuscript 06 of batch 85 (Part I, the base, whose abstract stands above) and manuscript 01 of

batch 85 (Part II). It is AI-assisted, unrefereed and not formalized. Where the two manuscripts prove the same statement, it is printed once, in Part I, in manuscript 06's form, with a bracketed note marked "[Added 3 October 2026, batch 85B]" that names manuscript 01's version and its route; Part II (Section 11 onwards) prints everything that manuscript 01 adds. *All of Part II assumes the column weight $u = 1$* : manuscript 01 never weights columns. The two manuscripts permute their Greek letters; everything here is printed in manuscript 06's letters, and the dictionary of Section 1.5 discloses every rename. Manuscript 01 presented its recurrence theorem as a proof of the OEIS order conjecture; Munarini, Poneti and Rinaldi had already proved a recurrence of that order with the same denominator [4, Proposition 25 and Remark 26, p. 16], so only its minimality is new (Section 11.2). Part I's labels carry the prefix `mxc:`, Part II's the prefix `mxc:tp:`; Appendix A records the provenance and every editorial change.

Editorial note (5 October 2026, batch 98). Since batch 98 the report has three Parts. Part III (Section 23 onwards) is built from manuscript 08 of batch 98, which answers Part I's Research question 7, on the limiting distribution of the zeros of the Hankel determinant of Theorem 3.1: after the factors at $u = 0$ and $u = -1$ are removed, the zeros converge to the image on $\operatorname{Re} u = -\frac{1}{2}$ of a Cauchy law of scale $\frac{1}{2}$, with explicit cyclotomic multiplicities, the sharp Kolmogorov constant $9/\pi^2$, an arithmetic edge law, the transition of the absolute moments and the limit under the full-degree normalization. Its re-derivation of Theorem 3.1 is not reprinted. Part III's labels carry the prefix `mxc:hz:`, and it renames manuscript 08's letters that clash with Parts I and II (Table 10). Dated notes in Sections 1, 1.5, 3 and 10 and in Appendix A record what has changed. Part III is AI-assisted, unrefereed and not formalized.

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Part I

Exact spectra, Hankel products, and uniform asymptotics

1 The problem, the sources, and the contribution

1.1 Why this family is worth studying

One generating function can behave quite differently when a parameter is fixed or grows with the coefficient index. Matrix compositions provide an especially explicit setting in which to study this difference. For fixed k , the sequence is a finite sum of exponential terms and has an exact linear recurrence. At $n \asymp k$, requiring every row to be used is a rare event, and a saddle point governs its count. At $n = k(\log k + O(1))$, the same requirement has a nondegenerate limiting probability. These regimes cannot be interchanged by simply substituting a growing k into a fixed- k asymptotic.

The current A261781 entry attributes to Václav Kotěšovec two conjectures dated October 14, 2017: a column recurrence order $k(k+1)/2$, and the limit $d(k+1) - d(k) \rightarrow 1/\log 2$ for its exponential growth constants [2]. Its companion A261780 counts the same matrices while allowing zero rows [1]. The diagonal $T(2N, N)$ is A261784 [3].

1.2 What was already known

The primary enumerative source is *Matrix Compositions* by Munarini, Poneti, and Rinaldi [4], preceded by a 2006 conference paper [5]. In the 2009 article, Proposition 1 gives the generating function, Propositions 12–13 give a root-of-unity formula and fixed-row asymptotics, Propositions 22–24 give the no-zero-row inclusion–exclusion formula, Proposition 25 gives a recurrence of order $\binom{k+1}{2}$, and Proposition 29 gives the positive Stirling/Fubini transform used below. Our elementary derivations of these identities make the article self-contained; the identities themselves are attributed to those authors.

Louchard [6] studies fixed-row probabilistic asymptotics, including the number of columns and column-size statistics. An empty *column size* in that work is distinct from a zero *row* here. The A261784 entry already records a leading equivalent with numerical amplitude and an explicit exponential constant, credited to Kotěšovec in 2017 and updated in 2024 [3].

Accordingly, this article does not claim the discovery of the fixed-row growth formula, recurrence existence, or the leading A261784 exponential rate. The first OEIS statement receives an explicit minimality proof and a careful indexing convention; the second follows from an already known closed form. The stronger developments concern exact determinants and uniform estimates as the number of rows grows.

1.3 Relation to ProveIt and the audit boundary

The target was selected after inspecting the OEIS research in ProveIt [7]. Searches for A261780, A261781, and A261784 found no indexed matching report. The relevant repository hit for matrix compositions was a citation in the report on A260700 and parabolic double cosets, which studies a different enumeration. The repository search exposed indexed commit `ce37e13f4aa16819c87c3ccc611362d15758f2ac`; the search result is not an exhaustive statement about every branch or historical archive. No unformalized repository theorem is assumed in our proofs.

Topic	Existing foundation	Result established here
Recurrences	Order- $\binom{k+1}{2}$ recurrence	Minimal eventual order; complete spectrum; initial impulse
Hankel determinants	Size- k identity for unrestricted matrices	Factored size- $\binom{k+1}{2}$ determinant with column weights
Fixed-row asymptotics	Dominant pole and amplitude	Explicit error uniform in all $k \geq 1$
Empty rows	Exact inclusion–exclusion	Poisson window, first correction, sharp total variation
Column count	Fixed-row Gaussian law	Joint Gaussian–Poisson limit; conditional limits
Diagonal A261784	Recorded leading equivalent	Exact amplitude, first correction, uniform all-orders ray expansion
Fixed column count	Stars-and-bars enumeration	Explicit shifted coverage threshold when columns grow slowly

Table 1: Attribution and the scope of the extensions. “Established here” describes the proofs in this article, not a claim of priority over all literature.

The source audit compared current OEIS records with the primary 2006, 2008, and 2009 works, and checked targeted searches for growing-row, empty-row, and coupon-collection results. It found no matching statement of the specific uniform expansions below. This is a bounded audit, so the responsible claim is a set of fully specified results and proofs, rather than a declaration of a historical breakthrough.

[Added 3 October 2026, batch 85B: the search statement describes the pinned index `ce37e13f4`. At that commit manuscript 01, the source of Part II, had not yet arrived; it now forms Part II of this report, which is filed at `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a261781-matrix-compositions`. Three repository neighbours bear on the method, and none is a dependency. (i) The report on A260700 [10] uses the same two devices as Section 7, a uniform extraction of the Fubini pole at L (its Section `pdcc:sec:poles`) followed by an outer Stirling transform (its Section `pdcc:sec:outer`), for a different sequence; it cites [4] as related work only. (ii) The canonical transseries volume proves the exact pole-lattice formula for the Fubini numbers, its Theorem `q2:thm:fubini` [11], of which the estimate (7.5) is the one-pole truncation. (iii) The only ingredient of this report that is formalized anywhere in the repository is the Fubini exponential generating function used in Section 7: the Lean declarations `Fabius.fubini` (the definition $F_q = \sum_b b! \left\{ \begin{smallmatrix} q \\ b \end{smallmatrix} \right\}$), `Fabius.egfA_fubini` and `Fabius.two_sub_exp_mul_egfA_fubini`, the identity $(2 - e^t) \sum_q F_q t^q / q! = 1$, in `Analysis/FabiusFunction/Lean/FabiusFunction/OrderedBell.lean` [12]. Nothing else stated in either Part is formalized, and placement beside the repository’s formal projects confers no formal status.] [Added 5 October 2026, batch 98: nor is anything stated in Part III.]

1.4 Notation and conventions

Throughout, n is total matrix mass, k is the number of rows, and C is the number of columns. Unless stated otherwise, M denotes the number of zero rows in an unrestricted uniformly chosen matrix. Write

$$L = \log 2, \quad a_k = 2^{-1/k}, \quad r_k = 1 - a_k, \quad d_k = r_k^{-1}, \quad c_k = \frac{a_k}{2kr_k}.$$

The empty-row window uses $h = n/k$ and $\eta = ke^{-h}$. The dense-ray regime uses $\lambda = n/k$ and a saddle $r > 0$; these are separate local notations. The falling factorial is $(x)_j = x(x-1) \cdots (x-j+1)$. All logarithms are natural. Constants in uniform big- O notation may depend on the fixed compact sets specified in each theorem.

We include one empty matrix at $n = 0$ for each fixed height when zero rows are allowed. Thus $A(0, k) = 1$, $A(n, 0) = 0$ for $n > 0$, $T(0, 0) = 1$, and $T(0, k) = 0$ for $k > 0$. These conventions matter for recurrences and Hankel rank at the initial index.

1.5 Notation dictionary for the two manuscripts

[Added 3 October 2026, batch 85B.] Manuscripts 06 (Part I) and 01 (Part II) use the same objects but different letters, and three Greek letters are cyclically permuted between them: manuscript 06's η , λ , ρ are manuscript 01's λ , ρ , η . Copying a formula from one manuscript into the other without translation therefore gives a formula that is wrong, not merely differently named. This report prints *everything* in manuscript 06's letters. Table 2 lists every symbol of manuscript 01 that was renamed for Part II and Part I's notes, with the tempting false reading where one exists; manuscript 01's other symbols are unchanged. Normalizations are unchanged throughout: every rename is a change of letter only. All of Part II is at column weight $u = 1$, so its A , T , c_k , d_k are the unweighted ones of Part I. [Added 5 October 2026, batch 98: the renames of manuscript 08, the source of Part III, are in its own Table 10; this table is unchanged.]

Table 2: Manuscript 01's symbols and the letters used for them in this report. “False reading” gives what the manuscript-01 letter means in this report.

Notion	This report (ms. 06)	Ms. 01	False reading if copied
Matrices allowing zero rows	$A(n, k)$	$U(n, k)$	—
Its generating function	$F_k(z; 1)$	$U_k(z)$	—
Zero rows of an unrestricted matrix	M	$Z_{n,k}$	—
Poisson mean $ke^{-n/k}$	η	λ	λ is n/k on dense rays
Ratio n/k on compact rays	λ	ρ	ρ is the pole-bound ratio
Pole-bound ratio $L/\sqrt{L^2 + 8}$	ρ	η	η is the Poisson mean
Dense-ray saddle, $r/(1 - e^{-r}) = \lambda$	r	τ	τ is a formal variable in (7.16); ms. 01's r is $n - k$ or a moment index
Half-saddle mass $Lr/2$	v	μ	$\mu_k(u)$ is an amplitude in Theorem 4.1 and a column mean in (6.1)
Fixed- k constants	$c_k, d_k; d(x)$ at real x	$c(k), d(k),$ $d(x)$	—
Diagonal index	$N,$ $a_N = T(2N, N)$	$m, a(m)$	m indexes d_m and counts columns in Section 8
Diagonal constants	c_*, d_*	c, d	c_k, d_k are the fixed- k constants
Diagonal corrections	$\beta_1 = \beta, \beta_2, \beta_3$	$\delta_1, \delta_2, \delta_3$	ms. 01's “ β_1 ” is $\tilde{\beta}_1 = \beta_1 + \frac{1}{6}$, a different constant
Saddle correction coefficients	$E_j(\lambda)$	$A_m(\rho)$	$A(n, k)$ is a count
Leading saddle approximation	$\mathcal{B}(n, k)$	$B(n, k)$	$B, B(u)$ are pole-bound constants
Truncation order of an expansion	J	M	M is the zero-row count
Coverage correction polynomials	$\mathcal{Q}_j(h, \eta; L)$	$Q_j(h, \lambda; L)$	$Q_q(d), Q_{k,u}(z), Q$ are other objects

Notion	This report (ms. 06)	Ms. 01	False reading if copied
Coverage exponent	$\Psi(j, h, \epsilon),$ $\epsilon = 1/k$	$G(r, h, \epsilon)$	—
Moment index; point mass	$j; m$	$r; t$	r is the saddle
Reduced denominator; its degree	$Q_{k,1}(z); D$	$D_k(z); N_k$	D is also $n - Q$ in Section 7.6
Denominator coefficients	q_i	d_r	d_k is a growth constant
Roots and roots of unity	$a_j = 2^{-1/j}, z_{j,\ell},$ ω_j	$q_j, r_{j,h}, \zeta_j$	q_i are denominator coefficients
Fubini numbers	F_q	F_j	$F_k(z; u)$ is a generating function
Entire Stirling kernel; its coefficients	$W_n(w); \chi_{n,\ell}$	$W_n(v); w_{n,\ell}$	W_N is a diagonal diagnostic, $w_{j,\lambda}$ a Hankel weight
Kernel variable; Euler operator	$w = Lz; D_w$	$v; D_v$	$v = Lr/2$
Kernel correction polynomials	$P_r(w)$	$p_r(v)$	$P_j(t)$ in Section 3 is another polynomial
Kernel-proof variable $1/n$	ϵ	h	$h = n/k$
Kernel-proof auxiliaries	$\Sigma_j(n), \mathbf{a}_j, \mathbf{e}_\ell, \mathbf{r}_\ell,$ $\mathbf{b}_r, \Pi_r(\ell), \mathbf{B}_a, \delta_0$	$S_j(n), A_j,$ $E_\ell, R_\ell, B_r,$ $P_r(\ell), B_a, \delta$	$E_j(\lambda), R_p(w), B, \delta$ of Lemma 7.2
Gaussian functional	$\mathcal{G}_b$	$\mathfrak{G}_b$	—
Cumulant recursion	$\mathcal{K}_j(z, a)$	$K_j(z, a)$	K is a compact set of rays
Stirling index; Stirling defect	$Q_{n,k};$ $\Delta_{n,k} = n - Q_{n,k}$	$J_{n,k}; \Delta_{n,k}$	J_q is a block count; $\Delta_{n,k}$ is Section 7.6's D
Defect generating variable	ξ	u	u is the column weight
Smooth comparator; its root	$\mathcal{A}^{[J]}(x); x_J(Y)$	$A^{[M]}(x);$ $x_M(Y)$	—
Inverse core	$\Lambda, \varpi = W_0(\alpha\Lambda)$	X, w	X is a Poisson or normalized variable, w a PGF or kernel variable
Integer threshold; its bracket	$\mathcal{N}(Y); \epsilon_J, C_J$	$N(Y); \epsilon_M,$ C_M	N is the diagonal index
$\log 2$	L	L, macro $\backslash \text{Li}$	—

2 Exact enumeration and the complete spectrum

Definition 2.1. A matrix composition of mass n and height k is a $k \times C$ matrix of nonnegative integers with positive column sums and total entry sum n . Its columns and rows are ordered. Let $A(n, k)$ count these matrices, and let $T(n, k)$ count those with every row nonzero.

The word interpretation in A261781 is equivalent: a column of sum i specifies the multiplicities of the k letters in an alphabetically ordered word of length i . Every letter occurs somewhere precisely when every row is nonzero. For example, $T(2, 2) = 3$: the one-column matrix with entries 1, 1, and the two 2×2 permutation matrices.

2.1 Generating functions and a positive transform

Let $u > 0$ mark columns. Define

$$A(n, k; u) = \sum_B u^{C(B)}, \quad T(n, k; u) = \sum_{B: M(B)=0} u^{C(B)}.$$

At $u = 1$ these reduce to the unweighted counts. A nonzero column has generating function $(1 - z)^{-k} - 1$, since a column of sum i has $\binom{i+k-1}{i}$ possibilities. Taking an ordered sequence of columns gives the classical identity

$$F_k(z; u) := \sum_{n \geq 0} A(n, k; u) z^n = \frac{1}{1 - u((1 - z)^{-k} - 1)} = \frac{(1 - z)^k}{(1 + u)(1 - z)^k - u}. \quad (2.1)$$

This also gives the positive recurrence

$$A(n, k; u) = u \sum_{i=1}^n \binom{i+k-1}{i} A(n-i, k; u) \quad (n \geq 1, k \geq 1). \quad (2.2)$$

Deleting any designated set of zero rows proves inclusion–exclusion:

$$T(n, k; u) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} A(n, j; u). \quad (2.3)$$

Equations (2.1)–(2.3) are established enumeration, not new formulas [4, 1, 2].

For later use let $F_q = \sum_{b=0}^q b! \left\{ \begin{smallmatrix} q \\ b \end{smallmatrix} \right\}$ be the ordered Bell or Fubini number, with $F_0 = 1$. The two Stirling triangles satisfy

$$x(x+1) \cdots (x+n-1) = \sum_q \begin{bmatrix} n \\ q \end{bmatrix} x^q, \quad (e^z - 1)^k = k! \sum_{q \geq k} \left\{ \begin{smallmatrix} q \\ k \end{smallmatrix} \right\} \frac{z^q}{q!}.$$

Expanding (2.1) at $u = 1$ in powers of $(1 - z)^{-k}$ gives

$$A(n, k) = \frac{1}{2} \sum_{b \geq 0} 2^{-b} \binom{n+kb-1}{n} = \frac{1}{n!} \sum_{q=0}^n \begin{bmatrix} n \\ q \end{bmatrix} F_q k^q.$$

For $n > 0$ the $b = 0$ term is zero. The identity $F_q = \frac{1}{2} \sum_{b \geq 0} b^q 2^{-b}$ follows by comparing the exponential generating functions, with the usual $0^0 = 1$ convention. Applying the k th binomial difference to j^q yields

$$\boxed{T(n, k) = \frac{k!}{n!} \sum_{q=k}^n \begin{bmatrix} n \\ q \end{bmatrix} \left\{ \begin{smallmatrix} q \\ k \end{smallmatrix} \right\} F_q.} \quad (2.4)$$

This is Proposition 29 of [4]. Its nonnegative summands will allow controlled asymptotic replacement without alternating-sum cancellation. In particular $T(n, n) = F_n$.

2.2 Every pole survives

For fixed $u > 0$ put

$$b = \frac{u}{1+u}, \quad L_u = \log(1 + 1/u), \quad a_{j,u} = b^{1/j}, \quad \omega_j = e^{2\pi i/j}.$$

The denominator in $F_j(z; u)$ has simple zeros $z_{j,\ell} = 1 - a_{j,u} \omega_j^\ell$, $0 \leq \ell < j$. Their residues are $-a_{j,u} \omega_j^\ell / ((1+u)j)$. Consequently, for $n \geq 1$,

$$A(n, j; u) = \frac{a_{j,u}}{(1+u)j} \sum_{\ell=0}^{j-1} \omega_j^\ell (1 - a_{j,u} \omega_j^\ell)^{-n-1}. \quad (2.5)$$

The polynomial part of F_j is the constant $1/(1+u)$, so (2.5) is deliberately stated only for $n \geq 1$. Its unweighted specialization is the published Binet formula [4, Proposition 12].

Theorem 2.2 (Minimal recurrence and spectrum). *Fix $k \geq 1$ and $u > 0$, and put $D = \binom{k+1}{2}$. The reduced denominator of $\sum_{n \geq 0} T(n, k; u) z^n$ is*

$$Q_{k,u}(z) = \prod_{j=1}^k ((1+u)(1-z)^j - u). \quad (2.6)$$

Thus the minimal eventual homogeneous constant-coefficient recurrence has order exactly D , over $\mathbb{C}$ as well as over $\mathbb{R}$. Its characteristic polynomial is

$$P_{k,u}(x) = \prod_{j=1}^k ((1+u)(x-1)^j - ux^j). \quad (2.7)$$

All D characteristic roots are simple and nonzero. At $u = 1$ the polynomial is monic with integer coefficients.

Proof. Roots belonging to different factors in (2.6) cannot coincide: their distances from 1 are $b^{1/j}$, strictly increasing with j . Within each factor the roots are simple. At a root of the j th factor, only the j th term in (2.3) has a pole, and its coefficient $(-1)^{k-j} \binom{k}{j}$ and residue are nonzero. Therefore no pole cancels. This proves the exact reduced denominator and its degree.

If an eventual recurrence of smaller order existed, multiplying the generating function by its recurrence polynomial would give a polynomial; that recurrence polynomial would have to vanish at every one of these D poles. Its degree would be at least D , a contradiction. Reversing the factors in (2.6) gives (2.7); its leading coefficient is $(1+u) - u = 1$ in each factor. No pole is zero, since $b < 1$, and every corresponding characteristic root is finite and nonzero. $\square$

Remark 2.3 (Both manuscripts; the published part). [Added 3 October 2026, batch 85B.] Manuscript 01 proves Theorem 2.2 at $u = 1$ by the same argument: the roots of $2(1-z)^j - 1$ satisfy $|1-z| = 2^{-1/j}$, which determines j , so the root sets are disjoint; at a root only the j th inclusion–exclusion summand is singular, with a nonzero coefficient; and an eventual recurrence of order D' would give a denominator of degree at most D' . It writes the recurrence as $T(n, k) = -\sum_{i=1}^D q_i T(n-i, k)$ for $n > D$, with $Q_{k,1}(z) = \sum_{i=0}^D q_i z^i$ normalized by $q_0 = 1$. The weighted statement (every $u > 0$) and Remark 2.4 are manuscript 06's alone.

What is new in either manuscript is the *minimality* (no cancellation, so no shorter recurrence even over $\mathbb{C}$) and the indexing of Remark 2.4. The existence of a recurrence of order $\binom{k+1}{2}$ is Proposition 25 of [4] (printed page 16, with m rows there): its proof displays the generating function of the no-zero-row column as $x^m F_m(x)$ over $(1-2x)(1-4x+2x^2) \cdots (2(1-x)^m - 1)$ with $\deg F_m \leq \binom{m}{2}$, which is $Q_{k,1}$ in our letters. Munarini, Poneti and Rinaldi do not claim minimality. Their Remark 26 gives the cases of two and three rows; for $k = 2$ it is the example that manuscript 01 prints,

$$\sum_{n \geq 0} T(n, 1) z^n = \frac{z}{1-2z}, \quad \sum_{n \geq 0} T(n, 2) z^n = \frac{z^2(3-2z)}{1-6z+10z^2-4z^3}, \quad (2.8)$$

whence $T(n, 2) = 6T(n-1, 2) - 10T(n-2, 2) + 4T(n-3, 2)$ for $n \geq 4$. The recurrence fails at $n = 3$ with the zero initial terms: there $T(3, 2) - 6T(2, 2) = 16 - 18 = -2 = (-1)^{k+D} 2^{k-1}$, the impulse of Remark 2.4 at $k = 2$, $D = 3$. Manuscript 01 presented its theorem as a proof of the OEIS order conjecture without citing Proposition 25; Section 11.2 records the correction.

Remark 2.4 (The initial impulse). The rational function for $T(n, k; u)$ tends to $(-1)^k u / (1+u)$ as $z \rightarrow \infty$: the $j = 0$ term of inclusion–exclusion is $(-1)^k$, and every other F_j tends to $1/(1+u)$. Hence its numerator has degree D , not less than D . The denominator recurrence begins at $n = D + 1$ for

the full column indexed from zero; it is an order- D recurrence on the tail $T(1, k; u), T(2, k; u), \dots$. At $u = 1$, writing $Q_{k,1}(z) = \sum_{i=0}^D q_i z^i$, one has

$$\sum_{i=0}^D q_i T(D-i, k) = (-1)^{k+D} 2^{k-1} \neq 0.$$

If a convention insists on an order- r recurrence starting already at index r on the unshifted column, its minimum is $D+1$, obtained by adjoining a zero characteristic root. This explains why the word “eventual” is part of the theorem.

2.3 The growth-constant conjecture

The smallest-modulus pole of $F_k(z; 1)$ is $r_k = 1 - 2^{-1/k}$, uniquely. It is also the smallest-modulus pole across all terms of (2.3), since $r_j > r_k$ for $j < k$. Thus, for each fixed k ,

$$T(n, k) \sim c_k d_k^n, \quad d_k = \frac{1}{1 - 2^{-1/k}}, \quad c_k = \frac{1}{2k(2^{1/k} - 1)}. \quad (2.9)$$

The underlying pole and unrestricted asymptotic are classical [4, Proposition 13]; inclusion–exclusion transfers them to T . These constants agree with the numerical columns in A261781.

Corollary 2.5. *The second conjecture recorded in A261781 holds:*

$$d_{k+1} - d_k \longrightarrow \frac{1}{L}.$$

More precisely,

$$d_k = \frac{k}{L} + \frac{1}{2} + \frac{L}{12k} - \frac{L^3}{720k^3} + O(k^{-5}), \quad (2.10)$$

$$d_{k+1} - d_k = \frac{1}{L} - \frac{L}{12k(k+1)} + O(k^{-4}), \quad (2.11)$$

$$c_k = \frac{1}{2L} - \frac{1}{4k} + \frac{L}{24k^2} - \frac{L^3}{1440k^4} + O(k^{-6}). \quad (2.12)$$

Proof. Use the Taylor expansion at zero of $1/(1 - e^{-x}) - 1/x$ with $x = L/k$. It is analytic in a neighborhood of zero; Taylor’s theorem supplies the stated remainders. Then subtract adjacent values and use $c_k = (d_k - 1)/(2k)$. $\square$

[Added 3 October 2026, batch 85B: manuscript 01 proves (2.9) and the expansions (2.10) and (2.12) in the same way, by expanding $x/(e^x - 1)$, and states, like manuscript 06, that the limit is immediate from the closed form of [4, Proposition 13]. The increment expansion (2.11) is manuscript 06’s alone. Manuscript 01 adds that $d_{k+1} - d_k$ *increases strictly* to $1/L$ and stays below it (Theorem 12.1).]

3 A factored Hankel determinant and its polynomial refinement

Let $T_u(n, k)$ be the polynomial that counts k -row matrix compositions of mass n , with no zero row or zero column, assigning weight u to each column. Thus $T_1(n, k) = T(n, k)$ is the triangle A261781. For $k \geq 1$ we set $T_u(0, k) = 0$. In the proofs we initially take $u > 0$; the resulting determinant identity is a polynomial identity and hence extends to every complex u .

Put

$$a = 1 + u, \quad b = u, \quad d = \binom{k+1}{2}, \quad c = \binom{k+1}{3}.$$

For $s \geq 1$ define the d -by- d Hankel determinant

$$H_{k,s}(u) = \det[T_u(s+p+q, k)]_{p,q=0}^{d-1}.$$

Theorem 3.1 (Maximal Hankel determinant). *For all integers $k, s \geq 1$, the following identity holds in $\mathbb{Z}[u]$:*

$$H_{k,s}(u) = u^{d+2c} (1+u)^{k(s-1)+2c} \prod_{j=1}^k \binom{k}{j}^j \prod_{1 \leq i < j \leq k} \left((1+u)^{(j-i)/g_{ij}} - u^{(j-i)/g_{ij}} \right)^{2g_{ij}}, \quad (3.1)$$

$$g_{ij} = \gcd(i, j).$$

Here $d + 2c = \sum_{j=1}^k j^2 = k(k+1)(2k+1)/6$. In particular, $H_{k,s}(u) > 0$ for every real $u > 0$, and

$$H_{k,s+1}(u) = (1+u)^k H_{k,s}(u).$$

The determinant is maximal for the positive-index tail: the infinite matrix $[T_u(1+p+q, k)]_{p,q \geq 0}$ has rank exactly d for $u > 0$. Every minor of order greater than d in that infinite matrix vanishes. The initial term at $n = 0$ requires separate treatment, given below.

3.1 Exact spectrum and residue weights

Use the column-weight notation $T_u(n, k) = T(n, k; u)$ from Section 2. The generating function and inclusion–exclusion have already been proved in (2.1)–(2.3).

For $j \geq 1$, set

$$P_j(t) = a(t-1)^j - bt^j.$$

Since $a - b = 1$, this is a monic polynomial of degree j . Its roots can be written

$$\lambda = \frac{1}{1-r}, \quad r^j = \frac{b}{a}.$$

For $u > 0$ all such roots are finite and nonzero. They are simple, and the root sets for distinct j are disjoint: a common root would give an r with two distinct prescribed moduli $(b/a)^{1/j}$. There are therefore exactly d roots in the union of the root sets.

Partial fractions yield, for $n \geq 1$,

$$T_u(n, k) = \sum_{j=1}^k \sum_{P_j(\lambda)=0} w_{j,\lambda} \lambda^n, \quad w_{j,\lambda} = (-1)^{k-j} \binom{k}{j} \frac{\lambda - 1}{aj}. \quad (3.2)$$

All the weights are nonzero. For completeness, at a pole $z_0 = 1/\lambda$, with $r = 1 - z_0$, the numerator of the rational generating function is b/a and the derivative of its denominator is $-ajr^{j-1}$. Its coefficient of $(1 - \lambda z)^{-1}$ is thus

$$\frac{-\lambda(b/a)}{-ajr^{j-1}} = \frac{\lambda r}{aj} = \frac{\lambda - 1}{aj}.$$

The polynomial part of the unrestricted generating function is the constant $1/a$, so it does not affect any $n \geq 1$ coefficient.

3.2 Root products, discriminants, and resultants

The following identities are enough to evaluate the determinant:

$$\prod_{P_j(\lambda)=0} \lambda = a, \quad (3.3)$$

$$\prod_{P_j(\lambda)=0} (\lambda - 1) = (-1)^{j+1} b, \quad (3.4)$$

$$\text{Disc}(P_j) = (-1)^{(j-1)(j-2)/2} j^j (ab)^{j-1}, \quad (3.5)$$

$$\text{Res}(P_i, P_j) = (-1)^i (ab)^i \left(a^{(j-i)/g} - b^{(j-i)/g} \right)^g, \quad 1 \leq i < j, \quad g = \gcd(i, j). \quad (3.6)$$

The first two identities follow from the constant coefficient of P_j and from $P_j(1) = -b$. At a root λ ,

$$P'_j(\lambda) = j \left(a(\lambda - 1)^{j-1} - b\lambda^{j-1} \right) = \frac{jb\lambda^{j-1}}{\lambda - 1}.$$

Taking products and using the root products gives

$$\prod_{P_j(\lambda)=0} P'_j(\lambda) = (-1)^{j+1} j^j (ab)^{j-1}.$$

For a monic degree- j polynomial, multiplication by $(-1)^{j(j-1)/2}$ converts this product into its discriminant. Since

$$\frac{j(j-1)}{2} + j + 1 \equiv \frac{(j-1)(j-2)}{2} \pmod{2},$$

this proves (3.5).

To prove the resultant formula, put $h = j - i$ and $g = \gcd(i, h)$. For a root of P_i write $r = 1 - 1/\lambda$, so that $r^i = b/a$. Then

$$P_j(\lambda) = \lambda^j (ar^j - b) = b\lambda^j (r^h - 1).$$

As r ranges through the i roots of $r^i = b/a$, its h th powers are the i/g roots of

$$v^{i/g} = (b/a)^{h/g},$$

each occurring with multiplicity g . Consequently

$$\begin{aligned} \prod_{r^i=b/a} (r^h - 1) &= (-1)^i \left(1 - (b/a)^{h/g} \right)^g \\ &= (-1)^i a^{-h} \left(a^{h/g} - b^{h/g} \right)^g. \end{aligned}$$

Because P_i is monic, its resultant with P_j is the product of $P_j(\lambda)$ over its roots. Using $\prod \lambda = a$ gives

$$\begin{aligned} \text{Res}(P_i, P_j) &= b^i a^j (-1)^i a^{-h} \left(a^{h/g} - b^{h/g} \right)^g \\ &= (-1)^i (ab)^i \left(a^{h/g} - b^{h/g} \right)^g, \end{aligned}$$

as asserted.

3.3 Vandermonde evaluation

Enumerate all d roots in (3.2) as $\lambda_1, \dots, \lambda_d$, keeping the j -blocks together, and write w_ν for the corresponding weights. The Hankel matrix factors as

$$[T_u(s + p + q, k)]_{p,q=0}^{d-1} = V \operatorname{diag}(w_\nu \lambda_\nu^s)_{\nu=1}^d V^\top, \quad V_{p,\nu} = \lambda_\nu^p.$$

Hence

$$H_{k,s}(u) = \left(\prod_{\nu=1}^d w_\nu \lambda_\nu^s \right) \prod_{\mu < \nu} (\lambda_\nu - \lambda_\mu)^2. \quad (3.7)$$

Splitting the squared Vandermonde product into within-block and between-block factors gives

$$\prod_{\mu < \nu} (\lambda_\nu - \lambda_\mu)^2 = \prod_{j=1}^k \operatorname{Disc}(P_j) \prod_{1 \leq i < j \leq k} \operatorname{Res}(P_i, P_j)^2.$$

For one block, the weights, the root power, and the discriminant combine to

$$\begin{aligned} & \left(\prod_{P_j(\lambda)=0} w_{j,\lambda} \lambda^s \right) \operatorname{Disc}(P_j) \\ &= (-1)^{j(k-j)+\binom{j}{2}} \binom{k}{j}^j b^j a^{s-1}. \end{aligned}$$

There is no residual sign after multiplying over j , because

$$\sum_{j=1}^k \left(j(k-j) + \binom{j}{2} \right) = 2 \binom{k+1}{3} = 2c.$$

The squared resultants supply

$$(ab)^{2 \sum_{1 \leq i < j \leq k} i} \prod_{i < j} \left(a^{(j-i)/g_{ij}} - b^{(j-i)/g_{ij}} \right)^{2g_{ij}} = (ab)^{2c} \prod_{i < j} \left(a^{(j-i)/g_{ij}} - b^{(j-i)/g_{ij}} \right)^{2g_{ij}}.$$

Substitution in (3.7) proves (3.1) for $u > 0$. Both sides are polynomials in u , so agreement on this interval proves the identity in $\mathbb{Z}[u]$.

3.4 The unweighted specialization

At $u = 1$ the formula becomes

$$\det[T(s + p + q, k)]_{p,q=0}^{d-1} = 2^{k(s-1)+2\binom{k+1}{3}} \prod_{j=1}^k \binom{k}{j}^j \prod_{1 \leq i < j \leq k} \left(2^{(j-i)/\gcd(i,j)} - 1 \right)^{2\gcd(i,j)}. \quad (3.8)$$

The first few values of $H_{k,1}(1)$ are

$$H_{1,1}(1) = 1, \quad H_{2,1}(1) = 8, \quad H_{3,1}(1) = 62208.$$

For example, the complete $k = 3$ polynomial is

$$H_{3,s}(u) = 27u^{14}(1+u)^{3s+5}(1+2u)^2.$$

This extends the Cassini-type phenomenon already observed for unrestricted matrix compositions. In [4, Proposition 21], the unrestricted height- k sequence has a k -by- k determinant $(-1)^{\lfloor k/2 \rfloor} 2^{s-1}$. Enforcing all rows to be used mixes the root sets for heights $1, \dots, k$. The determinant above has order $d = k(k+1)/2$ and includes the explicit pairwise resultant factors caused by that mixture. The primary paper does not state this order- d determinant; the proof here is independent of any priority claim.

3.5 Recurrence minimality and the initial-term correction

The nonzero determinant is an independent proof of the exact tail rank d and the minimality in Theorem 2.2. It also makes the initial impulse of Remark 2.4 quantitative.

At $n = 0$ the spectral sum has the value

$$\sum_{\nu=1}^d w_{\nu} = (-1)^{k+1} \frac{u}{1+u},$$

whereas $T_u(0, k) = 0$. Thus the all-index identity is

$$T_u(n, k) = \sum_{\nu=1}^d w_{\nu} \lambda_{\nu}^n + (-1)^k \frac{u}{1+u} \mathbf{1}_{\{n=0\}}. \quad (3.9)$$

In particular, the minimal forward-shift annihilator valid from $n = 0$ is $tP_{k,u}(t)$; the extra root is zero. In the conventional eventual-recurrence language the order is still d . This distinction also accounts for the rank of the unshifted infinite Hankel matrix when $u > 0$:

$$\text{rank}[T_u(p+q, k)]_{p,q \geq 0} = d + 1.$$

Indeed, adjoining the zero root to the Vandermonde factorization gives the exact exceptional determinant

$$\det[T_u(p+q, k)]_{p,q=0}^d = (-1)^k \frac{u}{1+u} H_{k,2}(u). \quad (3.10)$$

The extra squared Vandermonde factors are $\prod_{\nu} \lambda_{\nu}^2$, which change $H_{k,0}$ for the pure spectral extension into $H_{k,2}$. At $u = 1$,

$$\det[T(p+q, k)]_{p,q=0}^d = (-1)^k 2^{k-1} H_{k,1}(1).$$

3.6 Coefficient positivity and the location of the zeros

Every nonconstant difference factor on the right of (3.1) is

$$(1+u)^m - u^m = \sum_{r=0}^{m-1} \binom{m}{r} u^r,$$

which has positive coefficients. Therefore $H_{k,s}(u) \in \mathbb{Z}_{\geq 0}[u]$. This statement concerns these maximal determinants. It does not assert total positivity of the Hankel matrix: for $k = 2$, for example,

$$\det \begin{pmatrix} T_u(1, 2) & T_u(2, 2) \\ T_u(2, 2) & T_u(3, 2) \end{pmatrix} = -u^2(1+2u)^2 < 0 \quad (u > 0).$$

The factorization also determines the geometry of all complex zeros. Apart from $u = 0$ and $u = -1$, every zero lies on the vertical line $\text{Re } u = -1/2$. Indeed, a zero of $(1+u)^m - u^m$ has the form

$$u = \frac{1}{\zeta - 1}, \quad \zeta^m = 1, \quad \zeta \neq 1,$$

and

$$\frac{1}{e^{i\theta} - 1} = -\frac{1}{2} - \frac{i}{2} \cot \frac{\theta}{2}.$$

The multiplicity at $u = 0$ is exactly $\sum_{j=1}^k j^2$. When its displayed exponent is positive, the multiplicity at $u = -1$ is exactly $k(s-1) + 2\binom{k+1}{3}$. All other multiplicities follow by collecting

the cyclotomic difference factors in the product. [Added 5 October 2026, batch 98: Part III collects them. Each zero coming from a primitive m -th root of unity, $2 \leq m \leq k-1$, has the multiplicity $E_m(k) = 2 \sum_{i < j, m|q_{ij}} g_{ij}$, with $q_{ij} = (j-i)/g_{ij}$; every such order occurs ((25.1)–(25.3)), and $E_m(k) = k^2 \log k / (\zeta(2)\psi(m)) + O(k^2)$ uniformly in m (Proposition 25.2).]

4 An exponential error bound uniform in the row count

The essential analytic input for the growing-row results is stronger than a fixed-parameter pole asymptotic. The bound below does not deteriorate when arbitrarily many poles are present.

Theorem 4.1 (Uniform dominant-pole estimate). *For every $n, k \geq 1$,*

$$A(n, k) = c_k d_k^n (1 + \varepsilon_{n,k}), \quad |\varepsilon_{n,k}| \leq B \rho^{n-1}, \quad (4.1)$$

where

$$B = \frac{\pi^2 L^2}{24}, \quad \rho = \left(1 + \frac{8}{L^2}\right)^{-1/2} < 0.239. \quad (4.2)$$

More generally, with $b = u/(1+u)$, $L_u = \log(1+1/u)$, $a_{k,u} = e^{-L_u/k}$, $r_{k,u} = 1 - a_{k,u}$, and

$$d_k(u) = r_{k,u}^{-1}, \quad \mu_k(u) = \frac{a_{k,u}}{(1+u)k r_{k,u}},$$

one has

$$A(n, k; u) = \mu_k(u) d_k(u)^n (1 + \varepsilon_{n,k}(u)), \quad |\varepsilon_{n,k}(u)| \leq B(u) \rho(u)^{n-1}, \quad (4.3)$$

where

$$B(u) = \frac{\pi^2 L_u^2}{48b}, \quad \rho(u) = \left(1 + \frac{16b}{L_u^2}\right)^{-1/2}.$$

For u in a fixed compact subset of $(0, \infty)$ the constants can be replaced by common $B_I < \infty$ and $\rho_I < 1$.

Proof. In (2.5), divide by the positive $\ell = 0$ term. Its coefficient is exactly $\mu_k(u) d_k(u)^n$. If $j = \min(\ell, k - \ell)$, then $\sin(\pi\ell/k) \geq 2j/k$, $a_{k,u} \geq b$, and $r_{k,u} \leq L_u/k$. Therefore

$$\begin{aligned} |1 - a_{k,u} \omega_k^\ell|^2 &= r_{k,u}^2 + 4a_{k,u} \sin^2(\pi\ell/k) \\ &\geq r_{k,u}^2 \left(1 + \frac{16b}{L_u^2} j^2\right). \end{aligned}$$

Put $D_u = 16b/L_u^2$. There are at most two roots for each j , so

$$\begin{aligned} |\varepsilon_{n,k}(u)| &\leq 2 \sum_{j \geq 1} (1 + D_u j^2)^{-(n+1)/2} \\ &\leq 2(1 + D_u)^{-(n-1)/2} \sum_{j \geq 1} \frac{1}{D_u j^2} = \frac{\pi^2}{3D_u} \rho(u)^{n-1}. \end{aligned}$$

For $k = 1$ the omitted sum is empty and the error is exactly zero. On a compact interval use its minimum b and maximum L_u in the same inequalities. At $u = 1$, $b = 1/2$, giving (4.2). $\square$

Remark 4.2 (A second route with a k -dependent constant). [Added 3 October 2026, batch 85B.] At $u = 1$ manuscript 01 proves the same one-pole dominance with a different constant:

$$|\varepsilon_{n,k}| \leq (k-1)\rho^{n+1} \quad (n, k \geq 1), \quad (4.4)$$

with the same $\rho = L/\sqrt{L^2 + 8}$ (manuscript 01 calls it η ; see Table 2). Its proof bounds each of the $k-1$ nonprincipal terms of (2.5) separately: with $a_k \geq \frac{1}{2}$, $r_k \leq L/k$ and $\sin(\pi/k) \geq 2/k$ one gets $r_k/|1 - a_k\omega_k^\ell| \leq \rho$ for every $\ell \neq 0$, and each ratio is at most ρ^{n+1} . Since $\rho^2 = L^2/(L^2 + 8) < 0.057$ and $B = \pi^2 L^2/24 > 0.197$, the bound (4.4) is the sharper one for $k \leq 4$ and Theorem 4.1 for $k \geq 5$; only Theorem 4.1 is uniform in k , which the growing-row results need. Manuscript 01 says so itself: its relative bound is useful only when $k\rho^n$ is small, and is not claimed to be small for large k and fixed n ; both of its growing regimes satisfy the condition by a wide margin.]

Remark 4.3. The theorem controls the unrestricted count $A(n, k)$ uniformly. It does not say that $T(n, k) \sim c_k d_k^n$ uniformly: the subtraction enforcing all rows nonzero becomes significant when $n \approx k \log k$ and exponentially significant when $n \asymp k$. The next sections quantify these effects.

5 The empty-row transition and its first correction

Put $L = \log 2$ and suppose that $A(n, k)$ counts unrestricted objects on k labelled rows, with

$$\sum_{n \geq 0} A(n, k) z^n = \frac{(1-z)^k}{2(1-z)^k - 1}.$$

For $m \geq 1$ write

$$d_m = (1 - e^{-L/m})^{-1}, \quad c_m = \frac{d_m - 1}{2m}.$$

Theorem 4.1 supplies the estimate

$$A(n, m) = c_m d_m^n (1 + \varepsilon_{n,m}), \quad |\varepsilon_{n,m}| \leq C_0 \rho^{n-1}, \quad 0 < \rho < 1, \quad (5.1)$$

uniformly for $m \geq 1$ and $n \geq 2$.

Choose an unrestricted matrix of mass n and height k uniformly and let M be its number of unused rows. Its factorial moments are exactly

$$b_j := \mathbb{E}[(M)_j] = (k)_j \frac{A(n, k-j)}{A(n, k)} \quad (0 \leq j \leq k). \quad (5.2)$$

Here $(x)_j = x(x-1)\cdots(x-j+1)$, $b_0 = 1$, and $b_k = 0$ when $n > 0$.

Theorem 5.1 (Coverage-window expansion). *Fix $S, R < \infty$. Set*

$$h = \frac{n}{k}, \quad s = h - \log k, \quad \eta = ke^{-h} = e^{-s}.$$

As $k \rightarrow \infty$, uniformly for integers n satisfying $|s| \leq S$ and complex w satisfying $|w| \leq R$,

$$\mathbb{E}(1+w)^M = e^{\eta w} \left\{ 1 - \frac{(1+h)\eta^2 w^2 + (1-L)h\eta w}{2k} + O_{S,R}\left(\frac{h^2}{k^2}\right) \right\}. \quad (5.3)$$

In particular,

$$\mathbb{P}(M=0) = e^{-\eta} \left\{ 1 + \frac{(1-L)h\eta - (1+h)\eta^2}{2k} + O_S\left(\frac{(\log k)^2}{k^2}\right) \right\}. \quad (5.4)$$

The definition $s = n/k - \log k$ includes the integer rounding of n ; there is no unstated choice of rounding convention.

Proof. We first obtain a bound valid for all inclusion–exclusion indices, not only fixed j . The elementary inequalities

$$\frac{m}{L} \leq d_m \leq \frac{m}{L} + 1$$

follow respectively from $1 - e^{-x} \leq x$ and $e^x \geq 1 + x$. Also, c_m is bounded above and bounded away from zero uniformly for $m \geq 1$. Equation (5.1) therefore gives, for all sufficiently large k in the indicated window and $0 \leq j \leq k - 1$,

$$\begin{aligned} b_j &\leq C(k)_j \left(\frac{d_{k-j}}{d_k} \right)^n \\ &\leq Ck^j \left(1 - \frac{j-L}{k} \right)^n \\ &\leq Ce^{Lh} (ke^{-h})^j = Ce^{Lh} \eta^j. \end{aligned} \quad (5.5)$$

The same bound holds for $j = k$ since $b_k = 0$.

Let $J = \lceil 4 \log k \rceil$. Since $\eta \leq e^S$ and $e^{Lh} \leq e^{LS} k^L$, the factorial tail bound gives

$$\sum_{j>J} \frac{|w|^j b_j}{j!} \leq Ck^L \sum_{j>J} \frac{(Re^S)^j}{j!} = O_{S,R,B}(k^{-B}) \quad (5.6)$$

for every fixed $B > 0$. Indeed the logarithm of the last factorial tail is at most $-J \log J + O_{S,R}(J)$. This bound directly justifies absolute truncation of the exact factorial-moment polynomial.

For the remaining indices $1 \leq j \leq J$, the needed expansion is uniform. Define

$$g_0(x) = \log \frac{Lx/2}{\sinh(Lx/2)}, \quad g_0(0) = 0.$$

Near zero, g_0 is analytic, $g_0(x) = O(x^2)$ and $g'_0(x) = O(x)$. The exact identities

$$\log d_m = \log \frac{m}{L} + \frac{L}{2m} + g_0(1/m), \quad \log c_m = -\log(2L) - \frac{L}{2m} + g_0(1/m)$$

yield

$$\begin{aligned} \log \frac{d_{k-j}}{d_k} &= -\frac{j}{k} - \frac{j^2}{2k^2} + \frac{Lj}{2k^2} + O\left(\frac{j^3}{k^3}\right), \\ \log \frac{c_{k-j}}{c_k} &= O\left(\frac{j}{k^2}\right), \\ \log(k)_j &= j \log k - \frac{j(j-1)}{2k} + O\left(\frac{j^3}{k^2}\right). \end{aligned}$$

All constants here are absolute, once $J \leq k/2$. Consequently

$$\log b_j = j \log \eta + \frac{-j(j-1) + h(Lj - j^2)}{2k} + O\left(\frac{(1+h)j^3}{k^2} + \rho^{n-1}\right). \quad (5.7)$$

Since $hJ^2/k \rightarrow 0$, exponentiation is uniform and gives

$$b_j = \eta^j \left\{ 1 + \frac{-j(j-1) + h(Lj - j^2)}{2k} + O\left(\frac{h^2 j^4}{k^2} + \rho^{n-1}\right) \right\}. \quad (5.8)$$

The index $j = 0$ is exact and is treated separately.

Now use the exact identity

$$\mathbb{E}(1+w)^M = \sum_{j=0}^k \frac{b_j w^j}{j!}.$$

The error in (5.8) sums to $O_{S,R}(h^2/k^2)$, because $\sum_{j \geq 0} j^4 (Re^S)^j / j! < \infty$ and ρ^{n-1} is smaller than every fixed power of k^{-1} in this window. Equation (5.6) permits extension of all the main sums to infinity. With $x = \eta w$,

$$\sum_{j \geq 0} \frac{[-j(j-1) + h(Lj - j^2)]x^j}{j!} = -[(1+h)x^2 + (1-L)hx]e^x.$$

This proves (5.3); the additive and relative forms of its error are equivalent on each fixed disk because $|e^{\eta w}|$ is bounded away from zero there. Set $w = -1$, with the polynomial interpretation $0^0 = 1$, to obtain (5.4). $\square$

[Added 3 October 2026, batch 85B: manuscript 01 proves the case $w = -1$, that is (5.4) with error $O((1+h)^2/k^2)$, which is the same order, and the Poisson limit of M , by a second route. It expands the factorial moments exactly as in (5.8) (its form is $\log(\mathbb{E}(M)_j/\eta^j) = -(h+1)j^2 + (Lh+1)j/(2k) + O((1+h)(1+j)^4/k^2)$ for $j \leq C \log k$, the same polynomial), then truncates the inclusion–exclusion series for $\mathbb{P}(M=0)$ by the Bonferroni inequalities after $R \asymp \log k$ terms, the next term being at most $C^{R+1}/(R+1)!$, and sums with $\sum_j (-\eta)^j j / j! = -\eta e^{-\eta}$ and $\sum_j (-\eta)^j j^2 / j! = (\eta^2 - \eta)e^{-\eta}$. The same device gives every fixed order (Theorem 13.1). Manuscript 06 proves the whole generating function in complex w , and from it the probability-mass and total-variation statements below, which manuscript 01 does not have.]

5.1 A summable expansion of the entire missing-row law

For $\eta > 0$ put $\pi_\eta(r) = e^{-\eta} \eta^r / r!$ for $r \geq 0$, and set $\pi_\eta(r) = 0$ for negative integers r . Define

$$C_2(r; \eta) = r(r-1) - 2\eta r + \eta^2 = (r-\eta)^2 - r.$$

Corollary 5.2 (Probability-mass and total-variation expansion). *Under the assumptions of Theorem 5.1, uniformly for $r \geq 0$,*

$$\mathbb{P}(M=r) = \pi_\eta(r) \left\{ 1 + \frac{(1-L)h(\eta-r) - (1+h)C_2(r; \eta)}{2k} \right\} + \mathcal{R}_{k,r}, \quad (5.9)$$

where

$$|\mathcal{R}_{k,r}| \leq C_S 2^{-r} \frac{h^2}{k^2}, \quad \sum_{r \geq 0} |\mathcal{R}_{k,r}| = O_S \left(\frac{h^2}{k^2} \right).$$

In particular, if $X \sim \text{Poisson}(\eta)$, then

$$d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(\eta)) = \frac{1}{4k} \mathbb{E} |(1-L)h(\eta-X) - (1+h)C_2(X; \eta)| + O_S \left(\frac{h^2}{k^2} \right). \quad (5.10)$$

The total-variation distance is $\Theta_S((\log k)/k)$, uniformly for $|s| \leq S$. If $s \rightarrow c$ and $\eta_0 = e^{-c}$, its leading constant is given by

$$\frac{k}{\log k} d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(ke^{-n/k})) \rightarrow \frac{1}{4} \mathbb{E} |(1-L)(\eta_0 - X_0) - C_2(X_0; \eta_0)| > 0, \quad (5.11)$$

where $X_0 \sim \text{Poisson}(\eta_0)$.

Proof. Substitute $w = z - 1$ in (5.3). Its error is uniformly $O_S(h^2/k^2)$ on $|z| \leq 2$, using $R = 3$. Cauchy's coefficient bound gives $|\mathcal{R}_{k,r}| \leq C_S 2^{-r} h^2/k^2$. The coefficients of $(z - 1)e^{\eta(z-1)}$ and $(z - 1)^2 e^{\eta(z-1)}$ are respectively

$$\pi_\eta(r - 1) - \pi_\eta(r), \quad \pi_\eta(r - 2) - 2\pi_\eta(r - 1) + \pi_\eta(r).$$

Writing these as multiples of $\pi_\eta(r)$ proves (5.9). The reverse triangle inequality and summability of $\mathcal{R}_{k,r}$ prove (5.10).

The function

$$D(\eta) = \mathbb{E} |(1 - L)(\eta - X) - C_2(X; \eta)|, \quad X \sim \text{Poisson}(\eta),$$

is continuous for $\eta > 0$ and strictly positive: its integrand is the absolute value of a nonzero quadratic polynomial and a Poisson law with positive mean assigns positive mass to every nonnegative integer. Therefore D has a positive minimum and a finite maximum on $[e^{-S}, e^S]$. The difference between the expectation in (5.10) and $hD(\eta)$ is bounded by $\mathbb{E}|C_2(X; \eta)| = O_S(1)$. This proves the uniform two-sided rate and the claimed limit. $\square$

5.2 Mean and variance interpretation

The same expansion gives

$$\nu_k := \mathbb{E}M = \eta - \frac{(1 - L)h\eta}{2k} + O_S\left(\frac{h^2}{k^2}\right),$$

and

$$\text{Var}(M) - \mathbb{E}M = -\frac{(1 + h)\eta^2}{k} + O_S\left(\frac{h^2}{k^2}\right).$$

Thus the correction includes both a shift of the Poisson mean and a negative second factorial cumulant. Adjusting only the mean does not remove the leading dependence correction. More precisely,

$$d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(\nu_k)) = \frac{1 + h}{4k} \mathbb{E}|C_2(X; \eta)| + O_S\left(\frac{h^2}{k^2}\right), \quad X \sim \text{Poisson}(\eta).$$

To see this, replace the linear term in (5.3) by $e^{(\nu_k - \eta)w}$, with uniform error $O_S(h^2/k^2)$, and apply the same coefficient argument. Replacing η by ν_k in the expectation changes the displayed answer by only $O_S(h^2/k^2)$: the expectation is locally Lipschitz as a function of its positive Poisson parameter, by dominated summation of the polynomially weighted Poisson series on a compact parameter interval.

Corollary 5.3 (The coverage threshold). *If $n/k - \log k \rightarrow s \in \mathbb{R}$, then*

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \frac{T(n, k)}{A(n, k)} \longrightarrow \exp(-e^{-s}).$$

For bounded $s_k = n/k - \log k$, with $\eta = ke^{-n/k}$ and $h = n/k$,

$$T(n, k) = c_k d_k^m e^{-\eta} \left[1 + \frac{(1 - L)h\eta - (1 + h)\eta^2}{2k} + O\left(\frac{(\log k)^2}{k^2}\right) \right]. \quad (5.12)$$

Proof. The generating-function expansion implies the Poisson distributional limit; alternatively use the summable probability-mass expansion. The event $M = 0$ is precisely membership in the class counted by T . Multiply its probability by Theorem 4.1; the exponential error is negligible at the displayed scale. $\square$

[Added 3 October 2026, batch 85B: the limit statements are also manuscript 01's (its Theorem on the Poisson transition), proved there by marked inclusion–exclusion with $\mathbb{E}(M)_{j+m}/(m!j!)$ for each fixed point mass m , or equivalently by the factorial-moment method. Table 8 in Part II gives manuscript 01's coverage probabilities as ratios of exact integers for $k = 10, \dots, 120$; at $k = 20$, $n = 60$ they agree with Table 3 in every printed digit.]

Remark 5.4 (Moving versus fixed Poisson mean). The sharp rate in (5.11) compares the actual law with $\text{Poisson}(ke^{-n/k})$. If only $n/k - \log k \rightarrow s$ is assumed, replacing this moving mean by e^{-s} incurs an additional error governed by the rate of that convergence. For $n = \text{round}(k(\log k + s))$, the additional error is $O(1/k)$ and does not change the leading $(\log k)/k$ constant. No such rate is asserted for an arbitrary unspecified $o(1)$ perturbation.

6 Column counts and joint Gaussian–Poisson independence

The row-coverage law survives a fixed positive weight on the number of columns. This permits a useful independence theorem by exponential tilting. Under the probability measure proportional to $u^{C(B)}$, denote expectation and probability by $\mathbb{E}_u$ and $\mathbb{P}_u$.

Lemma 6.1 (Uniformity under column weights). *If $n/k - \log k \rightarrow s \in \mathbb{R}$, then, uniformly for u in each fixed compact interval $I \subset (0, \infty)$,*

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \mathbb{P}_u(M = m) \rightarrow e^{-e^{-s}} \frac{e^{-ms}}{m!} \quad (m \geq 0 \text{ fixed}).$$

The same conclusion holds along any sequence of weights in I .

Proof. Deleting designated zero rows gives the exact identity

$$\mathbb{E}_u[(M)_j] = (k)_j \frac{A(n, k-j; u)}{A(n, k; u)}.$$

Use Theorem 4.1. Uniformly for $u \in I$, with $L_u = \log(1 + 1/u)$ and $q = 1/(1 + u)$,

$$\log d_k(u) = \log k - \log L_u + \frac{L_u}{2k} + O_I(k^{-2}), \quad \log \mu_k(u) = \log \frac{q}{L_u} - \frac{L_u}{2k} + O_I(k^{-2}).$$

For each fixed j this gives

$$\log \mathbb{E}_u[(M)_j] = -j(n/k - \log k) + O_{I,j}((\log k)/k).$$

Thus all fixed factorial moments tend uniformly to e^{-js} . To make the distributional step explicit, expand $\mathbb{E}_u z^M$ at $z = 1$ for $0 \leq z \leq 1$. Taylor's remainder after $J - 1$ terms is bounded by $\mathbb{E}_u[(M)_J]/J!$. First pass to the limit with J fixed, then let $J \rightarrow \infty$; the limiting error $e^{-Js}/J!$ tends to zero. This identifies the Poisson probability generating function. The first moment bounds give tightness. Every subsequential limit is therefore Poisson, uniformly in the allowed weights. On the nonnegative integers, weak convergence gives convergence of each point probability by taking an interval with half-integer endpoints. $\square$

Fix a baseline $u_0 > 0$. In the formulas below evaluate

$$b = \frac{u_0}{1 + u_0}, \quad q = \frac{1}{1 + u_0}, \quad L_{u_0} = \log(1 + 1/u_0), \quad a = e^{-L_{u_0}/k}, \quad r = 1 - a$$

at this baseline. Define

$$\mu_k(u_0) = \frac{qa}{kr}, \quad \sigma_k^2(u_0) = \frac{\mu_k(u_0)^2}{a} - b\mu_k(u_0) = \frac{qa}{k^2r^2}(q - bkr). \quad (6.1)$$

These are the first two derivatives of $\ell_k(v) = \log d_k(u_0 e^v)$ at $v = 0$. Indeed logarithmic differentiation gives $q' = -bq$, $L'_u = -q$, $a' = aq/k$, and $r' = -aq/k$. The variance constant is strictly positive, since $q/b = e^{L_{u_0}} - 1 > L_{u_0} \geq kr$.

Theorem 6.2 (Joint limit and conditional Gaussian laws). *Suppose $n/k - \log k \rightarrow s \in \mathbb{R}$, and choose matrix compositions of mass n and height k with probabilities proportional to u_0^C , for fixed $u_0 > 0$. Then*

$$\left(\frac{C - n\mu_k(u_0)}{\sqrt{n\sigma_k^2(u_0)}}, M \right) \xrightarrow{d} (Z, P), \quad (6.2)$$

where Z is standard normal, P has law $\text{Poisson}(e^{-s})$, and Z, P are independent. For each fixed $m \geq 0$, the same Gaussian limit holds conditional on $M = m$. In particular it holds for the A261781 class, which requires $M = 0$.

Proof. Put $s_k = \sqrt{n\sigma_k^2(u_0)}$, $X = (C - n\mu_k(u_0))/s_k$, and $u_v = u_0 e^{v/s_k}$ for a fixed real v . The exact tilting identity is

$$\mathbb{E}_{u_0} e^{vX} = e^{-vn\mu_k(u_0)/s_k} \frac{A(n, k; u_v)}{A(n, k; u_0)}.$$

The two weights lie in a common compact interval. Applying (4.3) and Taylor's theorem gives

$$\begin{aligned} \log \mathbb{E}_{u_0} e^{vX} &= n [\ell_k(v/s_k) - \ell_k(0) - \mu_k(u_0)v/s_k] + \log \frac{\mu_k(u_v)}{\mu_k(u_0)} + O(\rho_I^{n-1}) \\ &= \frac{v^2}{2} + O_v(n^{-1/2}) + O(\rho_I^{n-1}). \end{aligned}$$

The derivatives used here are bounded uniformly in k : write

$$\log d_k(u) = \log k - \log L_u + g(L_u/k), \quad g(x) = \log \frac{x}{1 - e^{-x}},$$

where g is smooth at zero. The variance constants have a positive limit and remain bounded away from zero and infinity. The logarithmic derivatives of μ_k are likewise uniformly bounded. Convergence of moment generating functions on a neighborhood of zero proves the marginal normal limit.

For every fixed m , the conditional tilting identity is also exact:

$$\mathbb{E}_{u_0} [e^{vX} \mid M = m] = \mathbb{E}_{u_0} e^{vX} \frac{\mathbb{P}_{u_v}(M = m)}{\mathbb{P}_{u_0}(M = m)}.$$

Both probabilities tend to the same strictly positive Poisson probability by Lemma 6.1. Their ratio tends to one, proving the conditional Gaussian limit. Summing the conditional limits over finitely many m and controlling the remaining mass by tightness of M proves the joint limit and independence. $\square$

For the unweighted model these constants simplify to

$$\mu_k = c_k, \quad \sigma_k^2 = \frac{a_k(1 - kr_k)}{4k^2r_k^2}, \quad \mu_\infty = \frac{1}{2L}, \quad \sigma_\infty^2 = \frac{1 - L}{4L^2}. \quad (6.3)$$

In the coverage window, using $n\mu_\infty$ instead of $n\mu_k$ also gives the same limit: their difference is $O(n/k) = O(\log k)$, which is $o(\sqrt{n})$. The finite- k centering keeps this correction visible. Louchard's fixed-row Gaussian theorem is an antecedent [6]; the new conclusion here is uniformity with growing rows and its joint and conditional form.

7 Uniform asymptotics on dense rays

We now require every row to be nonzero and let n/k stay bounded above and away from 1. The Poisson-window expansion does not apply here. The positive transform (2.4) gives a different route.

7.1 The saddle and the first correction

Put $f(z) = e^z - 1$. For $\lambda > 1$, let $r = r(\lambda) > 0$ solve

$$\frac{r}{1 - e^{-r}} = \lambda. \quad (7.1)$$

With $\mathcal{D} = r \, d/dr$, define

$$\kappa_j = \mathcal{D}^j \log f(r), \quad b = \kappa_2 = \lambda(1 + r - \lambda), \quad v = \frac{Lr}{2}. \quad (7.2)$$

In particular $\kappa_1 = \lambda$ and $b > 0$.

Theorem 7.1 (Uniform ray expansion). *Let K be any fixed compact subset of $(1, \infty)$. Uniformly over integers $n, k \rightarrow \infty$ with $\lambda = n/k \in K$,*

$$T(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}} \left[1 + \frac{E_1(\lambda)}{k} + O_K(k^{-2}) \right], \quad (7.3)$$

where

$$E_1(\lambda) = \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b} + \frac{v^2/3 - v}{\lambda}. \quad (7.4)$$

The expansion continues to every fixed order in integral powers of $1/k$, uniformly on K . Section 7.4 gives an explicit finite coefficient-generation rule at each order.

The derivative definitions in (7.2) are exact finite formulas, not additional implicitly determined constants. If expanded,

$$\begin{aligned} \kappa_3 &= \lambda(2\lambda^2 - 3\lambda r - 3\lambda + r^2 + 3r + 1), \\ \kappa_4 &= \lambda(-6\lambda^3 + 12\lambda^2 r + 12\lambda^2 - 7\lambda r^2 - 18\lambda r - 7\lambda \\ &\quad + r^3 + 6r^2 + 7r + 1). \end{aligned}$$

7.2 Reducing the positive sum to an analytic amplitude

The Fubini generating function is

$$\sum_{q \geq 0} F_q \frac{z^q}{q!} = \frac{1}{2 - e^z}.$$

Its poles are $L + 2\pi i j$, $j \in \mathbb{Z}$, with residue $-1/2$. Subtracting the pole at L and applying Cauchy's coefficient estimate on a circle between it and the next poles proves

$$F_q = \frac{q!}{2L^{q+1}}(1 + O(\vartheta^q)) \quad (7.5)$$

for any fixed ϑ strictly between $L/\sqrt{L^2 + 4\pi^2}$ and 1. Every term in (2.4) is nonnegative and $q \geq k$, so this replacement incurs a relative error $O(\vartheta^k)$ in the entire sum.

[Added 3 October 2026, batch 85B: manuscript 01 proves the same estimate (its Lemma on the Fubini approximation) by subtracting $1/(2(L - z))$ from $1/(2 - e^z)$ and applying Cauchy's bound on a circle of radius between L/ϑ and $\sqrt{L^2 + 4\pi^2}$. Both are classical. The repository's transseries volume has the exact convergent pole-lattice formula $F_q = \frac{q!}{2} \sum_{j \in \mathbb{Z}} (L + 2\pi i j)^{-q-1}$, $q \geq 1$ (Theorem `q2:thm:fubini` of [11]), of which (7.5) keeps the term $j = 0$; and the generating function $1/(2 - e^z)$ is formalized in Lean as `Fabius.two_sub_exp_mul_egfA_fubini` [12].]

Define the polynomial

$$H_{n,k}(z) = \sum_{d=0}^{n-k} L^d \begin{bmatrix} n \\ n-d \end{bmatrix} \left(\frac{(n-d)!}{n!} \right)^2 z^d. \quad (7.6)$$

Using $[z^q]f(z)^k = k! \{q \atop k\} / q!$, including both factorials from (2.4) and (7.5), gives

$$T(n, k) = \frac{n!}{2L^{n+1}} [z^n] f(z)^k H_{n,k}(z) (1 + O(\vartheta^k)). \quad (7.7)$$

Terms with $d > n - k$ would have zero coefficient contribution since $f(z)^k$ begins at degree k . Notice that the numerator in this formula is $n!$, not $1/n!$; the square of the factorial ratio in (7.6) is essential.

Lemma 7.2 (Uniform analytic amplitude). *Fix $R > 0$ and $\delta > 0$. If $k \geq \delta n$ and $n - k \geq \delta n$, then, uniformly on $|z| \leq R$,*

$$H_{n,k}(z) = e^{Lz/2} \left[1 + \frac{L^2 z^2 / 12 - Lz / 2}{n} + O_{R,\delta}(n^{-2}) \right]. \quad (7.8)$$

The same expansion may be differentiated any fixed number of times, with the corresponding differentiated error bound.

Proof. The unsigned Stirling coefficient is the elementary symmetric function $[n-d] = e_d(1, \dots, n-1)$. Therefore

$$e_d(1, \dots, n-1) \leq \frac{(n(n-1)/2)^d}{d!} \leq \frac{n^{2d}}{2^d d!}.$$

Because $n - d \geq k \geq \delta n$, the absolute value of the d th term in (7.6) is at most $(LR/(2\delta^2))^d / d!$. Its tail for $d > n^{1/10}$ is smaller than every fixed inverse power of n .

For $d \leq n^{1/10}$ we claim the uniform expansions

$$\begin{bmatrix} n \\ n-d \end{bmatrix} = \frac{n^{2d}}{2^d d!} \left[1 - \frac{d(2d+1)}{3n} + O\left(\frac{(1+d)^4}{n^2}\right) \right], \quad (7.9)$$

$$\left(\frac{(n-d)!}{n!}\right)^2 = n^{-2d} \left[1 + \frac{d(d-1)}{n} + O\left(\frac{(1+d)^4}{n^2}\right)\right]. \quad (7.10)$$

For the first, set $S_1 = n(n-1)/2$ and sample d independent indices in $\{1, \dots, n-1\}$ with probabilities $p_j = j/S_1$. The probability of no equal pair is $d!e_d/S_1^d$. The first two inclusion–exclusion bounds for pair coincidences give

$$\mathbb{P}(\text{no coincidences}) = 1 - \binom{d}{2} \sum_j p_j^2 + O\left(d^4 \left(\sum_j p_j^2\right)^2 + d^3 \sum_j p_j^3\right).$$

Here $\sum p_j^2 = 4/(3n) + O(n^{-2})$ and $\sum p_j^3 = O(n^{-2})$. Expanding S_1^d proves (7.9). For the second, expand $-2 \sum_{j=0}^{d-1} \log(1 - j/n)$ with its uniformly bounded Taylor remainder.

Multiplication gives

$$L^d \binom{n}{n-d} \left(\frac{(n-d)!}{n!}\right)^2 = \frac{(L/2)^d}{d!} \left[1 + \frac{d(d-4)}{3n} + O\left(\frac{(1+d)^8}{n^2}\right)\right].$$

The factorial majorant permits summation of the errors and extension of the main terms to all $d \geq 0$. Since $\sum d(d-4)x^d/d! = (x^2 - 3x)e^x$, this proves (7.8). Establishing the same uniform estimate on a slightly larger disk and using Cauchy’s derivative estimate proves the differentiated versions. $\square$

Remark 7.3 (The hypothesis δ is not needed). [Added 3 October 2026, batch 85B.] The coefficient of z^d in (7.6) is $\chi_{n,d} L^d$, where $\chi_{n,d} = \lfloor \frac{n}{n-d} \rfloor / (n)_d^2$ does not depend on k ; so $H_{n,k}(z)$ is the truncation at degree $n-k$ of the entire polynomial $W_n(Lz)$ of Part II, $W_n(w) = \sum_{d=0}^{n-1} \chi_{n,d} w^d$. Since $f(z)^k$ begins in degree k , the truncation does not change the coefficient in (7.7), which therefore holds with $W_n(Lz)$ in place of $H_{n,k}(z)$; this is manuscript 01’s form of the reduction. Manuscript 01 proves the expansion of W_n to every order, uniformly on every compact subset of $\mathbb{C}$ and *with no hypothesis on k* , together with an exact differential recurrence in n (Theorem 14.3 and Lemma 14.2). That is the general form of Lemma 7.2; the conditions $k \geq \delta n$, $n-k \geq \delta n$ above serve only the truncated polynomial, and on compact rays, where both apply, the two statements coincide. The polynomials P_1, P_2 of (7.15) are manuscript 01’s p_1, p_2 ; Part II adds P_3 .

7.3 The saddle calculation and its error

The function $r/(1 - e^{-r})$ increases from 1 to ∞ , because its derivative is

$$\frac{1 - (1+r)e^{-r}}{(1 - e^{-r})^2} > 0.$$

Hence the positive saddle is unique. For $\lambda \in K$ it lies in a compact positive interval and $b = r \, d\lambda/dr$ is bounded away from zero.

On the circle $z = re^{i\theta}$, the coefficient integral has phase

$$\log \frac{f(re^{i\theta})}{f(r)} - i\lambda\theta = -\frac{b}{2}\theta^2 - \frac{i\kappa_3}{6}\theta^3 + \frac{\kappa_4}{24}\theta^4 + O_K(\theta^5). \quad (7.11)$$

Every coefficient of f in positive degree is strictly positive. Equality in $|f(re^{i\theta})| \leq f(r)$ can therefore occur only at $\theta = 0$ modulo 2π . Compactness gives a uniform strict gap on every fixed complementary arc. Near zero, the modulus is bounded by $f(r)e^{-c\theta^2}$ for some uniform $c > 0$.

For $g(z) = e^{Lz/2}$,

$$\frac{g(re^{i\theta})}{g(r)} = 1 + iv\theta - \frac{v + v^2}{2}\theta^2 + O_K(\theta^3).$$

Expanding (7.11), multiplying by this amplitude, and setting $\theta = x/\sqrt{k}$ gives the order-1/ k contribution

$$\frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b}.$$

These constants follow from the Gaussian moments $\mathbb{E}X^2 = b^{-1}$, $\mathbb{E}X^4 = 3b^{-2}$, and $\mathbb{E}X^6 = 15b^{-3}$ for $X \sim N(0, b^{-1})$. The order-1/ n term of Lemma 7.2, evaluated at the saddle, is $(v^2/3 - v)/n$. Since $n = \lambda k$, this proves the coefficient in (7.4).

For completeness, the error is relative $O_K(k^{-2})$. Expand the analytic phase and amplitude through sufficiently many terms, for example through degree eight, on a shrinking central arc. Odd terms integrate to zero by symmetry, and the Gaussian modulus bound dominates the integrated remainders. The fixed complementary arc is exponentially small; the intervening arc is controlled by the same Gaussian bound. Finally, an analytic amplitude error uniformly $O(n^{-2})$ contributes at most

$$O(n^{-2})f(r)^k r^{-n} \int_{-\pi}^{\pi} \left| \frac{f(re^{i\theta})}{f(r)} \right|^k d\theta = O(n^{-2}k^{-1/2})f(r)^k r^{-n}.$$

Retaining this Gaussian factor proves the required relative error; a bare maximum-modulus estimate would lose an unnecessary factor $\sqrt{k}$. Together with (7.7), this proves the first-order assertion of Theorem 7.1.

[Added 3 October 2026, batch 85B: Theorem 7.1, including its all-orders form (7.17), is also manuscript 01's main theorem on compact rays, with the leading approximation $\mathcal{B}(n, k)$ of (15.1) and the same first coefficient: its $A_1(\rho)$ is $E_1(\lambda)$ of (7.4), term for term, after the renames $\rho \rightarrow \lambda$, $\tau \rightarrow r$, $\mu \rightarrow v$. Its route differs in two places. It integrates the untruncated kernel $W_n(Lz)$ on $|z| = r$ and controls the tails with $|W_n(w)| \leq e^{|w|}$ (Lemma 14.2); on the central arc $|\theta| \leq k^{-2/5}$ it expands two orders beyond the target and bounds the intermediate region by $\exp(-ck^{1/5})$. It also proves the uniqueness of the maximum on the circle directly: equality in the triangle inequality for $\sum_{j \geq 1} r^j e^{ij\theta}/j!$ forces all $e^{ij\theta}$ to agree, and consecutive powers force $e^{i\theta} = 1$.]

7.4 A constructive expansion to every fixed order

Here is a precise algorithm for the higher coefficients, together with the uniform remainder argument. Put $w = Lz$ and $\epsilon = 1/n$. The exact polynomial

$$E_n(w) = \prod_{j=1}^{n-1} \left(1 + \frac{jw}{n^2} \right)$$

has, on every fixed disk, an expansion $E_n(w) \sim e^{w/2} \sum_{p \geq 0} R_p(w) n^{-p}$, with polynomial R_p . Compute them from

$$\log E_n(w) = \sum_{m \geq 1} \frac{(-1)^{m-1} w^m}{m n^{2m}} \sum_{j=1}^{n-1} j^m. \quad (7.12)$$

The inner power sum is a finite polynomial in n ; at order n^{-p} only $m \leq p + 1$ is needed. For example,

$$R_0 = 1, \quad R_1 = -\frac{w}{2} - \frac{w^2}{6}, \quad R_2 = \frac{w^2}{4} + \frac{w^3}{12} + \frac{1}{2} \left(\frac{w}{2} + \frac{w^2}{6} \right)^2.$$

Define polynomials $Q_q(d)$ by the formal identity

$$\exp \left(2 \sum_{m \geq 1} \frac{\epsilon^m}{m} \sum_{j=0}^{d-1} j^m \right) = \sum_{q \geq 0} Q_q(d) \epsilon^q. \quad (7.13)$$

Thus $Q_0 = 1$, $Q_1 = d(d-1)$, and

$$Q_2 = \frac{d(d-1)(2d-1)}{6} + \frac{d^2(d-1)^2}{2}.$$

With $D_w = w \, d/dw$, put

$$P_j(w) = e^{-w/2} \sum_{p+q=j} Q_q(D_w) \left[e^{w/2} R_p(w) \right]. \quad (7.14)$$

These are polynomials, and

$$H_{n,k}(z) \sim e^{Lz/2} \sum_{j \geq 0} \frac{P_j(Lz)}{n^j}, \quad P_0 = 1, \quad P_1(w) = \frac{w^2}{12} - \frac{w}{2}, \quad P_2(w) = \frac{w^4}{288} - \frac{w^2}{12}. \quad (7.15)$$

To justify this algorithm to any fixed order p , set $F_d(x) = \prod_{j=0}^{d-1} (1-jx)^{-2}$. For $d \leq n-k \leq (1-\delta)n$ and $0 \leq x \leq 1/n$, every $1-jx \geq \delta$. The m th derivative of its logarithm is

$$2(m-1)! \sum_{j < d} \frac{j^m}{(1-jx)^m} = O_{m,\delta}((1+d)^{m+1}).$$

The finite product rule for derivatives of an exponential then gives

$$|F_d^{(p+1)}(x)| \leq C_{p,\delta} \delta^{-2d} (1+d)^{2p+2}.$$

Taylor's theorem bounds the remainder after the p th term by $C_{p,\delta} n^{-p-1} \delta^{-2d} (1+d)^{2p+2}$. Multiplication by the exact coefficient $e_d(1, \dots, n-1)/n^{2d} \leq 1/(2^d d!)$ makes this error summable on every fixed w -disk. Multiplying its d th coefficient by $Q_q(d)$ is exactly the operation $Q_q(D_w)$. Extending the sum beyond $n-k \geq \delta n$ adds only a factorially small tail. Independently, expanding (7.12) and its derivatives on a slightly larger disk has a uniform Taylor remainder. Combining the two steps proves (7.15), with error $O(n^{-p-1})$ after any fixed order p .

The remaining Gaussian integrations are also explicit. Define the linear functional on polynomials in y by

$$\mathcal{G}_b(y^{2m}) = (-1)^m (2m-1)!! \, b^{-m}, \quad \mathcal{G}_b(y^{2m+1}) = 0,$$

with $(-1)!! = 1$. It is integration after the substitution $y = iX$, $X \sim N(0, b^{-1})$. Introduce a formal variable τ and set

$$\begin{aligned} E_j(\lambda) = \mathcal{G}_b \left([\tau^{2j}] \exp \left\{ \sum_{m \geq 3} \frac{\kappa_m y^m \tau^{m-2}}{m!} + v(e^{\tau y} - 1) \right\} \right. \\ \left. \times \sum_{p \geq 0} \frac{\tau^{2p}}{\lambda^p} P_p(Lre^{\tau y}) \right). \end{aligned} \quad (7.16)$$

At each order this is a finite polynomial calculation: it uses $p \leq j$ and $m \leq 2j + 2$. Odd powers vanish after applying $\mathcal{G}_b$. It gives $E_0 = 1$ and the E_1 in (7.4). Therefore for every fixed $J \geq 0$,

$$T(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}} \left[\sum_{j=0}^J \frac{E_j(\lambda)}{k^j} + O_{K,J}(k^{-J-1}) \right]. \quad (7.17)$$

The same arc estimates as above justify termwise finite integration. This is an asymptotic expansion to each finite order; convergence of the infinite displayed series is not asserted.

[Added 3 October 2026, batch 85B: manuscript 01 gives the same finite prescription with its expansion parameter called ε instead of τ : its $H_M(\varepsilon, y)$ is the expression inside $\mathcal{G}_b$ in (7.16), truncated at $m \leq 2J + 2$ and $p \leq J$, because $P_p(Lre^{\tau y})$ is its $p_p(2\mu e^{\varepsilon y})$ with $2\mu = Lr$. It stresses the order of operations: first expand and truncate in τ , then apply $\mathcal{G}_b$ to each polynomial coefficient; no Gaussian expectation of an untruncated exponential of a high-degree polynomial is intended, and such an integral need not exist. It computes the kernel polynomials by a different finite route, (14.11)–(14.12), and adds a polynomial recursion for the κ_j ((15.2)); both are in Part II.]

7.5 The A261784 diagonal

Put $n = 2N$, $k = N$, and

$$r = 2 + W_0(-2e^{-2}) = 1.5936242600400400923230 \dots \quad (7.18)$$

The principal real Lambert branch is necessary; the other real branch gives the excluded boundary value $r = 0$. Equation (7.1) becomes $r/(1 - e^{-r}) = 2$. Define

$$d_* = \frac{4(e^r - 1)}{L^2 r^2} = \frac{4}{L^2 r(2 - r)} = 12.8556891736661213193266 \dots, \quad (7.19)$$

$$c_* = \frac{2^{r/2}}{4\pi L \sqrt{r - 1}} = 0.2588649231114655502517752 \dots \quad (7.20)$$

Corollary 7.4 (Exact amplitude and correction for A261784). *The sequence $a_N = T(2N, N)$ satisfies*

$$a_N = c_* d_*^N \frac{(N!)^2}{N} \left(1 + \frac{\beta}{N} + O(N^{-2}) \right), \quad (7.21)$$

where, with $v = Lr/2$,

$$\begin{aligned} b &= 2(r - 1), \quad \kappa_3 = 2(r^2 - 3r + 3), \quad \kappa_4 = 2(r^3 - 8r^2 + 19r - 13), \\ \beta &= \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b} + \frac{v^2/3 - v}{2} - \frac{1}{8} \\ &= -0.5218067015564826638891441 \dots \end{aligned}$$

Proof. Specialize Theorem 7.1 at $\lambda = 2$ and use

$$\frac{(2N)!}{(N!)^2} = \frac{4^N}{\sqrt{\pi N}} \left(1 - \frac{1}{8N} + O(N^{-2}) \right).$$

The saddle prefactor has $b = 2(r - 1)$. Collecting the two square roots gives (7.20), and the central-binomial correction accounts for the final $-1/8$ in β . $\square$

The OEIS entry already gives the leading equivalent in (7.21), with d_* in Lambert- W form and c_* numerically [3]. The contribution here is its derivation from the positive transform, the exact expression (7.20), the explicit correction, and its place in the uniform all-orders theorem.

[Added 3 October 2026, batch 85B: manuscript 01 proves the same corollary to every fixed order, with the same r , d_* , c_* and first correction; written out with $\kappa_3 = 2q$, $\kappa_4 = 2s$, $q = r^2 - 3r + 3$, $s = r^3 - 8r^2 + 19r - 13$, its closed form is $\beta_1 = \frac{s}{16(r-1)^2} - \frac{5q^2}{48(r-1)^3} - \frac{v^2+v}{4(r-1)} + \frac{qv}{4(r-1)^2} + \frac{v^2}{6} - \frac{v}{2} - \frac{1}{8}$, which is the display above with $b = 2(r-1)$ substituted. Both manuscripts note that the expression for d_* agrees algebraically with the one posted in A261784 and that c_* reproduces its decimal prefactor. The two computations of r , d_* , c_* , β_1 and β_2 were made by independent programs and agree in every printed digit. Manuscript 01 adds the third correction β_3 , the recipe for every further one, and fifty-digit values (Section 17).]

7.6 Column counts under the dense row condition

There is a further consequence: enforcing every row to be nonzero in this regime still leaves a universal leading Gaussian column law.

Corollary 7.5. *For a uniformly chosen matrix counted by $T(n, k)$, with $n/k \in K \Subset (1, \infty)$, its number of columns satisfies*

$$\frac{C - n/(2L)}{\sqrt{n(1-L)/(4L^2)}} \xrightarrow{d} N(0, 1). \quad (7.22)$$

Uniformly in this regime,

$$\mathbb{E}C = \frac{n+1}{2L} - \frac{1}{2} - \frac{r}{4} + O_K(k^{-1}), \quad (7.23)$$

$$\text{Var}(C) = \frac{(n+1)(1-L)}{4L^2} - \frac{1}{4} + \frac{r}{8} + O_K(k^{-1}). \quad (7.24)$$

Proof. The weighted version of (2.4) replaces F_q by $F_q(u) = \sum_c c! \{c^q\} u^c$, whose exponential generating function is $1/(1+u-ue^z)$. Its dominant pole is $L_u = \log(1+1/u)$ with residue $-1/(1+u)$; the other poles are $L_u + 2\pi i j$. The same positive-sum and saddle proof works uniformly for real u in a compact interval around 1. It gives

$$T(n, k; u) = \frac{n! f(r)^k e^{L_u r/2}}{(1+u) L_u^{n+1} r^n \sqrt{2\pi k b}} (1 + O_K(k^{-1})). \quad (7.25)$$

In the ratio of (7.25) to its value at $u = 1$, put $u = e^{t/\sqrt{n}}$. Direct Taylor expansion of the elementary function $\log(L/L_{e^v})$ has first derivative $1/(2L)$ and second derivative $(1-L)/(4L^2)$ at zero. Hence

$$\log \mathbb{E} \exp \left(\frac{t(C - n/(2L))}{\sqrt{n}} \right) \rightarrow \frac{t^2(1-L)}{8L^2}.$$

This proves the Gaussian limit by real moment generating functions; no differentiation of an unspecified asymptotic error is used.

For the sharper moment statements, use an exact auxiliary mixture. Let Q have probabilities proportional to $\binom{n}{q} \{q\} F_q$, $k \leq q \leq n$, and put $D = n - Q$. Conditional on $Q = q$, the distribution of C is the block-count distribution J_q in a uniform ordered set partition of q elements, because its probability generating polynomial is $F_q(u)/F_q$. This specifies an exact distributional mixture; no canonical matrix statistic is being claimed for D .

Marking D with a fresh variable ξ replaces $H_{n,k}(z)$ by $H_{n,k}(\xi z)$. Applying the coefficient proof separately to $zH'_{n,k}(z)$ and $z^2H''_{n,k}(z)$ gives

$$\mathbb{E}D = \frac{Lr}{2} + O_K(k^{-1}), \quad \text{Var}(D) = \frac{Lr}{2} + O_K(k^{-1}).$$

Here the differentiated analytic kernel estimates follow from Lemma 7.2; the exponentially small Fubini replacement error is also valid for each of these nonnegative weighted sums. The exact identities

$$F'_q(1) = \frac{F_{q+1} - F_q}{2}, \quad F''_q(1) = \frac{F_{q+2} - 5F_{q+1} + 2F_q}{4}$$

follow from the Stirling recurrence or direct differentiation of the generating function. Using (7.5), they yield

$$\mathbb{E}J_q = \frac{q+1}{2L} - \frac{1}{2} + O(q\vartheta^q), \quad \text{Var}(J_q) = \frac{(q+1)(1-L)}{4L^2} - \frac{1}{4} + O(q^2\vartheta^q).$$

Since $Q \geq k$, the exponential errors stay negligible after mixing. The laws of total expectation and total variance now give (7.23)–(7.24). $\square$

[Added 3 October 2026, batch 85B: the auxiliary variable $D = n - Q$ of this proof is manuscript 01's *Stirling defect* $\Delta_{n,k}$ (with $Q = Q_{n,k}$). Manuscript 01 proves more than its two moments: $\Delta_{n,k}$ converges in law to $\text{Poisson}(Lr/2)$, and its probability generating function is $e^{v(\xi-1)}(1 + O(k^{-1}))$ uniformly for complex ξ in compact sets, with an expansion to every fixed order (Theorem 16.2). The two displayed moments follow from it. Both manuscripts state that no canonical matrix statistic is claimed for this variable.]

The compact restriction $K \Subset (1, \infty)$ is essential in all these statements. At $k = n$ the Stirling kernel collapses and $T(n, n) = F_n$ exactly. At the coverage window $n/k \sim \log k$ the saddle leaves compact sets. Neither boundary is covered by the uniformity asserted in Theorem 7.1.

8 Prescribing the number of columns changes the threshold

The preceding models allow the column count to fluctuate. A different coverage scale appears when it is prescribed to be much smaller than the number of rows. The exact stars-and-bars identities underlying this model are classical [4, Proposition 28]; the following limit follows from direct uniform estimates on them.

Theorem 8.1 (Fixed-column coverage). *Let $k \rightarrow \infty$, let $m = m(k) \geq 1$ be integers with $m = o(k)$, and let $n = n(k) \geq 1$. Choose uniformly a $k \times m$ matrix of nonnegative integers of total sum n , conditioned on no zero column. Let M count its zero rows and put*

$$h = \frac{n}{k}, \quad \theta = m \log \left(1 + \frac{n}{km} \right). \quad (8.1)$$

If $\theta - \log k \rightarrow s \in \mathbb{R}$, then

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \mathbb{P}(M = 0) \longrightarrow \exp(-e^{-s}).$$

Thus an exact parametrization of the critical mass, up to integer rounding, is

$$n = km \left(\exp \left(\frac{\log k + s}{m} \right) - 1 \right). \quad (8.2)$$

Proof. First allow zero columns, and call the zero-row count M_0 . There are $\binom{n+km-1}{n}$ matrices. If j designated rows are zero, deleting them and comparing these binomial coefficients gives, with $N = km$,

$$\mathbb{E}[(M_0)_j] = (k)_j \prod_{a=1}^{jm} \frac{N-a}{N+n-a}. \quad (8.3)$$

Fix j and take $k \geq 2j$. For $f(a) = \log(N-a) - \log(N+n-a)$ one has $f(0) = -\log(1+n/N)$ and

$$f'(a) = -\frac{n}{(N-a)(N+n-a)}.$$

The mean value theorem and summation imply

$$\sum_{a=1}^{jm} (f(a) - f(0)) = O_j \left(\frac{nm^2}{N(N+n)} \right) = O_j \left(\frac{mh}{k(m+h)} \right).$$

Also $\log(k)_j = j \log k + O_j(k^{-1})$. Therefore

$$\log \mathbb{E}[(M_0)_j] = j(\log k - \theta) + O_j \left(\frac{1 + \min(m, h)}{k} \right).$$

For $x \geq 0$, $\log(1+x) \geq (\log 2) \min(1, x)$, so $\theta \geq (\log 2) \min(m, h)$. At the assumed critical scale, the error is $O_j((\log k)/k)$. The factorial moments tend to e^{-js} ; the finite Taylor/Bonferroni argument in Lemma 6.1 proves the Poisson limit.

It remains to justify conditioning. If $m = 1$, the sole column cannot be zero because $n > 0$. If $m \geq 2$, the probability that one designated column is zero is at most

$$\prod_{a=1}^k \frac{N-a}{N+n-a} \leq \left(\frac{N}{N+n} \right)^k = \exp(-k\theta/m).$$

The union bound gives probability at most $m \exp(-k\theta/m)$ for any zero column. Its logarithm tends to $-\infty$, since $m = o(k)$ and $\theta = \log k + O(1)$. Conditioning on an event whose probability tends to one changes the law of any statistic in total variation by a quantity tending to zero. Thus M and M_0 have the same limit. Rounding n in (8.2) changes θ by at most $1/k$. $\square$

Corollary 8.2 (A shifted window). *Suppose $m/(\log k)^2 \rightarrow \gamma \in (0, \infty)$ and $n/k - \log k \rightarrow s$. In the preceding fixed-column model,*

$$M \xrightarrow{d} \text{Poisson}\left(e^{-s+1/(2\gamma)}\right), \quad \mathbb{P}(M=0) \longrightarrow \exp\left(-e^{-s+1/(2\gamma)}\right). \quad (8.4)$$

Proof. Expanding (8.1) gives

$$\theta = h - \frac{h^2}{2m} + O(h^3/m^2) = \log k + s - \frac{1}{2\gamma} + o(1).$$

Apply Theorem 8.1 with this shifted constant. $\square$

Other regimes follow from the exact parametrization. If $(\log k)^2 = o(m)$ and $m = o(k)$, the usual critical mass $k(\log k + s) + o(k)$ returns. If $m \sim \gamma \log k$, the leading critical mass is $\gamma(e^{1/\gamma} - 1)k \log k$; the smaller order- k centering depends on the actual integer sequence $m(k)$, so the exact formula should be retained for that precision. For a fixed positive integer m , the critical mass is

$$n = me^{s/m} k^{1+1/m} - mk + O(1).$$

In particular, one column requires mass of order k^2 for coverage.

This is a theorem about a prescribed column count. It is not a theorem about a column weight $u = u_k \rightarrow 0$; relating those two ensembles is a separate research problem.

9 Exact checks and numerical diagnostics

The accompanying programs reproduce the integer checks and numerical tables independently of the typesetting. Integer recurrences, polynomial identities, and determinants are evaluated exactly. The subsequent asymptotic comparisons use 120 decimal digits of working precision. Their role is to detect implementation errors and illustrate the proved estimates; the numerical comparisons do not constitute proofs.

[Added 3 October 2026, batch 85B: this section uses manuscript 06's delivery names. In the report, `code/verify.py` is `code/06-hankel-verify.py`, `code/make_figures.py` is `code/06-hankel-make_figures.py`, each `data/X` is `data/06-hankel-X`, `code/requirements.txt` is `data/06-hankel-requirements.txt`, the figure is `figures/06-hankel-coupon_window.pdf`, and `README.txt` is replaced by the report's `README.md`, which gives rerun instructions that work with these names. Manuscript 01's checks are described in Section 19.]

9.1 Independent exact calculations

The program `code/verify.py` constructs $A(n, k)$ from the nonzero-column composition recurrence (2.2) and obtains $T(n, k)$ by binomial inversion (2.3). Separately, it constructs both Stirling triangles and the Fubini numbers by their integer recurrences, and evaluates the positive transform (2.4). These two routes agree for all $0 \leq n \leq 30$, $0 \leq k \leq 10$: 341 comparisons for each of A and T . All nine supplied rows $n = 0, \dots, 8$ of A261781 also agree exactly [2].

The recurrence checks cover $k = 1, \dots, 10$, with coefficient arrays computed through $n = 85$. They verify coprimality of the exact numerator and denominator, the nonzero impulse at degree $D = k(k+1)/2$, and the homogeneous recurrence beginning at $D+1$, as in Theorem 2.2. The determinant checks verify Theorem 3.1 for $k = 1, \dots, 5$ and shifts 1, 2, 3; the next-size tail determinants vanish for $k \leq 4$. They also check the initial size- $(D+1)$ identity (3.10) at $u = 1$ for $k \leq 4$. Five further cases are exact polynomial identities in the column weight u , including

$$H_{3,1}(u) = 27u^{14}(1+u)^8(1+2u)^2.$$

These finite checks agree with the general algebraic proofs; they do not replace the uniform arguments in those proofs.

9.2 The coverage window

Write $p_{n,k} = T(n, k)/A(n, k)$ and, using the actual integer value of n , set

$$h = \frac{n}{k}, \quad \eta = ke^{-h}, \quad p_0 = e^{-\eta}, \quad p_1 = e^{-\eta} \left(1 + \frac{(1-L)h\eta - (1+h)\eta^2}{2k} \right).$$

Thus p_0 is the moving-mean Poisson approximation and p_1 includes the first correction in Theorem 5.1. Table 3 gives the signed relative errors $\delta_0 = p_{n,k}/p_0 - 1$ and $\delta_1 = p_{n,k}/p_1 - 1$. Here n is the nearest integer to $k \log k$, so the target window coordinate is zero but its actual value $s = n/k - \log k$ is retained. This avoids assigning integer-rounding effects to the asymptotic error.

The probability calculation uses explicit error accounting. Put $E_n = B\rho^{n-1}$, with the constants of Theorem 4.1. For $1 \leq j < k$ define

$$\tilde{m}_j = \binom{k}{j} \frac{c_{k-j} d_{k-j}^m}{c_k d_k^m}, \quad m_j = \mathbb{E} \binom{M}{j}.$$

k	n	Actual s	$p_{n,k}$	δ_0	δ_1
20	60	+0.004268	0.339714114626	-8.04855×10^{-2}	-4.60223×10^{-3}
50	196	+0.007977	0.356935471459	-3.74272×10^{-2}	-9.74476×10^{-4}
100	461	+0.004830	0.361879493451	-2.10377×10^{-2}	-3.02538×10^{-4}
500	3107	-0.000608	0.365694896898	-5.33335×10^{-3}	-1.86151×10^{-5}
1000	6908	+0.000245	0.366903156844	-2.89783×10^{-3}	-5.38847×10^{-6}

Table 3: Coverage diagnostics at the rounded central window. The displayed probabilities and errors are rounded from 120-digit evaluations; both approximations use the actual n .

Whenever $E_n < 1$, the dominant-pole estimate gives

$$|m_j - \tilde{m}_j| \leq \frac{2E_n}{1 - E_n} \tilde{m}_j.$$

Odd and even Bonferroni sums, enlarged by the summed error on their terms, therefore give analytic lower and upper bounds. The computation stops when the last retained moment is below 10^{-70} , or at $j = k$, where the finite inclusion–exclusion sum is complete. All 15 tabulated cases at target $s = -1, 0, 1$ are supplied in the data files. Five cases with $n \leq 200$ were also compared against independently computed exact integer ratios. Thirty additional evaluations check the uniform pole bound against exact unrestricted coefficients.

The truncation and pole-replacement errors are thus bounded analytically. The decimal endpoints themselves use high-precision floating arithmetic and have *not* been certified by directed rounding. The stated numerical precision is a diagnostic setting, not an interval-arithmetic certificate.

9.3 The dense diagonal and two correction coefficients

For A261784 define the normalized exact count

$$\mathcal{R}_N = \frac{T(2N, N)}{c_* d_*^N (N!)^2 / N},$$

with c_*, d_* from (7.20) and (7.19). The all-orders expansion (7.17), specialized at $\lambda = 2$, gives

$$\mathcal{R}_N = 1 + \frac{\beta_1}{N} + \frac{\beta_2}{N^2} + O(N^{-3}), \quad \beta_1 = \beta.$$

There is an exact coefficient rule for the second constant:

$$\beta_1 = E_1(2) - \frac{1}{8}, \quad \beta_2 = E_2(2) - \frac{E_1(2)}{8} + \frac{1}{128}. \quad (9.1)$$

Here E_1, E_2 are obtained from the finite Gaussian-moment rule (7.16); the last two terms arise from the central-binomial expansion. Evaluating that rule, independently of the diagonal integer data, gives

$$\begin{aligned} \beta_1 &= -0.521806701556482663889144103069 \dots, \\ \beta_2 &= 0.128288828088705778459796704234 \dots \end{aligned}$$

Thus β_2 is a derived coefficient rather than a fitted constant. The finite arithmetic expression used for its evaluation is included in `code/verify.py`.

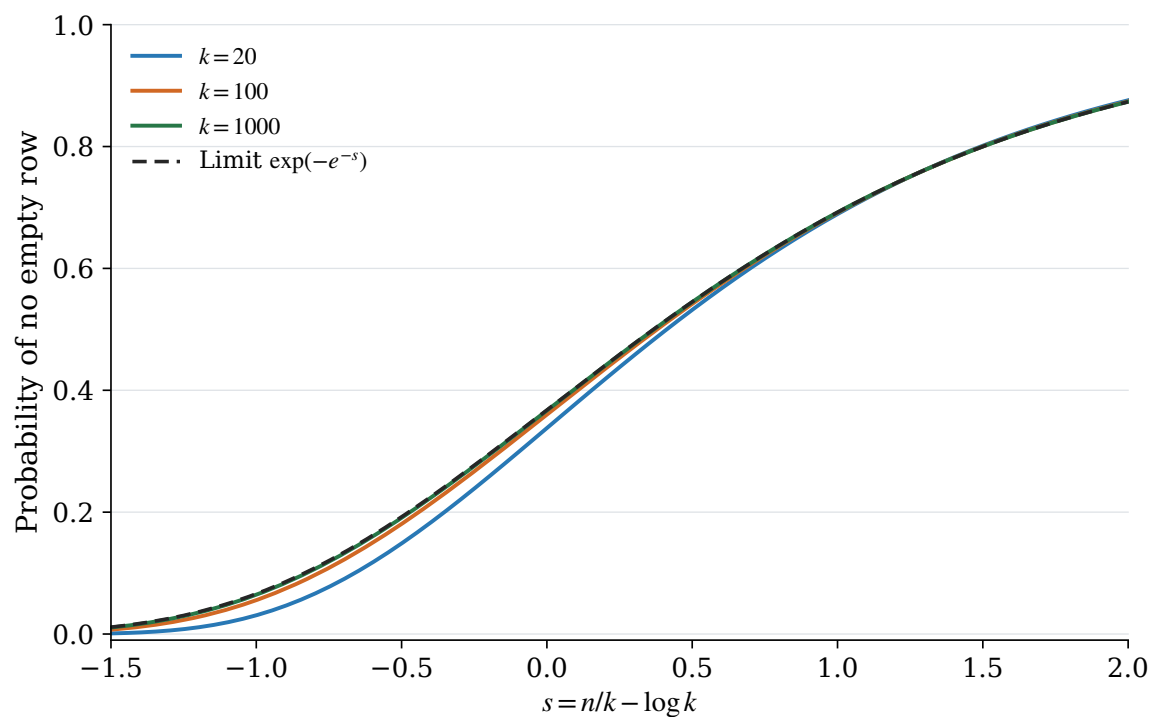


Figure 1: The empty-row transition. Finite-height curves connect evaluations at integer masses for $k = 20, 100, 1000$; the dashed curve is $\exp(-e^{-s})$. The horizontal coordinate always uses the actual $s = n/k - \log k$. The probabilities are evaluated by the bounded Bonferroni calculation described above.

To display the convergence, define

$$U_N = N(\mathcal{R}_N - 1), \quad V_N = N^2 \left(\mathcal{R}_N - 1 - \frac{\beta_1}{N} \right),$$

$$W_N = N^3 \left(\mathcal{R}_N - 1 - \frac{\beta_1}{N} - \frac{\beta_2}{N^2} \right).$$

Theorem 7.1 and its all-orders refinement imply $U_N \rightarrow \beta_1$, $V_N \rightarrow \beta_2$, and boundedness of W_N . The numerator $T(2N, N)$ in every row of Table 4 is computed as an exact integer from (2.4); only the normalization uses floating arithmetic.

N	U_N	V_N	W_N
5	-0.49620219	0.12802256	-0.00133133
10	-0.50899740	0.12809304	-0.00195793
20	-0.51539780	0.12817802	-0.00221621
30	-0.51753296	0.12821232	-0.00229519
50	-0.51924187	0.12824171	-0.00235602
75	-0.52009661	0.12825702	-0.00238568
100	-0.52052405	0.12826482	-0.00240032

Table 4: Dense-diagonal diagnostics from exact integer counts. The first two columns approach the independently derived correction coefficients; the final column records the next scaled remainder.

At $N = 100$, the signed relative error after including β_1/N is 1.28938×10^{-5} ; after also including β_2/N^2 it is -2.41288×10^{-9} . These are finite numerical comparisons, not uniform error constants for the asymptotic theorem.

[Added 3 October 2026, batch 85B: manuscript 01’s Table 7 reports the same comparisons with the opposite sign convention, approximation divided by exact count minus one; at $N = 100$ it lists -1.289×10^{-5} and $+2.413 \times 10^{-9}$, the same two numbers to first order. Its exact counts a_N , $N \leq 200$, come from its own program.]

9.4 Reproduction and data

The package includes the complete L^AT_EX source, compiled article, both Python programs, their pinned dependencies, the figure in PDF and PNG form, and CSV/JSON outputs. From the package root, `python3 code/verify.py` regenerates the checks and tables; `python3 code/make_figures.py` regenerates the plot. The `README.txt` gives the dependency and L^AT_EX build commands. The programs use no network access after their dependencies are installed. The supplied run records report Python 3.12.14, SymPy 1.14.0, mpmath 1.3.0, and matplotlib 3.10.8.

[Added 3 October 2026, batch 85B: in this report the compiled article is a build of the merged text, not manuscript 06’s delivered PDF, and the names are those listed at the start of this section. Both programs write unprefixed files under their output root, which defaults to the directory above `code/`, and `06-hankel-make_figures.py` imports its sibling as `verify`; they must therefore be rerun on a copy with the delivered names restored, as the report README describes, never in the report directory. At the intake’s rerun on a copy (Python 3.13.5) `verify.py` passed in about 5 s of Python time and the figure program took about 74 s, against the 0.3 s and 2.3 s of the delivered README; the six CSV files were byte-identical, five JSON files equal apart from line endings, and the PNG figure pixel-identical.]

10 Further research: precise questions and useful next steps

The following are proposed research problems motivated by the results above. They are not assertions that every formulation is open throughout the literature. The source audit was narrower than that. Each question identifies a statement not proved in this article and a concrete way in which its resolution would extend the present results.

Research question 1 (A uniform approximation across the triangle). Can one obtain relative asymptotics for $T(n, k)$ that remain valid from the near-diagonal boundary $n - k = o(k)$ through the dense rays and into the coverage window $n/k = \log k + O(1)$?

There are two distinct boundaries to resolve. As $n/k \downarrow 1$, the saddle r tends to zero, the variance parameter b tends to zero, and the compact-set saddle proof no longer applies. At the endpoint, $T(n, n) = F_n$ exactly. As n/k grows, the saddle escapes to infinity and the kernel estimates must hold on expanding disks. A useful first deliverable is a uniform formula for $n = k + \ell$ with ℓ growing through the range where a fixed-defect expansion ceases to be accurate. A second is a proved overlap region in which the positive-transform saddle estimate and the empty-row probability estimate agree, including their relative errors. Such a result would turn the separate formulas in this article into a controlled asymptotic description of a much larger part of A261781.

Research question 2 (Wider windows and higher coverage corrections). For which explicit ranges of $s_k = n/k - \log k$ can the Poisson–Charlier expansion of Section 5 be extended, and can all its correction polynomials be generated with a uniform summable remainder?

The theorem here keeps $\eta = ke^{-n/k}$ in a fixed compact subset of $(0, \infty)$. If η grows, the factorial-moment sums are concentrated at growing indices, so the same fixed-disk argument does not directly give a useful relative estimate for the small probability $M = 0$. If η tends to zero, a different error scale is appropriate. Expanding the exact ratios $c_{k-j}d_{k-j}^n/(c_k d_k^n)$ suggests a constructive series in Poisson–Charlier polynomials, but the admissible joint growth of j , η , and h must be proved. A satisfactory result would state both an explicit range and a remainder that remains smaller than the probabilities or distances being approximated.

[Added 3 October 2026, batch 85B: for the single event $M = 0$ and bounded s , the second half of this question is answered by manuscript 01: Theorem 13.1 gives every correction polynomial of $\mathbb{P}(M = 0)$ by a finite algorithm, with a uniform remainder $O((1+h)^{J+1}/k^{J+1})$ after order J . The question stays open for the whole law of M (the generating function (5.3) and the probability-mass expansion beyond the first correction) and for every window wider than bounded s .]

Research question 3 (Joint local limits and residual dependence). Can the Gaussian–Poisson convergence in Section 6 be strengthened to a joint local limit theorem with an explicit first correction?

For example, one can ask for a uniform approximation to $\mathbb{P}(C = c, M = m)$ for integer c in a growing central range and m in a specified range, and for the corresponding law conditioned on $M = 0$. The present moment-generating-function proof establishes distributional independence after normalization, but it does not control the Fourier integral away from the origin. A local theorem requires both this off-axis control and a lattice analysis. The first mixed correction would quantify the small dependence that survives before taking the limit, and would refine the order-one column-moment formulas on dense rays. This is a natural setting in which to compare joint Edgeworth and Poisson–Charlier expansions.

Research question 4 (Small column weights and equivalence of ensembles). How does the coverage transition change when the column weight $u = u_k$ tends to zero, and when does conditioning its weighted model on $C = m$ preserve the corresponding asymptotic laws?

For any fixed m , conditioning on $C = m$ removes the factor u^m , so the resulting distribution is exactly the prescribed-column model. The analytic difficulty is making that conditioning asymptotically uniform when its probability may be small. The weighted pole estimate in Section 4 is uniform on compact positive weight intervals; it does not supply a uniform spectral gap as $u \downarrow 0$. The explicit critical parameter $m \log(1 + n/(km))$ and its shift at $m \asymp (\log k)^2$ provide testable targets for a joint saddle analysis in z and u . Extending the prescribed-column theorem from $m = o(k)$ to comparable or larger column counts is another concrete part of this problem.

Research question 5 (A combinatorial explanation of determinant positivity). Is there a direct combinatorial model for the coefficients of $H_{k,s}(u)$, or a sign-reversing involution proving the product in Theorem 3.1?

The entries of the determinant count matrices, but the determinant itself is a signed sum of products of such counts. Its coefficientwise positivity is presently explained by spectral factorization. A direct model would explain the factors $\binom{k}{j}^j$ and the powers of $(1+u)^m - u^m$, and might expose refinements that the one-variable spectrum does not reveal. The negative smaller minor exhibited in Section 3 rules out a blanket appeal to total positivity. One should seek a model specific to these maximal minors, or a precise classification of which other minors have a definite sign.

[Added 5 October 2026, batch 98: manuscript 08, the source of Part III, asks the same question (its question 8, printed only here) and adds which arithmetic factors a successful model would have to explain: the gcd multiplicities, the degree lost at the trivial root of unity, and the positive factors $(1+u)^q - u^q$ (Section 25); it stresses that the target is the maximal determinant itself, since positivity of all smaller Hankel minors is not available as a general shortcut. Its zero analysis does not answer the question, which stays open. The classification of other minors is continued by Research question 15.]

Research question 6 (Recurrences in finite characteristic). What is the minimal eventual recurrence of the column $T(n, k)$ over $\mathbb{F}_p$, and how does it depend on k and the prime p ?

Over characteristic zero every pole survives and all root sets are disjoint. Reduction modulo a prime can merge roots, make multiplicities inseparable, or cancel numerator and denominator factors. The determinant product isolates the possible exceptional primes through binomial coefficients, powers of 2, and factors $2^m - 1$. It therefore supplies an arithmetic guide, but does not by itself classify the remaining rank or the reduced rational function at each exceptional prime. An exact classification could yield new divisibility and periodicity statements for the triangle, with fast recurrence-based proofs suitable for individual OEIS entries.

Research question 7 (The limiting distribution of determinant zeros). After removing the factors at $u = 0$ and $u = -1$, what is the limiting zero-counting measure of $H_{k,s}(u)$ as $k \rightarrow \infty$?

Every remaining zero has real part $-1/2$ and is the image of a root of unity under $\zeta \mapsto 1/(\zeta - 1)$. Its multiplicity is controlled by the weighted gcd sum over pairs $1 \leq i < j \leq k$. Determining the normalized measure therefore reduces to a specific arithmetic counting problem, rather than an unknown root-location problem. Useful targets include an explicit density along the vertical line, tail estimates for zeros with large imaginary part, and the contribution of roots of small order. The answer should specify whether the zero measure is normalized by the full degree or by the degree remaining after the two exceptional factors are removed.

[Added 5 October 2026, batch 98: answered in Part III. Normalized by the degree remaining after the two exceptional factors are removed, the zero measure converges to the image on $\operatorname{Re} u = -\frac{1}{2}$ of the Cauchy law of scale $\frac{1}{2}$, with density $2/(\pi(1+4y^2))$ in the imaginary coordinate, and the Kolmogorov distance is $\sim (9/\pi^2)(\log k)/k$ (Theorem 24.1, Corollary 28.3). Roots of order m carry the normalized mass $3 \log k/(\zeta(2)\psi(m)k) + O_m(1/k)$ each (Corollary 25.4); the tails at height $\asymp k$ follow a divisor-sum law, matched to the Cauchy tail at every intermediate height (Theorems 29.2 and 30.2); and under the full degree the limit is a three-component mixture determined by $\lim s/k^2$ (Theorem 24.3). Finer questions, among them the second term of the distance and the last few zeros, are Research questions 9–15.]

Research question 8 (Which column models have the same coverage law?). Which replacements for the unrestricted column generating function $(1-z)^{-k} - 1$ retain the window $n = k(\log k + s)$, and which alter its first correction or its scale?

Natural examples impose bounded entries, bounded column sums, or a weight depending on column size. The row symmetry needed for exact factorial moments may survive even when the root-of-unity spectrum does not. One can aim for sufficient analytic conditions on the dominant pole and its row-deletion ratios that imply a Poisson law, then identify which derivatives determine the first correction. A theorem formulated this way would separate properties special to matrix compositions from properties shared by a wider class of symmetric occupancy models.

10.1 A practical order of attack

For a continuation in ProveIt, the most direct algebraic project is to formalize the reduced denominator and its no-cancellation proof, followed by the resultant and maximal Hankel determinant identities. These statements are finite algebra and have explicit base conventions. They provide a manageable route to a proof-assistant-checked contribution without first formalizing saddle-point analysis.

[Added 3 October 2026, batch 85B: manuscript 01 names the same first target (Section 20). Nothing of either manuscript is formalized; the only formal neighbour is the Lean Fubini generating function cited in Section 1, and this paragraph is a proposal, not a plan of the repository.] [Added 5 October 2026, batch 98: manuscript 08 extends the same first stage by the cyclotomic multiplicities, the exact degree identities and the full-support CDF comparisons, then the primitive-lattice error and the elementary CDF bound (Research question 16); also a proposal only.]

For analytic work, the highest-value next targets are the near-diagonal boundary and a proved overlap with the coverage regime. The positive Stirling transform avoids cancellation and gives an explicit starting point in both cases. The small-weight problem is a second substantial direction, because the prescribed-column result already identifies a change of scale that a uniform two-variable theory must explain. The computation scripts in the accompanying package can supply regression examples for these projects, but every extension needs a new proof of its claimed uniformity and error bounds.

Part II

Finite spectra, uniform asymptotics, and two Poisson limits

11 Provenance, scope and notation of Part II

[Added 3 October 2026, batch 85B.] Part II was added in the write of batch 85B. It is built from manuscript 01 of batch 85 and prints everything that manuscript adds to Part I. Where the two manuscripts prove the same statement, the statement is printed once, in Part I, with a bracketed note naming manuscript 01’s version; Table 5 lists those second routes. Everything in Part II is written in Part I’s letters (Table 2).

Standing assumption: $u = 1$. Manuscript 01 never weights columns. *Every statement of Part II is at column weight $u = 1$:* its $A(n, k)$, $T(n, k)$, c_k , d_k , $F_k(z; 1)$ and M are the unweighted objects of Part I, and its generating variable ξ in Section 16 marks the Stirling defect, not columns. No weighted analogue of a Part II result is claimed. Several of its proofs use $u = 1$ explicitly: the residue $-a_k/(2k)$, the bound $a_k \geq \frac{1}{2}$, the Fubini pole at $L = \log 2$ and the constants of Section 13 are the $u = 1$ values of Part I’s $-a_{k,u}/((1+u)k)$, $a_{k,u} \geq b$, L_u and $\mu_k(u)$, $d_k(u)$.

11.1 The manuscript

Manuscript 01 of batch 85 is the archive `OEIS_Matrix_Compositions_Research.zip` (wrapper directory `matrix_compositions_research/`, 26 files), which arrived in commit `317c1ce2e` and was placed in this report by commit `fa2f3e419`. Its article is titled *Matrix compositions: finite spectra, uniform asymptotics, and two Poisson limits*, subtitled *Proofs for OEIS A261781 and all-orders refinements for A261784* (22 pages, dated 3 October 2026). Its title page and PDF metadata read “Research report prepared for Vladimir Reshetnikov”; it names no author and no tool. Its pin is the repository tree revision `d68b65ea064084fb121df9bb431b21d086f7be81` (2026-10-03, 17:08), at which it read the series-and-transseries overview `Analysis/Transseries/docs/series-and-transseries/README.md` (blob `7bdc83aeb`) as motivation for its inversion section; its indexed searches for A261781 and A261784 returned nothing and exposed a different revision, `ce37e13f4` [9].

It was written without sight of manuscript 06, the source of Part I, which arrived thirteen minutes later in commit `345284ac8`. The two texts share 0.7% (01 in 06) and 0.5% (06 in 01) of their word 8-grams, almost all of them bibliography entries. Every number they share agrees: the constants r , b , v , d_* , c_* , β_1 , β_2 in about seventy printed digits, computed by two independent programs, and the coverage probabilities they both tabulate.

Status of manuscript 01, as delivered. It is a proof-bearing research report, not a Lean-certified or independently refereed publication. The two OEIS conjecture labels were present in the entry it inspected. The A261784 leading asymptotic was already posted; it is not claimed as a new prediction, and the entry is not said to label it a conjecture. Priority for the additional uniform results is not established by its bounded literature search. It separates the *proof* of a statement from *priority* for that statement, and says that no uninspected repository theorem is assumed and no Lean certification is inherited merely by association.

11.2 What is new, and a correction

Manuscript 01's abstract says that it proves the two conjectures posted in A261781, the recurrence order $k(k+1)/2$ of column k and the limit $d_{k+1} - d_k \rightarrow 1/L$; its delivered README and its OEIS draft (Section 22) say the same. That claim is narrowed here.

- A homogeneous constant-coefficient recurrence of order $\binom{k+1}{2}$ for column k , with the product denominator $Q_{k,1}(z) = \prod_{j=1}^k (2(1-z)^j - 1)$, is Proposition 25 of Munarini, Poneti and Rinaldi [4] (printed page 16), published in 2009, eight years before the conjecture; their Remark 26 prints the cases of two and three rows, the first of which is manuscript 01's example (2.8). Manuscript 01 cites Propositions 1, 12, 13 and 29 of that paper, but not Proposition 25 or Remark 26. What it proves that is new is the *minimality*: the product is the reduced denominator, so no eventual constant-coefficient recurrence of smaller order exists, even over $\mathbb{C}$. Manuscript 06 states this correctly (Section 1 and Remark 2.3).
- The limit $d_{k+1} - d_k \rightarrow 1/L$ is immediate from the closed form of [4, Proposition 13], as both manuscripts say. Manuscript 01's genuine strengthening is that the increments *increase strictly* and stay below $1/L$ (Theorem 12.1).
- Manuscript 01 says that Louchard's paper [6] was identified through A261780 but not retrieved. That remains a true statement about its own preparation and its non-claim stands; but the paper is retrievable, and manuscript 06 inspected the author's long version and found no matching coverage-window statement (Section 1). The intake confirmed that the author's publication list carries it as item 99 but did not read it.

The generating functions and the positive Stirling–Fubini transform are Munarini, Poneti and Rinaldi's, and both manuscripts credit them; the leading A261784 law, d_* and the decimal value of c_* are Kotěšovec's [3].

New in manuscript 01 relative to manuscript 06, and printed below: strict monotonicity of the increments (Section 12); every fixed order of the coverage probability (Section 13); the entire Stirling kernel with its differential recurrence, its third polynomial and a second exact algorithm (Section 14); a recursion for the cumulants (Section 15); the Poisson law of the Stirling defect (Section 16); the third diagonal correction and fifty-digit constants (Section 17); an asymptotic inversion with integer threshold brackets, which is an instance of the repository's inversion apparatus (Section 18); exact-ratio coverage and defect tables (Sections 16 and 19); and its research questions and proof audit (Sections 20–22).

11.3 Manuscript 01's second routes to Part I's results

Results of Part I that manuscript 01 does not have: the column weight u throughout; the initial impulse (Remark 2.4); the whole Hankel section (Theorem 3.1); the uniform constant of Theorem 4.1; the coverage generating function, the probability-mass and total-variation expansions and the moving-mean remark of Section 5; the joint Gaussian–Poisson law of Section 6; the column central limit theorem on dense rays (Corollary 7.5); and the prescribed-column threshold of Section 8.

12 Strict monotonicity of the growth-constant increments

Extend d_k to real $x > 0$ by $d(x) = (1 - 2^{-1/x})^{-1}$, so that $d_k = d(k)$.

Theorem 12.1 (Strengthened increment statement). *The sequence $d_{k+1} - d_k$, $k \geq 1$, is strictly increasing, is always below $1/L$, and tends to $1/L$.*

Part I	Manuscript 01's version and route
(2.1)–(2.3), (2.4)	the same classical identities at $u = 1$, credited to [4]; the Stirling transform proved by the same geometric expansion and finite difference
Theorem 2.2	the same theorem at $u = 1$, same proof (Remark 2.3); the recurrence written from $n > D$
(2.9), Corollary 2.5	the same constants and expansions of d_k, c_k ; adds strict monotonicity (Theorem 12.1); lacks (2.11)
Theorem 4.1	the bound $(k - 1)\rho^{n+1}$, $u = 1$ only (Remark 4.2)
(5.4), Corollary 5.3	the same expansion of $\mathbb{P}(M = 0)$ and Poisson limit, by Bonferroni truncation of inclusion–exclusion (note after Theorem 5.1); no generating function in complex w , no probability-mass or total-variation statement
(7.5)	the same lemma, same proof
(7.7), Lemma 7.2	the untruncated kernel W_n , with no hypothesis on k (Remark 7.3, Section 14)
Theorem 7.1, (7.16), (7.17)	the same theorem and prescription; tails by $ W_n(w) \leq e^{ w }$ (notes after the proof of Theorem 7.1 and after (7.17))
Corollary 7.4, (9.1)	the same constants and β_1, β_2 ; adds β_3 and all orders (Section 17)
ED, Var D in Corollary 7.5	implied by the Poisson law of $\Delta_{n,k} = D$ (Theorem 16.2)
Table 3, Table 4	Tables 8 and 7, from its own exact counts; equal where they overlap

Table 5: Results of Part I that manuscript 01 proves as well, and its version of each. None is reprinted in Part II.

Proof. Put $t = L/(2x)$. Since $d(x) = \frac{1}{2} + \frac{1}{2} \coth t$,

$$d'(x) = \frac{1}{L} \left(\frac{t}{\sinh t} \right)^2. \quad (12.1)$$

This lies strictly between 0 and $1/L$. The function $t/\sinh t$ is strictly decreasing for $t > 0$, because $\sinh t - t \cosh t < 0$; the latter expression vanishes at 0 and has derivative $-t \sinh t < 0$. As x increases, t decreases; thus d' increases strictly to $1/L$. Integrating d' over $[k, k + 1]$ proves the claim. $\square$

Remark 12.2 (Further details from manuscript 01). (i) The inverse of (2.3) is $A(n, k) = \sum_{j=0}^k \binom{k}{j} T(n, j)$: choose the nonzero rows. Also $T(n, k) = 0$ for $k > n$, and the rational expression (2.1) at $u = 1$ equals 1 for $k = 0$, as it must. (ii) The bijection with the sorted-word compositions of A261780 and A261781 lists the row label i as many times as the entry in row i of a column, in increasing order; column boundaries become part boundaries, and the construction is reversible. (iii) At the unique smallest-modulus pole $1 - a_k$ only the $j = k$ summand of (2.3) contributes, with residue $-a_k/(2k)$, so its coefficient contribution is $\frac{a_k}{2k}(1 - a_k)^{-n-1} = c_k d_k^n$. (iv) The alternating sum (2.3) explains why cancellation is a numerical concern when all rows must be occupied; the positive transform (2.4) is an exact computational alternative to it.

13 Every fixed order in the coverage window

For real $x \rightarrow \infty$ the exact constants have the expansions

$$\log d(x) = \log \frac{x}{L} + \frac{L}{2x} - \frac{L^2}{24x^2} + O(x^{-4}), \quad (13.1)$$

$$\log c(x) = -\log(2L) - \frac{L}{2x} - \frac{L^2}{24x^2} + O(x^{-4}), \quad (13.2)$$

where $c(x) = (d(x)-1)/(2x)$; these are the identities for $\log d_m$ and $\log c_m$ in the proof of Theorem 5.1, with $g_0(x) = -L^2x^2/24 + O(x^4)$, and both differences from the leading logarithms are analytic in $1/x$ at 0. Let $\mathcal{T}_a(x)$ denote the Touchard polynomial, defined by $\sum_{j \geq 0} j^a x^j / j! = e^x \mathcal{T}_a(x)$.

Theorem 13.1 (All fixed orders for coverage). *Let $k \rightarrow \infty$ through integers, with $h = n/k = \log k + s$ and $\eta = ke^{-h} = e^{-s}$, where s stays in a fixed compact real interval. For every fixed $J \geq 0$ there are polynomials $\mathcal{Q}_1, \dots, \mathcal{Q}_J$ in h and η , with coefficients polynomial in L over $\mathbb{Q}$, such that*

$$\frac{T(n, k)}{A(n, k)} = \mathbb{P}(M = 0) = e^{-\eta} \left[1 + \sum_{j=1}^J \frac{\mathcal{Q}_j(h, \eta; L)}{k^j} + O\left(\frac{(1+h)^{J+1}}{k^{J+1}}\right) \right], \quad (13.3)$$

uniformly in such s . Each $\mathcal{Q}_j$ has degree at most j in h , and

$$\mathcal{Q}_1(h, \eta; L) = \frac{1}{2}((1-L)h\eta - (h+1)\eta^2).$$

For $J = 1$ this is (5.4).

Proof. Put $\epsilon = 1/k$ (a local letter; Part I's ϵ in Section 7.4 is $1/n$). Using the analytic expansions behind (13.1)–(13.2), form the formal power series in ϵ

$$\Psi(j, h, \epsilon) = \sum_{i=0}^{j-1} \log(1 - i\epsilon) + \log \frac{c(\epsilon^{-1} - j)}{c(\epsilon^{-1})} + \frac{h}{\epsilon} \left(\log \frac{d(\epsilon^{-1} - j)}{d(\epsilon^{-1})} + j\epsilon \right). \quad (13.4)$$

By (5.2) and Theorem 4.1, $\mathbb{E}(M)_j = \eta^j \exp \Psi(j, h, 1/k)$ up to the exponentially small spectral errors. The cancellation in the last parentheses removes its linear leading term. Every coefficient of Ψ is a polynomial in j, h, L , and its dependence on h is at most linear. Hence $[\epsilon^i]e^\Psi$ is a polynomial in j, h, L of degree at most i in h . Replace each monomial j^a in that coefficient by $\mathcal{T}_a(-\eta)$ to obtain $\mathcal{Q}_i$; this is the summation $\sum_j (-\eta)^j j^a / j! = e^{-\eta} \mathcal{T}_a(-\eta)$ of the inclusion–exclusion series $\mathbb{P}(M = 0) = \sum_j (-1)^j \mathbb{E}(M)_j / j!$.

To justify the operation, truncate at $j \leq C \log k$. Taylor's formula applied to the finitely many logarithms and to (13.1)–(13.2), continued to the required order, bounds the moment remainder by

$$\frac{\eta^j}{k^{J+1}} C_J (1+h)^{J+1} (1+j)^{C'_J}$$

for constants C_J, C'_J on the chosen s -interval. The exponential of the small discarded logarithm is uniformly bounded in this range. Summing against $1/j!$ preserves the stated error, and the Bonferroni inequalities bound the tail superalgebraically, as in the second route noted after Theorem 5.1. This proves both the algorithm and the uniform remainder. $\square$

The appearance of $\log k$ inside the coefficients, through h , is intrinsic to the transition scale. Theorem 13.1 is an all-orders Poincaré expansion of the single probability $\mathbb{P}(M = 0)$; it does not assert convergence as $J \rightarrow \infty$, nor a classification of exponentially small terms, and it does not extend Part I's expansion of the whole law of M beyond the first correction (see the note to Research question 2 in Section 10).

14 The entire Stirling kernel

For $n \geq 1$ define, with the falling factorial $(n)_\ell$ of Part I and $(n)_0 = 1$,

$$\chi_{n,\ell} = \frac{\left[\begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right]}{(n)_\ell^2}, \quad W_n(w) = \sum_{\ell=0}^{n-1} \chi_{n,\ell} w^\ell, \quad (14.1)$$

and set $\chi_{n,\ell} = 0$ outside $0 \leq \ell < n$. By Remark 7.3, Part I's kernel is the truncation $H_{n,k}(z) = \sum_{\ell \leq n-k} \chi_{n,\ell} (Lz)^\ell$.

Proposition 14.1 (Positive coefficient representation). *Let $f(z) = e^z - 1$ and*

$$\tilde{T}(n, k) = \frac{n!}{2L^{n+1}} [z^n] f(z)^k W_n(Lz). \quad (14.2)$$

For $n \geq k \geq 1$, $T(n, k) = \tilde{T}(n, k)(1 + O_\vartheta(\vartheta^k))$.

Proof. In (2.4) replace F_q using (7.5), use $\{q_k\} = q! [z^q] f(z)^k / k!$ and put $q = n - \ell$. The resulting exact main sum is

$$\frac{1}{2n! L^{n+1}} \sum_{\ell=0}^{n-1} \left[\begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right] ((n-\ell)!)^2 L^\ell [z^{n-\ell}] f(z)^k,$$

which is (14.2). Positivity of every summand transports the relative error. $\square$

Lemma 14.2 (Bounds and an exact differential recurrence). *For $0 \leq \ell < n$, $0 \leq \chi_{n,\ell} \leq 1/\ell!$; consequently $|W_n(w)| \leq e^{|w|}$ for every complex w . With $D_w = w d/dw$,*

$$(n+1)^2 W_{n+1}(w) = ((n+1 - D_w)^2 + nw) W_n(w), \quad W_1(w) = 1. \quad (14.3)$$

Proof. The first-kind coefficient $\left[\begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right]$ is the ℓ th elementary symmetric polynomial of $1, 2, \dots, n-1$. Each of its $\binom{n-1}{\ell}$ summands is at most $(n-1)_\ell$. Hence

$$\chi_{n,\ell} \leq \frac{(n-1)_\ell^2}{\ell! (n)_\ell^2} \leq \frac{1}{\ell!}.$$

The first-kind recurrence gives

$$\chi_{n+1,\ell} = \frac{(n+1-\ell)^2}{(n+1)^2} \chi_{n,\ell} + \frac{n}{(n+1)^2} \chi_{n,\ell-1}.$$

Multiplying by w^ℓ and summing proves (14.3). $\square$

Theorem 14.3 (Uniform entire kernel expansion). *For each fixed $J \geq 0$ there are rational polynomials $P_0, \dots, P_J$, with $P_0 = 1$ and $\deg P_r \leq 2r$, such that*

$$W_n(w) = e^{w/2} \left(\sum_{r=0}^J \frac{P_r(w)}{n^r} + O_{J,K}(n^{-J-1}) \right) \quad (14.4)$$

uniformly on every compact $K \subset \mathbb{C}$, with no condition relating n to any k . The expansion may be differentiated any fixed number of times, uniformly on compact sets. The polynomials are those of (7.15):

$$\begin{aligned} P_1(w) &= \frac{w^2}{12} - \frac{w}{2}, & P_2(w) &= \frac{w^4}{288} - \frac{w^2}{12}, \\ P_3(w) &= \frac{w^6}{10368} + \frac{w^5}{576} + \frac{w^4}{480} - \frac{w^3}{24} - \frac{w^2}{12}. \end{aligned} \quad (14.5)$$

Proof. This is a coefficient-level proof with a uniform remainder. Let $\Sigma_j(n) = \sum_{a=1}^{n-1} a^j$, regarded as its Faulhaber polynomial in n . With $\epsilon = 1/n$ put $\mathbf{a}_j(\epsilon) = \epsilon^{2j} \Sigma_j(1/\epsilon)$. This is a polynomial beginning in degree $j - 1$, and $\mathbf{a}_1(\epsilon) = (1 - \epsilon)/2$. For a fixed integer $\ell \geq 0$ define

$$\mathbf{e}_\ell(\epsilon) = [t^\ell] \exp \left(\sum_{j=1}^{\ell} \frac{(-1)^{j-1}}{j} \mathbf{a}_j(\epsilon) t^j \right), \quad \mathbf{r}_\ell(\epsilon) = \mathbf{e}_\ell(\epsilon) \prod_{a=0}^{\ell-1} (1 - a\epsilon)^{-2}. \quad (14.6)$$

Newton's identity for elementary symmetric functions shows that $\mathbf{r}_\ell(1/n) = \chi_{n,\ell}$ whenever $n > \ell$. In particular $\mathbf{r}_\ell(0) = 1/(2^\ell \ell!)$.

The exponent in (14.6) is $t/2 + \sum_{r \geq 1} \epsilon^r \mathbf{b}_r(t)$ with $\deg \mathbf{b}_r \leq r + 1$. Thus at total order ϵ^r the part multiplying $e^{t/2}$ has degree at most $2r$ in t , and extracting t^ℓ converts $t^q e^{t/2}$ into $2^q(\ell)_q/(2^\ell \ell!)$. The other factor of $\mathbf{r}_\ell$ has logarithm $2 \sum_{r \geq 1} \frac{\epsilon^r}{r} \sum_{a=0}^{\ell-1} a^r$, whose coefficient at order r is a polynomial in ℓ of degree $r + 1 \leq 2r$. Therefore

$$[\epsilon^r] \mathbf{r}_\ell(\epsilon) = \frac{\Pi_r(\ell)}{2^\ell \ell!}, \quad \Pi_r \in \mathbb{Q}[\ell], \quad \deg \Pi_r \leq 2r; \quad (14.7)$$

for example $\Pi_0 = 1$ and $\Pi_1(\ell) = \ell(\ell - 4)/3$, the polynomial $d(d - 4)/3$ of the proof of Lemma 7.2.

Next, the Taylor remainders are bounded uniformly in ℓ . There are absolute $c, C > 0$ such that, for $|\epsilon| \leq c/(\ell + 1)^2$ and $2 \leq j \leq \ell$,

$$|\mathbf{a}_j(\epsilon)| \leq C|\epsilon|^{j-1}. \quad (14.8)$$

For completeness: Faulhaber's formula writes $\mathbf{a}_j$ as $\frac{1}{j+1} \sum_{a=0}^j \binom{j+1}{a} \mathbf{B}_a \epsilon^{j-1+a}$, with Bernoulli numbers $\mathbf{B}_a$. The coefficients of $t/(e^t - 1)$ obey $|\mathbf{B}_a|/a! \leq C_0$ by a Cauchy bound on a fixed circle of radius 1. Since $j!/(j+1-a)! \leq j^{a-1}$ for $a \geq 1$, the sum of the terms after the first is bounded by a geometric series in $j|\epsilon|$.

For $\ell \geq 1$, apply Cauchy's coefficient bound in (14.6) on $|t| = 2\ell$. The $j = 1$ term contributes at most $\ell + O(1)$ to the real part of the exponent, while (14.8) bounds the sum of the terms $j \geq 2$ by a constant, because $|\epsilon|\ell^2$ is bounded; the product $\prod (1 - a\epsilon)^{-2}$ is bounded for the same reason. Stirling's factorial estimate then gives

$$\sup_{|\epsilon| \leq c/(\ell+1)^2} |\mathbf{r}_\ell(\epsilon)| \leq \frac{C\sqrt{\ell+1}}{2^\ell \ell!}; \quad (14.9)$$

for $\ell = 0$ this is immediate. Cauchy's Taylor-remainder estimate at $\epsilon = 1/n$ consequently yields

$$\chi_{n,\ell} = \frac{1}{2^\ell \ell!} \sum_{r=0}^J \frac{\Pi_r(\ell)}{n^r} + O_J \left(\frac{(\ell+1)^{2J+5/2}}{2^\ell \ell! n^{J+1}} \right) \quad (14.10)$$

whenever $\ell \leq \delta_0 \sqrt{n}$, for a sufficiently small absolute $\delta_0 > 0$ and all sufficiently large n .

Multiply (14.10) by w^ℓ and sum. On $|w| \leq V$ the remainder is summable uniformly. Beyond $\delta_0 \sqrt{n}$ the bound $\chi_{n,\ell} \leq 1/\ell!$ makes the original tail smaller than every power of n^{-1} , and the same holds for the polynomially weighted Taylor coefficients in (14.7). Hence

$$\mathbf{W}_n(w) = \sum_{r=0}^J \frac{1}{n^r} \sum_{\ell \geq 0} \frac{\Pi_r(\ell)}{\ell!} \left(\frac{w}{2} \right)^\ell + O_{J,V}(n^{-J-1}).$$

The inner sum is $e^{w/2}$ times a polynomial of degree at most $2r$, by the Touchard identity. This proves (14.4); expanding (14.6) gives P_1, P_2, P_3 . Cauchy's derivative estimate on a slightly larger compact disk gives the differentiated expansions. $\square$

A finite exact coefficient algorithm. The polynomials can be computed without a symbolic parameter ℓ . For a requested order J , compute $e_\ell(n) = \begin{bmatrix} n \\ n-\ell \end{bmatrix}$ for $0 \leq \ell \leq 2J$ by

$$e_0(n) = 1, \quad \ell e_\ell(n) = \sum_{j=1}^{\ell} (-1)^{j-1} e_{\ell-j}(n) \Sigma_j(n), \quad (14.11)$$

expand $e_\ell(n)/(n)_\ell^2$ in powers of $1/n$ to order J , and use $\deg P_r \leq 2r$:

$$P_r(w) = \sum_{m=0}^{2r} w^m \sum_{\ell=0}^m \frac{(-1/2)^{m-\ell}}{(m-\ell)!} [n^{-r}] \frac{e_\ell(n)}{(n)_\ell^2}. \quad (14.12)$$

Here $[n^{-r}]$ is the coefficient in the expansion at infinity, not a coefficient of the polynomial numerator alone. Manuscript 01's program (`code/01-two-poisson-verify.py`) runs (14.11)–(14.12) exactly over $\mathbb{Q}$ and checks (14.3) coefficientwise through $n = 80$. This is a second finite route to Part I's algorithm (7.12)–(7.14). [Added 3 October 2026, batch 85B: the intake checked P_3 numerically: at $w = 1.3$, $n^3(e^{-w/2}W_n(w) - 1 - P_1/n - P_2/n^2)$ is $-0.22070, -0.22010, -0.21981$ for $n = 400, 800, 1600$, against $P_3(1.3) = -0.21951$, with the expected $O(1/n)$ approach.]

15 Complements on compact rays

Manuscript 01 writes the leading factor of Theorem 7.1 as

$$\mathcal{B}(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}}, \quad T(n, k) = \mathcal{B}(n, k) \left[1 + \sum_{j=1}^J \frac{E_j(\lambda)}{k^j} + O_{J,K}(k^{-J-1}) \right]. \quad (15.1)$$

It adds two observations and a recursion.

The variance b . $b = \kappa_2 = \lambda(1 + r - \lambda)$ is the variance of the positive-integer random variable with probabilities proportional to $r^j/j!$, $j \geq 1$, which explains its positivity; the proof needs only the monotonicity of $r/(1 - e^{-r})$.

A recursion for the cumulants. With independent variables z, a , define

$$\mathcal{K}_1(z, a) = a, \quad \mathcal{K}_{j+1} = z \partial_z \mathcal{K}_j + a(1 + z - a) \partial_a \mathcal{K}_j. \quad (15.2)$$

Then $\kappa_j = \mathcal{K}_j(r, \lambda)$. This follows from $\mathcal{D}\lambda = \lambda(1 + z - \lambda)$ for $\lambda(z) = \mathcal{D} \log f(z) = z/(1 - e^{-z})$, before evaluation at the saddle. One must not differentiate a specialized formula while holding λ fixed. (For example $\mathcal{K}_3 = a(2a^2 - 3az - 3a + z^2 + 3z + 1)$, which is the expanded κ_3 printed after Theorem 7.1.) Together with (14.12) and (7.16), this specifies every E_j by finite rational operations and the evaluation of r ; in fact E_j is a rational function of r, λ, L whose denominators are powers of b and λ .

Remark 15.1 (Floors and endpoints). The ratio in Theorem 7.1 is the *actual* n/k . For $n = \lfloor \lambda_0 k \rfloor$, replacing it prematurely by λ_0 inside exponential factors can lose a nonvanishing factor. The theorem is not uniform as $\lambda \downarrow 1$ or $\lambda \rightarrow \infty$; at the boundary $n = k$ the exact identity is $T(n, n) = F_n$, and the coverage window, where $\lambda \sim \log k$, needs the separate proof of Section 5. Manuscript 01's optional program `code/01-two-poisson-ray_spotchecks.py` tests the expansion at $\lambda = 3/2, 3$ and 5 for $40 \leq k \leq 200$ with 75-digit arithmetic (`data/01-two-poisson-ray_spot_checks.csv`); the relative error after the third correction is between 2.38×10^{-12} and 1.63×10^{-9} in absolute value at the seven tested points.

16 The Stirling defect: a second Poisson limit

The positive identity (2.4) supplies an auxiliary probability space. It should not be confused with the zero-row statistic M , which vanishes identically on the objects counted by T .

Definition 16.1. For $n \geq k \geq 1$, define $Q_{n,k}$ on $\{k, \dots, n\}$ by

$$\mathbb{P}(Q_{n,k} = q) = \frac{[n] \{q\} F_q}{\sum_{p=k}^n [n] \{p\} F_p}, \quad \Delta_{n,k} = n - Q_{n,k}. \quad (16.1)$$

$\Delta_{n,k}$ is the *Stirling defect* of the transform. $Q_{n,k}$ and $\Delta_{n,k}$ are Part I's Q and D of Section 7.6.

Theorem 16.2 (Poisson defect and analytic generating functions). *Suppose $n, k \rightarrow \infty$ and $n/k \rightarrow \lambda_0 \in (1, \infty)$. Then*

$$\Delta_{n,k} \xrightarrow{d} \text{Poisson}\left(\frac{Lr(\lambda_0)}{2}\right). \quad (16.2)$$

More precisely, uniformly for complex ξ in compact sets and for n/k in compact subsets of $(1, \infty)$,

$$\mathbb{E}\xi^{\Delta_{n,k}} = e^{v(\xi-1)}(1 + O(k^{-1})), \quad v = \frac{Lr(n/k)}{2}. \quad (16.3)$$

There is an expansion through every fixed order, locally uniformly in ξ , and the same holds for each fixed derivative with respect to ξ .

Proof. Marking $n - q = \ell$ in the main transform replaces $W_n(Lz)$ by $W_n(\xi Lz)$. Lemma 14.2 and positivity bound the Fubini replacement error, also for bounded complex ξ by using absolute values. The proof of Theorem 7.1, with ξ in a compact set, now replaces $e^{Lr/2}$ by $e^{\xi Lr/2}$; this uses Theorem 14.3 on compact sets of $\mathbb{C}$. Division by the unmarked coefficient gives (16.3); the denominator's leading factor is positive and bounded away from zero after normalization, so the division is uniform. Replacing $P_p(Lre^{\tau y})$ by $P_p(\xi Lre^{\tau y})$ in (7.16) gives every higher order. Uniform holomorphy on a larger compact set permits differentiation. Coefficient extraction at $\xi = 0$ yields convergence of each probability to the corresponding Poisson mass, and these limiting masses sum to one. $\square$

Differentiating (16.3) once and twice at $\xi = 1$ gives $\mathbb{E}\Delta_{n,k} = v + O(k^{-1})$ and $\text{Var } \Delta_{n,k} = v + O(k^{-1})$, the two moments of Corollary 7.5. The theorem explains why only a bounded distance below the top Stirling index contributes at leading order, although the positive sum ranges from k to n : the factor $e^v = e^{Lr/2}$ in (15.1) is exactly the aggregate contribution of the limiting Poisson cloud. The report does not assert that $\Delta_{n,k}$ is a canonical statistic on unadorned matrices; finding such a statistic is a separate combinatorial problem (Section 20).

[Added 5 October 2026, batch 98: the collection's report `a122399-surjection-diagonal`, Part II (its Theorems 13.1 and 15.1, labels `a122:pd:thm:trichotomy` and `a122:pd:thm:allorders`), proves the analogous law for a different array: for a uniformly chosen pair of an ordered partition of an m -set and a map of an n -set to its blocks, the defect (the number of missing blocks, m minus the block count) tends to a Poisson law of parameter $e^{-c}/2$ at $n = m(\log m + c)$, with an expansion through every fixed order summed by Touchard polynomials. There the factor $1/2$ comes from defects that are pairs (its Proposition 16.1); its source note names the same mechanism, pairs weighted by $1/2!$, for the parameter $Lr/2$ of Theorem 16.2. No theorem is shared.]

For $n = 2N$, $k = N$ the limiting mean is $v = 0.55230808135934145323\dots$. Direct evaluation of the positive weights in (16.1) gives Table 6.

ℓ	$N = 10$	$N = 50$	$N = 200$	$e^{-v}v^\ell/\ell!$
0	0.592485680	0.578956893	0.576452261	0.575619699
1	0.314681461	0.317313214	0.317769736	0.317919411
2	0.078997588	0.086058333	0.087361728	0.087794730
3	0.012377926	0.015394446	0.015970554	0.016163246
4	0.001344681	0.002042736	0.002184003	0.002231773
5	0.000106192	0.000214396	0.000238309	0.000246525

Table 6: Convergence of $\mathbb{P}(\Delta_{2N,N} = \ell)$ to its Poisson limit (manuscript 01).

17 A261784: the third correction and every further one

Corollary 17.1 (Diagonal expansion to every order). *For every fixed $J \geq 0$,*

$$a_N = c_* d_*^N \frac{(N!)^2}{N} \left(1 + \sum_{j=1}^J \frac{\beta_j}{N^j} + O_J(N^{-J-1}) \right). \quad (17.1)$$

Every β_j is a rational function of L and r , specified exactly by the prescription (7.16) (with (15.2) and (14.12)) and the central binomial expansion. Besides $\beta_1 = \beta$ and β_2 of Part I,

$$\beta_3 = -0.00244353842284859500374038095590013033570627782694 \dots \quad (17.2)$$

Proof. At $\lambda = 2$ one has $f(r) = r/(2-r)$ and $b = 2(r-1)$. Substitute these in (15.1) and use

$$\frac{(2N)!}{(N!)^2} = \frac{4^N}{\sqrt{\pi N}} \left(1 - \frac{1}{8N} + \frac{1}{128N^2} + \frac{5}{1024N^3} + O(N^{-4}) \right). \quad (17.3)$$

With $E_j = E_j(2)$ this gives

$$\beta_1 = E_1 - \frac{1}{8}, \quad \beta_2 = E_2 - \frac{E_1}{8} + \frac{1}{128}, \quad \beta_3 = E_3 - \frac{E_2}{8} + \frac{E_1}{128} + \frac{5}{1024}, \quad (17.4)$$

the first two being (9.1). The same multiplication determines every further coefficient. All computations are finite; the displayed decimals are evaluations of these exact prescriptions, not fitted constants. $\square$

Manuscript 01's ninety-digit evaluation gives, truncated to fifty decimals,

$$\begin{aligned} r &= 1.59362426004004009232304187587516024178900242481885 \dots, \\ d_* &= 12.85568917366612131932660300054233483234414963116428 \dots, \\ c_* &= 0.25886492311146555025177523170232718427705044811049 \dots, \\ E_1(2) &= -0.39680670155648266388914410306946945484974842970590 \dots, \\ E_2(2) &= 0.07087549039414544547365369135067624979402488769624 \dots, \\ E_3(2) &= 0.00463313773232960649210026876816463100456049274216 \dots, \\ \beta_1 &= -0.52180670155648266388914410306946945484974842970590 \dots, \\ \beta_2 &= 0.12828882808870577845979670423435993165024344140947 \dots \end{aligned}$$

[Added 3 October 2026, batch 85B: the intake compared these with manuscript 06's independent seventy-digit values in `data/06-hankel-diagonal_constants.json`; r , b , v , d_* , c_* , β_1 and β_2 agree in every printed digit.]

Let $a_N^{[i]}$ denote (17.1) truncated after β_i/N^i , with $i = 0$ the leading approximation. Table 7 reports $a_N^{[i]}/a_N - 1$, using exact integer a_N and 90-digit working precision, and Figure 2 plots the same quantities for more N . (Part I's Table 4 uses the reciprocal convention.) The very small

N	$i = 0$	$i = 1$	$i = 2$	$i = 3$
10	$+5.363 \times 10^{-2}$	-1.350×10^{-3}	$+2.063 \times 10^{-6}$	-5.117×10^{-7}
20	$+2.645 \times 10^{-2}$	-3.289×10^{-4}	$+2.844 \times 10^{-7}$	-2.917×10^{-8}
50	$+1.049 \times 10^{-2}$	-5.183×10^{-5}	$+1.905 \times 10^{-8}$	-7.075×10^{-10}
100	$+5.232 \times 10^{-3}$	-1.289×10^{-5}	$+2.413 \times 10^{-9}$	-4.344×10^{-11}
200	$+2.613 \times 10^{-3}$	-3.215×10^{-6}	$+3.035 \times 10^{-10}$	-2.691×10^{-12}

Table 7: Relative errors of successive asymptotic approximations to A261784 (manuscript 01).

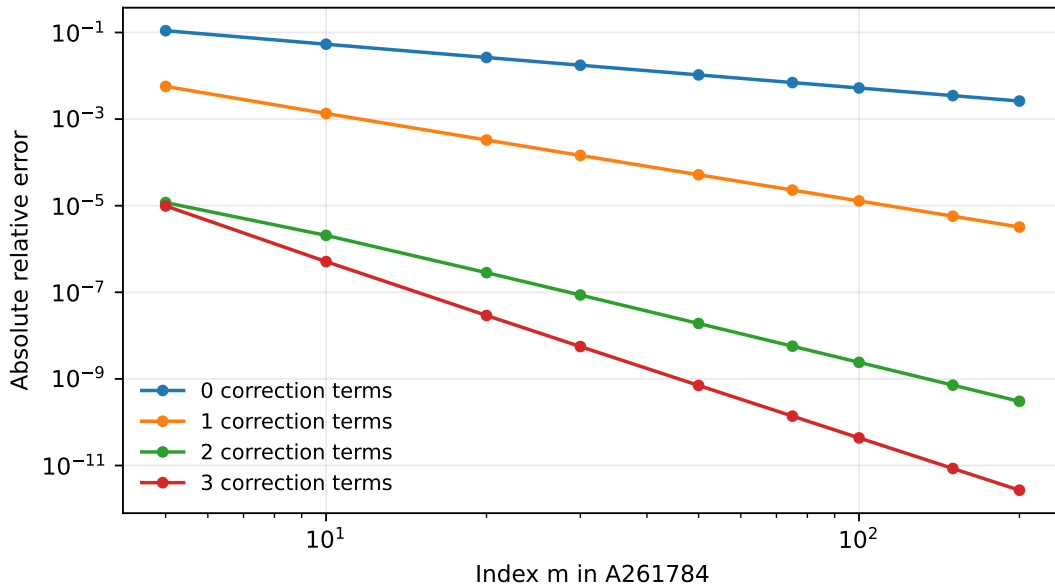


Figure 2: Absolute relative errors from the same exact data as Table 7, with additional intermediate indices (manuscript 01). These are numerical validations, not certified interval error bounds.

third-order residual at $N = 200$ is consistent with the proved $O(N^{-4})$ relative remainder; between $N = 100$ and $N = 200$ it falls by a factor of about 16. It does not by itself establish that order, which comes from the uniform analytic proof.

18 Asymptotic inversion and integer thresholds

[Added 3 October 2026, batch 85B: this section is an instance of the inversion apparatus of the repository's transseries volume [11]: Proposition 18.1 is a first step of its perturbed inversion, Theorem `p0:thm:perturbed-inversion`, for the core $2x \log(\alpha x)$, and Proposition 18.2 is a case of the separation condition of its staircase theorem, `p0:thm:staircase`. Manuscript 01 does not cite either; it was motivated by the volume's overview README, read at its pin. No novelty is claimed for the method; the constants are specific to A261784.]

A rapidly growing sequence has a natural discrete inverse, but no unique smooth interpolation; the two objects are kept separate.

A selected smooth comparator. For fixed $J \geq 1$ and sufficiently large real x define

$$\mathcal{A}^{[J]}(x) = c_* d_*^x \frac{\Gamma(x+1)^2}{x} \left(1 + \sum_{j=1}^J \beta_j x^{-j} \right). \quad (18.1)$$

It is positive and eventually strictly increasing, because the derivative of its logarithm is $2 \log x + \log d_* + O(x^{-1}) > 0$. Set

$$\alpha = \frac{\sqrt{d_*}}{e}, \quad \tilde{\beta}_1 = \beta_1 + \frac{1}{6} = -0.35514003488981599722247743640280278818 \dots \quad (18.2)$$

($\tilde{\beta}_1$ is manuscript 01's " β_1 "; it is not Part I's β_1 .) Stirling's expansion and (17.1) give, at integer N ,

$$\log a_N = 2N \log(\alpha N) + \log(2\pi c_*) + \frac{\tilde{\beta}_1}{N} + O(N^{-2}). \quad (18.3)$$

The cancellation of the $\log N$ term is specific to the $(N!)^2/N$ normalization.

Proposition 18.1 (Lambert- W inverse and first residual correction). *Let $Y \rightarrow \infty$ and put*

$$\Lambda = \frac{\log Y - \log(2\pi c_*)}{2}, \quad \varpi = W_0(\alpha \Lambda), \quad x_0 = \frac{\Lambda}{\varpi}. \quad (18.4)$$

If $x_J(Y)$ solves $\mathcal{A}^{[J]}(x_J) = Y$, then

$$x_J(Y) = x_0 - \frac{\tilde{\beta}_1}{2x_0(1+\varpi)} + O\left(\frac{x_0^{-2}}{\log x_0}\right). \quad (18.5)$$

Further terms are obtained by substituting a formal correction into the logarithm of (18.1) and cancelling successive residuals.

Proof. The definition of ϖ gives $\alpha x_0 = e^\varpi$, so $x_0 \log(\alpha x_0) = \Lambda$ and x_0 inverts the dominant core exactly. Substitution in the logarithm of (18.1) leaves the residual $\tilde{\beta}_1/x_0 + O(x_0^{-2})$. The derivative of the core is $2(1 + \log(\alpha x_0)) = 2(1 + \varpi)$, and the displayed correction cancels the residual to $O(x_0^{-2})$; quadratic errors in the correction are smaller. The derivative stays comparable to $\log x_0$, and the mean-value theorem transports the remaining logarithmic residual to the claimed root error. $\square$

The discrete threshold. Define

$$\mathcal{N}(Y) = \min\{N \geq 0 : a_N \geq Y\}. \quad (18.6)$$

The sequence a_N is strictly increasing. Indeed, pad every matrix counted by a_N with a new zero row and append two columns, each having a single 1 in that new row. This is an injection into the matrices counted by a_{N+1} . It is not surjective: a positive one-column matrix with $N+1$ rows and total $2N+2$ is not in its image.

Proposition 18.2 (Residual brackets for integer inversion). *For each fixed $J \geq 1$ there is a constant C_J such that, for all sufficiently large Y , if $\epsilon_J = C_J x_J(Y)^{-J-1} / \log x_J(Y)$, then*

$$\lceil x_J(Y) - \epsilon_J \rceil \leq \mathcal{N}(Y) \leq \lceil x_J(Y) + \epsilon_J \rceil. \quad (18.7)$$

In particular $\mathcal{N}(Y) = \Lambda/W_0(\alpha \Lambda) + O(1)$.

Proof. At integer N the logarithmic difference between a_N and $\mathcal{A}^{[J]}(N)$ is $O(N^{-J-1})$ by Corollary 17.1. The derivative of $\log \mathcal{A}^{[J]}$ is bounded below by a positive constant times $\log x_J$ in a fixed-width interval around x_J . Choosing C_J large enough, the model’s change across ϵ_J dominates that logarithmic error at every integer in the interval. Integers strictly below $x_J - \epsilon_J$ cannot attain Y ; every integer at least $x_J + \epsilon_J$ does. Monotonicity handles integers farther from the interval. Taking ceilings gives (18.7). $\square$

No unconditional formula $\mathcal{N}(Y) = \lceil x_J(Y) \rceil$ follows. When the smooth root lies extremely close to an integer, even a tiny asymptotic error can decide which of two consecutive integer thresholds is correct; the bracket is a valid statement that does not hide that ambiguity. Its constants are existential consequences of the asymptotic proof; no numerically certified value of C_J is claimed.

19 Manuscript 01’s checks

Manuscript 01’s program `code/01-two-poisson-verify.py` (delivered as `scripts/verify.py`) uses Python integers for exact combinatorics, SymPy for rational polynomial operations, and mpmath at 90 decimal digits for evaluation. It (1) matches eight displayed rows of A261781 and all fifteen displayed terms of A261784, finite transcriptions marked as such in the source; (2) compares the positive Stirling transform with a separately implemented generating-function recurrence followed by inclusion–exclusion for every $0 \leq k \leq n \leq 40$; (3) computes the reduced rational functions for $1 \leq k \leq 8$, checks the denominator degrees and coprimality, and verifies later terms against the extracted recurrence; (4) checks every coefficient of (14.3) through $n = 80$, constructs P_1, P_2, P_3 over $\mathbb{Q}$, and compares the finite Gaussian algorithm’s E_1 with the closed formula (7.4); and (5) computes $a_0, \dots, a_{200}$ exactly (`data/01-two-poisson-a261784_exact.txt`) and produces the diagonal, coverage and defect tables, which are floating-point tables, not interval-arithmetic certificates. Its record (`data/01-two-poisson-verification.json`) says “ALL CHECKS PASSED”. Finite tests do not prove the infinite statements; they check algebra, indexing, normalization and implementation.

The coverage table takes n to be the nearest integer to $k \log k$ and uses the actual $h = n/k$ and $\eta = ke^{-h}$ in both approximations; the probabilities in its third column are ratios of exact integers. The correction is not advertised as monotone in accuracy for every small input: the theorem is a

k	n	Exact coverage	$e^{-\eta}$	First correction
10	23	0.314344674	0.366928440	0.319053131
20	60	0.339714115	0.369449445	0.341284787
40	148	0.355320143	0.371970377	0.355818395
80	351	0.360509443	0.369892970	0.360669046
120	574	0.359614880	0.366349652	0.359697754

Table 8: The first correction of (5.4) near the centre of the coverage transition (manuscript 01). Rounding n explains why the leading values are not all exactly e^{-1} .

uniform asymptotic statement for bounded s , not an approximation inequality for all k .

[Added 3 October 2026, batch 85B: at the intake’s rerun on a copy, `verify.py -max-m 200` passed all checks in 162s (its README says about 16s), with every output byte-identical to the shipped one or equal apart from line endings and elapsed time; `ray_spotchecks.py`, which imports `verify` and so needs SymPy as well, reproduced its CSV byte for byte in 207s; and the

figure program reproduced the PNG pixel for pixel. The 201 computed terms of A261784 were not compared with the OEIS b-file [8], and manuscript 01 does not claim that they were.]

20 Further research questions of manuscript 01

Manuscript 01 orders these by proximity to its methods, not by a claim that any is easy or guaranteed to be new. The bracketed notes re-scope them against Part I.

Uniformity between compact rays and the coverage window. Prove a single relative asymptotic valid on an expanding interval such as $1 + \varepsilon \leq n/k \leq \log k + C$, with an error that remains informative when coverage is exponentially rare. The kernel suggests extending (14.4) from fixed w to growing $w = Lr$, while the saddle must be controlled as $r \rightarrow \infty$; a successful result should reproduce the polynomial corrections of Theorem 13.1, not only the limiting curve. A concrete first step is a bound for the kernel remainder on $|w| \leq C \log n$ that keeps its exponential dependence on $|w|$. [Added: this is the second deliverable of Research question 1 of Part I; open.]

The near-diagonal boundary. At $n = k$ the answer is F_n , whereas the saddle variance b vanishes as $n/k \downarrow 1$. Determine expansions uniform when $\ell = n - k$ is fixed, when $\ell \rightarrow \infty$ with $\ell = o(k)$, and across their transition. The positive Stirling sum has only $\ell + 1$ summands in the fixed- ℓ regime, so exact polynomial descriptions may be feasible before any saddle approximation; matching them to a large- ℓ saddle expansion would complete the present analysis at the endpoint. [Added: this is the first deliverable of Research question 1 of Part I; open.]

A combinatorial realization of the latent defect. Find a direct decorated-matrix construction whose forgetful map induces exactly the weights (16.1). Ideally the defect would count a familiar local feature and make the factor $e^{Lr/2}$ visible; such a construction could also explain the polynomials P_r and the negative first-order term in P_1 without Faulhaber expansion. [Added: open; both manuscripts decline to claim a canonical statistic.]

Joint laws with the number of columns. Mark the columns by u ; before requiring nonzero rows the bivariate generating function is $F_k(z; u)$ of (2.1). Study the joint distribution of the column count and the zero-row count in the coverage window, and of the column count together with a combinatorial defect on compact rays; the principal pole is governed by $\log(1 + 1/u)$, and uniform control near $u = 1$ is needed before claiming a central limit theorem or asymptotic independence. [Added 3 October 2026, batch 85B: the first half is answered by Part I. Theorem 4.1 gives the uniform control in u on compact intervals, Lemma 6.1 the Poisson law under column weights, and Theorem 6.2 the joint limit of the normalized column count and the zero-row count, with independence and the conditional Gaussian law given $M = m$. On compact rays Corollary 7.5 gives the column central limit theorem and its proof the exact mixture over $Q_{n,k}$; the joint law of the column count and the defect $\Delta_{n,k}$, or a combinatorial defect, remains open, as do the local refinements of Research question 3.]

Weighted rows, columns, and restricted alphabets. Replace the identical row weights or the unrestricted column entries by a specified weight system, and determine which features of the recurrence-order proof survive when distinct factors can share poles. The proof suggests a precise obstruction: cancellations and coincident spectra, rather than the mere presence of a rational generating function. A classification for finite sets of permitted column weights would be a useful

structural extension. [Added: related to Research question 8 of Part I, which asks the same about the coverage law; open.]

Beyond all algebraic orders. The Fubini generating function has further poles at $L + 2\pi ij$. The present proofs bound their total effect exponentially but do not extract their separate saddle contributions. Derive those sectors, with their oscillatory phases, and compare their scale with the late terms of the algebraic expansion; this is where a genuinely multi-sector transseries, rather than a Poincaré series, should enter, and optimal truncation and rigorous remainder bounds would be part of that problem. [Added 3 October 2026, batch 85B: for the Fubini numbers themselves the sectors are exact and convergent, Theorem `q2:thm:fubini` of [11]; the open problem is to carry them through the Stirling transform and the saddle. The report on A260700 [10] lists the same open problem for its sequence.]

Effective certified computation and formalization. Make the constants in the compact-ray error bound and in (18.7) explicit; interval estimates on the central arc and a certified minor-arc gap could turn the inversion into a proof-producing threshold algorithm. The exact recurrence theorem is a suitable first formalization target: its core is pairwise disjoint simple poles and the equivalence between rational generating functions and eventual linear recurrences. The analytic kernel and saddle theorems are substantially larger tasks and should be scoped separately. [Added: the same target as Part I’s “practical order of attack”; nothing is formalized yet.]

21 Hypotheses that cannot be dropped

Manuscript 01 ends with a proof audit, kept here in full.

Constant versus polynomial coefficients. The minimality assertion is for homogeneous recurrences with constant coefficients. It is not a lower bound for recurrences with polynomial coefficients in n , nonlinear recurrences, or recurrences involving several columns k at once.

Nonnegative versus strictly positive entries. Entries are nonnegative; the restrictions are on entire rows and columns, not on each cell. Replacing cells by positive integers changes the generating functions and the sequence.

The polynomial part of a rational function. The fixed-spectrum formula for $A(n, k)$ applies to $n \geq 1$, and the recurrence for $T(n, k)$ starts after the numerator has ceased contributing. Ignoring either point creates incorrect initial conditions.

Positivity before relative error transport. The approximation of F_q is transported through the positive identity (2.4), not through the alternating inclusion–exclusion sum: an exponentially accurate approximation term by term in an alternating sum need not be relatively accurate after cancellation. The coverage calculation instead controls factorial moments and then uses Bonferroni truncation.

Compactness in the ray theorem. The uniform constants require n/k to stay away from 1 and infinity. Neither endpoint is covered by substituting it in (15.1), and the actual rational n/k must be used when n is rounded.

Two different Poisson variables. M counts zero rows in uniformly sampled matrices that may have zero rows; $\Delta_{n,k}$ is an auxiliary variable defined by positive transform weights. Their means, parameter regimes and sample spaces are different.

Formal Gaussian algebra. The prescription (7.16) means coefficient extraction first, Gaussian evaluation second; it does not require the integral of the untruncated exponential expression to converge.

Asymptotic expansion versus convergent series. Every order J is fixed before the size parameter tends to infinity. Neither growth estimates for the coefficients as $J \rightarrow \infty$ nor optimal truncation are asserted.

Numerics and proof status. Exact integer checks certify their finite inputs. The 90-digit numerical calculations are independent consistency checks, not rigorous floating-point enclosures or a machine-checked proof of the theorems.

22 Manuscript 01's draft OEIS additions (not submitted)

Manuscript 01 includes draft additions to two OEIS entries, shipped as `data/01-two-poisson-OEIS_updates_draft.txt`. *They were not submitted*, by manuscript 01 or by this report; manuscript 01 itself asks for independent review and a stable public manuscript link first, and for the earlier source and conjecture attributions to be preserved. Their substance, in this report's letters:

A261781. For $k > 0$ the reduced denominator of the ordinary generating function of column k is $\prod_{j=1}^k (2(1-x)^j - 1)$, so its minimal eventual homogeneous constant-coefficient recurrence has order $k(k+1)/2$: inclusion–exclusion expresses the column generating function as the alternating binomial sum of $(1-x)^j/(2(1-x)^j - 1)$, whose j th summand has simple poles at $1 - 2^{-1/j}e^{2\pi i\ell/j}$, and different summands have disjoint pole sets. The fixed-column constants are $c_k = 1/(2k(2^{1/k} - 1))$ and $d_k = 1/(1 - 2^{-1/k})$, and $d_{k+1} - d_k$ increases strictly to $1/\log 2$, with $d'(x)$ as in (12.1). Attribution should retain the existing conjecture attribution and cite the generating-function source.

[Added 3 October 2026, batch 85B: *before any use* this draft must be corrected as in Section 11.2. Its sentence saying that these statements prove the two claims labelled conjectures overstates them: a recurrence of order $k(k+1)/2$ with this denominator is [4, Proposition 25], with the case $k = 2$ in Remark 26, and the limit follows from Proposition 13 there; what is new is the minimality and the strict monotonicity. The shipped file is kept byte-identical and still contains the overstated sentence.]

A261784. With $L = \log 2$ and $r = 2 + W_0(-2e^{-2})$, the posted prefactor is $c_* = 2^{r/2}/(4\pi L\sqrt{r-1})$ and the growth rate $d_* = 4/(L^2r(2-r))$; the relative correction begins $1 + \beta_1/N + \beta_2/N^2 + \beta_3/N^3$ with the values of Section 17 (the draft's index n is N here). The decimals evaluate finite exact prescriptions and are not fitted. The leading asymptotic already in the entry, Kotěšovec's (18 February 2017, updated 20 April 2024), is not to be presented as new, nor described as labelled a conjecture.

Part III

Cauchy limits, arithmetic atoms, and edge laws for the Hankel zeros

23 Provenance, scope and notation of Part III

[Added 5 October 2026, batch 98.] Part III was added in the write of batch 98. It is built from manuscript 08 of that batch, which answers Research question 7 of Part I, “The limiting distribution of determinant zeros” (Section 10). After the factors at $u = 0$ and $u = -1$ are removed, the zeros of the maximal Hankel determinant $H_{k,s}(u)$ of Theorem 3.1 converge to the image on the line $\operatorname{Re} u = -\frac{1}{2}$ of a Cauchy law of scale $\frac{1}{2}$. Manuscript 08 also determines the multiplicity of every root of small order, the sharp Kolmogorov rate, the tails at every height, the transition of the absolute moments and the limit under the full-degree normalization. Its Section 2 re-proves Theorem 3.1 by the argument of Section 3; that proof is not reprinted (Remark 24.4). Everything else of manuscript 08 is printed below, in the letters of Table 10. Bracketed notes marked “[Note of the batch-98 write]” and remarks marked “[write]” are the write’s; all other text of Part III is manuscript 08’s.

23.1 The manuscript

Manuscript 08 of batch 98 is the archive `a261781_hankel_zero_laws.zip` (30 files in a wrapper directory of the same name), which arrived in commit `2172df76a` and was placed in this report by commit `0fba5167f`. Its article is titled *Cauchy Limits, Arithmetic Atoms, and Edge Laws for Matrix-Composition Hankel Zeros*, subtitled “An explicit continuation of the ProveIt study of OEIS A261781” (29 pages; main file `a261781_hankel_zero_laws.tex`, 2,041 lines, 110 labels; dated October 5, 2026). Its author line and PDF metadata read “OpenAI ChatGPT”, with “An OEIS-related research article”. Its pin is the repository commit `d18416ec7e187a0948248cb3077e37d7b089a8f9` (2026-10-04, 19:16 –0700) [15], at which this report consisted of Parts I and II exactly as they stand: the text of this report did not change between the pin and the batch-98 write. Its source note, shipped as `07-zero-laws-PROVENANCE.md`, names the labels `mxc:thm:hankel` and `mxc:eq:weighted-hankel-product` and the question it answers.

Status and non-claims of manuscript 08, kept. Its proofs are mathematical proofs in an unrefereed research article; its computations are independent checks, not proofs of limits and not proof-assistant certificates. No exhaustive search for historical priority was made: targeted repository and public-source searches checked overlap and attribution only. Its proof tools are standard (finite spectral algebra, Möbius inversion, elementary Fourier approximation and real-variable estimates); its contribution is the family-specific theorem set, not those tools. The exact determinant product is prior repository mathematics, which it re-derives for self-containment and credits to this report; the enumeration and the existence of an order- $\binom{k+1}{2}$ recurrence are published ([4, Proposition 25]), and it does not present recurrence existence as an OEIS conjecture newly solved. It claims no Lean or Rocq formalization and no OEIS submission. Its further non-claims are kept where they stand: the error in (24.6) is absolute, not a uniform relative equivalent (Section 24); total variation is not the right distance (Remark 26.1); normalizing the reduced zeros by the full degree gives a subprobability (Section 26); the smooth correction does not extend to indicators (Section 27); the sharp constant comes without a quantitative error and the finite data lie far below it (Sections 28 and 32); the edge

law does not describe the finitely many largest zeros (Section 29); and its questions are “not claims that every formulation has been checked against all literature” (Section 33). Its source note adds that the statements were cross-checked by independent reviews in its own session, whose integration review corrected the range of a primitive angle, the absolute-value convention of the Kolmogorov distance, the nearest-integer notation and the domain of the primitive triangle count.

23.2 What Part III adds

Table 9 sets Part III against what Parts I and II already contain.

Table 9: What Part III adds. Research question 7 is answered; its finer continuations are Research questions 9 to 16.

Topic	Parts I and II	Part III (manuscript 08 of batch 98)
Location and multiplicities	zeros other than $0, -1$ on $\operatorname{Re} u = -\frac{1}{2}$; exact multiplicities at 0 and -1 ; the others “follow by collecting the cyclotomic difference factors” (Section 3)	the cyclotomic factorization (25.2), every order $2 \leq m \leq k-1$ present, the exact count of distinct zeros, and $E_m(k) = k^2 \log k / (\zeta(2)\psi(m)) + O(k^2)$ uniformly in m (Section 25)
Limit law	asked in Research question 7	the Cauchy law of scale $\frac{1}{2}$, with the elementary bound $2G_k/D_k$ and the sharp Kolmogorov constant $9/\pi^2$ (Theorem 24.1, Corollary 28.3)
Corrections	—	the first correction of smooth statistics (Theorem 27.1); arithmetic CDF corrections at fixed and moving endpoints (Theorems 28.1 and 28.2)
Tails	asked in Research question 7	the divisor-sum edge law at height $\asymp k$ (Theorem 29.2), the exact extreme zeros (Proposition 29.3), the expansion of the edge profile at the origin and a uniform transition formula for every height (Theorems 30.1 and 30.2)
Moments	—	the transition at $p = 1$, with the constant term at $p = 1$ and the constants above it (Theorem 24.2)
Normalization	asked in Research question 7	the three-component limit of the full zero measure as $s/k^2 \rightarrow \varkappa$ (Theorem 24.3)

23.3 Notation of Part III

Manuscript 08 uses many letters that Parts I and II use for other objects. Part III renames every such letter; the renames are changes of letter only, and no normalization is changed. Objects of manuscript 08 that collide with Latin letters of Parts I and II are written in sans-serif type, as Part II did for F_q and W_n ; the three zero measures carry a superscript z . Table 10 lists every rename with the false reading the delivered letter would have in this report. The figures of Section 32 are delivered files and keep manuscript 08’s letters; their captions translate them.

Table 10: Manuscript 08's symbols and the letters used for them in Part III. “False reading” gives what the manuscript-08 letter means elsewhere in this report.

Notion	Part III	Ms. 08	False reading if copied
Hankel size $\binom{k+1}{2}$	written out (d of Section 3)	d_k	d_k is the growth constant (2.9); in Part III a bare d is only a summation or divisor index
$\binom{k+1}{3}$	c , as in Section 3	c_k	c_k is the amplitude (2.9)
Reduced factor; its degree	$R_k(u); D_k$	$R_k(u); D_k$	$R_p(w)$ are kernel remainders; $D, D_u, D(\eta)$ are Part I objects
Exceptional exponents; constant	$A_k, B_{k,s}; C_k$	$A_k, B_{k,s}; C_k^{(0)}$	$A(n, k)$ is a count, $B, B(u)$ are constants, C counts columns
Full degree $\deg H_{k,s}$	$\deg H_{k,s}$	$L_{k,s}$	$L = \log 2$
Difference factor $(1+u)^q - u^q$	$U_q(u)$	$F_q(u)$	$F_k(z; u)$ is a generating function
Primitive multiplicity	$E_m(k)$	$E_m(k)$	$E_n(w), E_j(\lambda), E_n$
Grid weight	$\mathcal{W}_k(q)$	$W_k(q)$	$W_n(w)$ is the Stirling kernel, W_N a diagonal diagnostic
Zero measures on angles, on Y , on the line; full zero measure	$\rho_k^z, \nu_k^z, \widehat{\nu}_k^z; \mu_{k,s}^z$	$\rho_k, \nu_k, \widehat{\nu}_k; \mu_{k,s}$	ρ is the pole ratio, $\nu_k = \mathbb{E}M$ in Section 5, $\mu_k(u)$ an amplitude
Dirac masses	δ_0, δ_{-1}	δ_0, δ_{-1}	δ_k below is a displacement
Angular distribution function	$\mathcal{F}_k(t)$	$F_k(t)$	$F_k(z; u); F_q$ are Fubini numbers
Kolmogorov distance	Δ_k^K	Δ_k	$\Delta_{n,k}$ is the Stirling defect
$d_K(\nu_k^z, C)$			
Absolute moments; their constant	$M_p(k); \mathcal{M}_p$	$M_p(k); C_p$	M is the zero-row count, C_J a bracket constant
Euler's constant	γ_E	γ	$\gamma = \lim m/(\log k)^2$ in Corollary 8.2
von Mangoldt function	Λ_{vM}	Λ	Λ is Part II's inverse core
$\lim s_k/k^2; \lim s/k^{p+1}$	$\varkappa; \varkappa_p$	$\tau; a$	τ is a formal variable in (7.16); $a = 1 + u$ in Section 3
Sign of a displacement	ς	σ	$\sigma_{-1}(n), \sigma_1(n)$ are divisor functions
$\pi/\arctan(1/(2y));$ its reciprocal	$\Omega(y); 1/\Omega(y)$	$\beta(y); \delta(y)$	$\beta_1, \beta_2, \beta_3$ are diagonal corrections
$k/\Omega(y)$	$\mathbf{m}$	M	M is the zero-row count
Tail count	$Z_k(y)$	$\mathcal{T}_k(y)$	$\mathcal{T}_a(x)$ are Part II polynomials
Counts without coprimality	$P_\ell^*(Y), \mathcal{P}^*(K, Q)$	$B_\ell(Y), \mathcal{N}(K, Q)$	$\mathcal{N}(Y)$ is Part II's integer threshold
Primitive congruence count	$P_m(X)$	$P_m(X)$	$P_j(t), P_j(w), P_r(w)$ are polynomials
Fractional-part sum; logarithmic sums	$\mathcal{S}_k(t); \mathcal{S}_k^{\text{gap}}, \mathcal{S}_k^{\text{gcd}}$	$\mathcal{A}_k(t); \mathcal{A}_k^{\log}, \mathcal{B}_k^{\log}$	$\mathcal{A}^{[J]}(x)$ is a comparator, $\mathcal{B}(n, k)$ a saddle approximation
Harmonic sawtooth sum and auxiliaries; harmonic numbers	$Q_N(\theta), K_N(\theta), J_b(t); H_N$	$Q_N, K_N, J_b; H_N$	$Q_{n,k}; K$ a compact set; $J, J_q;$
Sawtooth; trigonometric bounds; Bernoulli polynomials	$\overline{B}_1(x); T^\pm; B_3(v)$	$s(x); P^\pm; B_3(v)$	$H_{k,s}$ the determinant s is the Hankel shift
Sawtooth lemma: displacement, exponent, constant, cutoff	$\widetilde{\delta}_k, \widetilde{\alpha}, \kappa, \mathbf{N}_k$	η_k, β, c, N_k	η is the Poisson mean, $\beta = \beta_1$, $c = \binom{k+1}{3}$, N the diagonal index
Edge density; scaled measures	$-\Xi'(x); \nu_k^e$	$\omega_{\text{edge}}(x); \lambda_k$	ω_j are roots of unity, $\lambda = n/k$

Notion	Part III	Ms. 08	False reading if copied
Grid error; cotangent sum; its regular part	$\text{err}_q; S_1(q); h(v)$	$\epsilon_q; S_1(q); h(v)$	$\epsilon = 1/k; h = n/k$
Bernoulli-proof auxiliaries	$\Upsilon(v), \mathcal{I}_3; n = d\ell$	$F(v), I; n = dm$	m is the cyclotomic order
Homogeneous cyclotomic polynomial	$\Phi_m(X_1, X_2)$	$\Phi_m(X, Y)$	Y is the imaginary coordinate

Unchanged, with their meaning in Part III: $g_{ij} = \gcd(i, j)$, as in Section 3; $q_{ij} = (j - i)/g_{ij}$ and a grid denominator q , not the denominator coefficients q_i of Section 2; $H_{k,s}$ and $T_u(n, k)$, the objects of Section 3; the zero $u(t)$ and its angle t ; the imaginary coordinate Y , not Part II's threshold argument; $\mathcal{C}, \hat{\mathcal{C}}, f_{\mathcal{C}}, F_{\mathcal{C}}$ for the Cauchy law; d_K for the Kolmogorov distance (Part I writes d_{TV} for total variation); ψ (Dedekind), φ (Euler) and $\mu(\cdot)$ (Möbius, only inside arithmetic sums, not the amplitude $\mu_k(u)$); $\sigma_{-1}(n)$ and $\sigma_1(n)$, divisor functions with their argument (n) , not the variance σ_k^2 of Section 6; $\Xi, \mathcal{H}$ and $\mathcal{P}(K, Q)$; and the local letters $a, b, h, r, \ell, p, x, X, \alpha, \theta$, whose meaning is fixed where they occur (a, b are coprime integers, not $1 + u$ and u of Section 3).

24 The zero measures and the main conclusions

24.1 An OEIS family with an arithmetic zero problem

Let $T_u(n, k)$ be the polynomial enumerating matrices with k ordered rows, an arbitrary number of ordered columns, nonnegative integer entries, total sum n , and no zero row or column, each column receiving weight u (Section 3). At $u = 1$ this is the triangle A261781 [2], and the diagonal $T_1(2n, n)$ is A261784 [3]. Matrix compositions were studied by Munarini, Poneti and Rinaldi [4], whose generating functions and recurrences are the enumerative foundation of the subject. Part I supplies the weighted maximal Hankel factorization and asks what the limiting zero-counting measure is after the factors at $u = 0$ and $u = -1$ have been removed, with tail information and the contribution of roots of small order. Part III answers those parts of the question. It does not relabel the known enumeration or the determinant product as a new result, and it is an explicit extension of this report, not a claim that an exhaustive search has established historical priority.

24.2 Notation and the three zero measures

Throughout, k, s are positive integers, $c = \binom{k+1}{3}$, $g_{ij} = \gcd(i, j)$ and $q_{ij} = (j - i)/g_{ij}$. With $H_{k,s}(u)$ the determinant of Theorem 3.1, define

$$R_k(u) = \prod_{1 \leq i < j \leq k} ((1 + u)^{q_{ij}} - u^{q_{ij}})^{2g_{ij}}, \quad (24.1)$$

$$G_k = \sum_{i < j \leq k} g_{ij}, \quad D_k = \deg R_k = 2(c - G_k). \quad (24.2)$$

The polynomials R_1 and R_2 are constant. Hence all probability measures normalized by D_k are defined only for $k \geq 3$.

Every zero of R_k has the form

$$u(t) = \frac{1}{e^{2\pi i t} - 1} = -\frac{1}{2} - \frac{i}{2} \cot(\pi t), \quad 0 < t < 1. \quad (24.3)$$

We use three measures, always counting multiplicity: ρ_k^z on angles $t \in (0, 1)$; ν_k^z on the imaginary coordinate $Y = -\frac{1}{2} \cot(\pi t)$; and $\widehat{\nu}_k^z$ on the complex line $\operatorname{Re} u = -\frac{1}{2}$. They are pushforwards of one another. The full zero measure of $H_{k,s}$, normalized by its full degree, is denoted by $\mu_{k,s}^z$.

Write $\mathcal{C}$ for the probability distribution on $\mathbb{R}$ with density and cumulative distribution function

$$f_{\mathcal{C}}(y) = \frac{2}{\pi(1+4y^2)}, \quad F_{\mathcal{C}}(y) = \frac{1}{2} + \frac{1}{\pi} \arctan(2y). \quad (24.4)$$

Its image on the vertical line is $\widehat{\mathcal{C}}$. For real probability laws, d_K is the supremum of the absolute difference of their cumulative distribution functions. We reserve $\mu(n)$ inside arithmetic sums for the Möbius function; this is distinct from the measure $\mu_{k,s}^z$.

24.3 The main conclusions

Theorem 24.1 (Bulk limit and arithmetic multiplicities). *As $k \rightarrow \infty$,*

$$D_k = \frac{k^3}{3} - \frac{k^2 \log k}{\zeta(2)} + O(k^2), \quad \nu_k^z \implies \mathcal{C}. \quad (24.5)$$

If $E_m(k)$ is the multiplicity of any zero associated with a primitive m -th root of unity, $m \geq 2$, then

$$E_m(k) = \frac{k^2 \log k}{\zeta(2) \psi(m)} + O(k^2), \quad \psi(m) = m \prod_{p|m} \left(1 + \frac{1}{p}\right), \quad (24.6)$$

with an absolute implied constant, uniform in m . Moreover,

$$d_K(\nu_k^z, \mathcal{C}) \sim \frac{3}{2\zeta(2)} \frac{\log k}{k} = \frac{9}{\pi^2} \frac{\log k}{k}. \quad (24.7)$$

The proofs are in Sections 25–28. The uniform error in (24.6) is an absolute error, not a uniform relative equivalent for all growing m . It implies a relative equivalent when $\psi(m) = o(\log k)$; for larger m it may be too large to reveal the leading term. The function ψ is itself OEIS A001615 [13].

Theorem 24.2 (Extreme tails and moments). *For each fixed $x > 0$,*

$$k \nu_k^z((kx, \infty)) \longrightarrow \Xi(x) := \frac{3}{\zeta(2)} \sum_{n \geq 1} \sigma_{-1}(n) (1 - 2\pi n x)_+^2, \quad \sigma_{-1}(n) = \sum_{d|n} \frac{1}{d}. \quad (24.8)$$

The sum is finite. For every sequence $y_k \rightarrow \infty$ with $y_k = o(k)$,

$$\nu_k^z((y_k, \infty)) \sim \frac{1}{2\pi y_k}. \quad (24.9)$$

For the absolute moments $M_p(k) = \int |Y|^p d\nu_k^z(Y)$,

$$M_p(k) \longrightarrow \frac{2^{-p}}{\cos(\pi p/2)}, \quad 0 < p < 1, \quad (24.10)$$

$$M_1(k) = \frac{1}{\pi} \left(\log k + \gamma_E - \log \pi - \frac{5}{6} + \frac{\zeta'(2)}{\zeta(2)} \right) + O\left(\frac{\log^2 k}{k}\right), \quad (24.11)$$

$$M_p(k) \sim \mathcal{M}_p k^{p-1}, \quad p > 1, \quad (24.12)$$

$$\mathcal{M}_p = \frac{12\zeta(p)\zeta(p+1)}{(2\pi)^p(p+1)(p+2)\zeta(2)}. \quad (24.13)$$

Here $(z)_+ = \max(z, 0)$, and γ_E is Euler's constant. Sections 29–31 give proofs, including a quantitative edge-to-bulk matching formula. The finite sum in (24.8) retains arithmetic structure that weak convergence cannot see.

Theorem 24.3 (Retaining the two exceptional factors). *If $s = s_k \geq 1$ and $s_k/k^2 \rightarrow \varkappa \in [0, \infty)$, then*

$$\mu_{k,s_k}^z \Rightarrow \frac{\delta_0}{3(1+\varkappa)} + \frac{1+3\varkappa}{3(1+\varkappa)} \delta_{-1} + \frac{\hat{C}}{3(1+\varkappa)}. \quad (24.14)$$

If $s_k/k^2 \rightarrow \infty$, the limit is δ_{-1} . For any fixed shift s , the limiting masses of the two point components and the Cauchy component are all $\frac{1}{3}$.

Scale or statistic	Result	Information retained
Fixed imaginary coordinate	Cauchy law	Overall distribution
Fixed cyclotomic order	Multiplicity proportional to $1/\psi(m)$	Arithmetic concentration
CDF error	Exact constant $9/\pi^2$	Sharp global convergence rate
Smooth statistic	Explicit $(\log k)/k$ correction	A universal first bias
Height proportional to k	Quadratic divisor-sum tail	Extreme zeros and cutoff
Moment order p	Transition at $p = 1$	Failure of moment convergence above the threshold
Shift s proportional to k^2	Three-component mixture	Dependence on full-degree normalization

Table 11: The results of Part III concern different asymptotic questions about the same exact polynomial (manuscript 08).

24.4 The determinant foundation

Manuscript 08's Section 2 proves the determinant identity with explicit conventions, to connect the arithmetic to the OEIS objects without depending on an unexamined factorization. In Part III's letters it reads

$$H_{k,s}(u) = C_k u^{A_k} (1+u)^{B_{k,s}} R_k(u), \quad A_k = \binom{k+1}{2} + 2c, \quad B_{k,s} = k(s-1) + 2c, \quad C_k = \prod_{j=1}^k \binom{k}{j}^j, \quad (24.15)$$

which is (3.1) of Theorem 3.1, with $A_k = \sum_{j \leq k} j^2$.

Remark 24.4 (Manuscript 08's re-proof of Theorem 3.1; a pointer). [Note of the batch-98 write.] Manuscript 08's Proposition 2.1 ("Prior maximal Hankel identity") and its proof are a second derivation of Theorem 3.1 by the same argument as Section 3, and it credits the identity to this report: the generating function and inclusion–exclusion (2.1)–(2.3); the spectrum and residue weights (3.2); the Vandermonde factorization (3.7); the root products, discriminant and resultant (3.4)–(3.6); and the sign count $\sum_j (j(k-j) + \binom{j}{2}) = 2c$. It is not reprinted. Also shared with Section 3: the remark that the size $\binom{k+1}{2}$ is maximal for the positive-index Hankel matrix at $u > 0$, that this does not assert positive definiteness of smaller minors, the reference to the different Cassini-type determinant of [4, Proposition 21], and $H_{3,s}(u) = 27u^{14}(1+u)^{3s+5}(1+2u)^2$. Manuscript 08 adds the small cases

$$H_{1,s}(u) = u(1+u)^{s-1}, \quad H_{2,s}(u) = 2u^5(1+u)^{2s},$$

which explain why reduced zeros first occur at $k = 3$, the example $T_u(2, 2) = u + 2u^2$, and the observation that the convention $s \geq 1$ is essential because the constant polynomial part of the generating function contributes at $n = 0$ (Part I's (3.9)). The write checked $H_{1,s}$ and $H_{2,s}$ against (24.15).

25 Cyclotomic multiplicities and the degree loss

25.1 An exact arithmetic description

Put $U_q(u) = (1 + u)^q - u^q$. For $q \geq 2$, it has exactly $q - 1$ simple roots $u(\ell/q)$, $1 \leq \ell < q$; $U_1 = 1$. Indeed $((1 + u)/u)^q = 1$, and the root of unity 1 has no finite preimage. Its absence is the degree cancellation in U_q .

For every integer $m \geq 1$, define

$$E_m(k) = 2 \sum_{\substack{1 \leq i < j \leq k \\ m | q_{ij}}} g_{ij}. \quad (25.1)$$

For $m \geq 2$, this is the multiplicity of each primitive order- m zero. The auxiliary value $E_1(k) = 2G_k$ counts the weighted degree lost at the excluded root of unity 1; it does not describe a zero at a finite point.

Let $\Phi_m(X_1, X_2)$ be the homogeneous cyclotomic polynomial

$$\Phi_m(X_1, X_2) = \prod_{\substack{1 \leq a \leq m \\ (a, m) = 1}} (X_1 - e^{2\pi i a/m} X_2).$$

Factoring $X_1^q - X_2^q$, and using $X_1 - X_2 = 1$ at $(X_1, X_2) = (1 + u, u)$, proves the exact identities

$$R_k(u) = \prod_{m=2}^{k-1} \Phi_m(1 + u, u)^{E_m(k)}, \quad (25.2)$$

$$D_k = \sum_{m=2}^{k-1} \varphi(m) E_m(k) = 2c - E_1(k). \quad (25.3)$$

The factors in (25.2) need not be monic: their leading coefficients are $\Phi_m(1, 1)$, which are nonzero. The degree is nevertheless $\varphi(m)$.

Every order $2 \leq m \leq k - 1$ occurs, since the pair $(i, j) = (1, m + 1)$ has $q_{ij} = m$. Thus the number of distinct reduced zeros, ignoring multiplicity, is exactly $\sum_{m=2}^{k-1} \varphi(m)$. This is much smaller than D_k , of cubic order, and explains why ignoring multiplicity changes the finite distribution substantially.

For computation, it is useful to group first by the unreduced grid denominator q . Write

$$\mathcal{W}_k(q) = \sum_{\substack{q < b \leq k \\ (b, q) = 1}} \left\lfloor \frac{k}{b} \right\rfloor \left(\left\lfloor \frac{k}{b} \right\rfloor + 1 \right). \quad (25.4)$$

Then

$$R_k = \prod_{q=1}^{k-1} U_q^{\mathcal{W}_k(q)}, \quad E_m(k) = \sum_{\substack{q \leq k-1 \\ m | q}} \mathcal{W}_k(q).$$

To see this, set $i = ga$, $j = g(a + q)$, where $(a, q) = 1$, and sum the multiplicity $2g$ over $1 \leq g \leq \lfloor k/(a + q) \rfloor$.

25.2 A primitive-lattice estimate with uniform error

Lemma 25.1. *For $X \geq 1$ and any integer $m \geq 1$, let*

$$P_m(X) = \#\{1 \leq a < b \leq X : (a, b) = 1, m \mid b - a\}.$$

Uniformly in m ,

$$P_m(X) = \frac{X^2}{2\zeta(2)\psi(m)} + O(X \log(2X)). \quad (25.5)$$

Proof. Without the coprimality condition, let

$$P_\ell^*(Y) = \#\{1 \leq a < b \leq Y : \ell \mid b - a\}.$$

Writing the difference as ℓh gives

$$P_\ell^*(Y) = \sum_{\ell h < \lfloor Y \rfloor} (\lfloor Y \rfloor - \ell h) = \frac{Y^2}{2\ell} + O(Y),$$

with an absolute error constant for all ℓ , including $\ell > Y$. Möbius inversion now gives

$$\begin{aligned} P_m(X) &= \sum_{d \leq X} \mu(d) P_{m/(m,d)}^*(X/d) \\ &= \frac{X^2}{2m} \sum_{d \leq X} \frac{\mu(d)(m, d)}{d^2} + O\left(X \sum_{d \leq X} \frac{1}{d}\right). \end{aligned}$$

The infinite-sum tail costs $O(X)$, uniformly in m , because $(m, d)/m \leq 1$. Its absolutely convergent Euler product is

$$\frac{1}{m} \sum_{d \geq 1} \frac{\mu(d)(m, d)}{d^2} = \frac{1}{m\zeta(2)} \prod_{p \mid m} \frac{1 - 1/p}{1 - 1/p^2} = \frac{1}{\zeta(2)\psi(m)}.$$

This proves the estimate. □

Proposition 25.2 (Uniform multiplicity asymptotic). *Equation (24.6) holds, including $m = 1$ under the auxiliary convention $E_1 = 2G_k$.*

Proof. Separate each pair by its gcd:

$$E_m(k) = 2 \sum_{g \leq k} g P_m(k/g).$$

Insert Lemma 25.1. The main term is

$$\frac{k^2}{\zeta(2)\psi(m)} \sum_{g \leq k} \frac{1}{g}.$$

The total error is bounded by a constant times

$$k \sum_{g \leq k} \log(2k/g) = k\{k \log(2k) - \log(k!)\} = O(k^2).$$

Replacing the harmonic sum by $\log k + O(1)$ preserves that error, uniformly in m . □

Remark 25.3 (write). [Note of the batch-98 write.] The uniformity in m includes the orders $m \geq k$, at which no pair has $m \mid q_{ij}$ and $E_m(k) = 0$: there $\psi(m) \geq m \geq k$, so the main term is at most $k \log k / \zeta(2)$ and is itself $O(k^2)$. The statement is true there, and empty.

The $m = 1$ case and (25.3) give (24.5). The underlying average-gcd phenomenon is classical; see Tóth [14] for context. The calculation above supplies the particular congruence refinement and all uniformity needed here.

Corollary 25.4 (Macroscopic multiplicity of a fixed arithmetic zero). *For each fixed $m \geq 2$ and integer a with $1 \leq a < m$ and $(a, m) = 1$,*

$$\rho_k^z(\{a/m\}) = \frac{3}{\zeta(2)} \frac{\log k}{\psi(m)} \frac{1}{k} + O_m(1/k). \quad (25.6)$$

In particular,

$$\nu_k^z(\{0\}) = \frac{1}{\zeta(2)} \frac{\log k}{k} + O(1/k). \quad (25.7)$$

Proof. Divide (24.6) by $D_k = k^3/3 + O(k^2 \log k)$. The extra $O(\log^2 k/k^2)$ term is $O(1/k)$. Use $\psi(2) = 3$ for the second assertion. $\square$

m	$\psi(m)$	Coefficient of $(\log k)/k$ per zero	Imaginary coordinates for small orders
2	3	$1/\zeta(2)$	0
3	4	$3/(4\zeta(2))$	$\pm 1/(2\sqrt{3})$
4	6	$1/(2\zeta(2))$	$\pm 1/2$
5	6	$1/(2\zeta(2))$	$-\frac{1}{2} \cot(\pi a/5), 1 \leq a < 5$
6	12	$1/(4\zeta(2))$	$\pm \sqrt{3}/2$

Table 12: Small-order arithmetic concentrations (manuscript 08). Multiplicity is stated per primitive zero, not as the sum over all $\varphi(m)$ such zeros.

26 The Cauchy limit and full normalization

26.1 A direct quantitative proof

Let $\mathcal{F}_k(t) = \rho_k^z((0, t])$, $0 \leq t \leq 1$. For $0 \leq t < 1$, the q -grid contains $\lfloor qt \rfloor$ points among $1/q, \dots, (q-1)/q$ at or below t . Thus

$$D_k(\mathcal{F}_k(t) - t) = \sum_{i < j} 2g_{ij}(t - \{q_{ij}t\}). \quad (26.1)$$

Since the absolute value of each bracket is at most one,

$$\sup_{0 \leq t \leq 1} |\mathcal{F}_k(t) - t| \leq \frac{2G_k}{D_k} = \frac{3}{\zeta(2)} \frac{\log k}{k} + O(1/k). \quad (26.2)$$

The identity at $t = 1$ is interpreted separately: both cumulative distribution functions equal one.

The function $Y(t) = -\frac{1}{2} \cot(\pi t)$ is a strictly increasing bijection from $(0, 1)$ to $\mathbb{R}$. Its inverse is $t = \frac{1}{2} + \pi^{-1} \arctan(2y)$, so it sends the uniform distribution exactly to $\mathcal{C}$. It also preserves the Kolmogorov discrepancy. Equation (26.2) therefore proves the weak limit and an effective, finite- k bound.

The logarithm cannot be removed from the rate. Symmetry about $t = \frac{1}{2}$, with the central atom retained, gives

$$\mathcal{F}_k(1/2) - 1/2 = \frac{E_2(k)}{2D_k} = \frac{1}{2\zeta(2)} \frac{\log k}{k} + O(1/k).$$

The exact optimal leading constant requires a more delicate argument, given in Section 28. In particular the central atom supplies a useful lower bound, but it does not locate the largest discrepancy.

Remark 26.1 (The mode of convergence). Every ν_k^z is discrete, whereas $\mathcal{C}$ has a density. Their total-variation distance, defined as the supremum over measurable events, is therefore exactly one. Weak convergence and uniform cumulative-distribution convergence are the appropriate statements here. Similarly, weak convergence alone gives no convergence of the first or higher absolute moments.

26.2 Exceptional roots and growing Hankel shifts

By (24.15), the full degree is

$$\deg H_{k,s} = A_k + B_{k,s} + D_k = \binom{k+1}{2} + 6c + k(s-1) - 2G_k. \quad (26.3)$$

The full zero measure satisfies the exact decomposition

$$\mu_{k,s}^z = \frac{A_k}{\deg H_{k,s}} \delta_0 + \frac{B_{k,s}}{\deg H_{k,s}} \delta_{-1} + \frac{D_k}{\deg H_{k,s}} \hat{\nu}_k^z. \quad (26.4)$$

For $k \geq 3$, the three components have disjoint supports. For $k = 1, 2$, the last component is absent; a zero exponent in an exceptional factor simply contributes no root.

Proof of Theorem 24.3. We have

$$A_k/k^3 \rightarrow 1/3, \quad D_k/k^3 \rightarrow 1/3, \quad B_{k,s}/k^3 = 1/3 + s/k^2 + o(1).$$

If $s/k^2 \rightarrow \varkappa < \infty$, divide (26.4) by its total degree and use $\hat{\nu}_k^z \implies \hat{\mathcal{C}}$. This gives (24.14). If $s/k^2 \rightarrow \infty$, the coefficient of δ_{-1} tends to one, which proves the remaining assertion. $\square$

Thus the shift has no effect whatever on the reduced zero measure, but has a genuine effect on full-degree normalization. In particular, normalizing only the retained zeros by the original full degree would produce a subprobability measure, with mass $1/[3(1 + \varkappa)]$, rather than a Cauchy probability measure.

27 The first correction for smooth statistics

The arithmetic jumps in Table 12 coexist with a particularly simple first correction after smooth averaging.

Theorem 27.1. *If f is a twice continuously differentiable periodic function on $\mathbb{R}/\mathbb{Z}$, and $I_f = \int_0^1 f(t) dt$, then*

$$\int f d\rho_k^z = I_f + \frac{3}{\zeta(2)} \frac{\log k}{k} (I_f - f(0)) + O_f(1/k). \quad (27.1)$$

Consequently, if $\phi \in C_c^2(\mathbb{R})$, then

$$\int \phi(Y) d\nu_k^z(Y) = \left(1 + \frac{3}{\zeta(2)} \frac{\log k}{k}\right) \int_{\mathbb{R}} \phi(y) \frac{2 dy}{\pi(1 + 4y^2)} + O_\phi(1/k). \quad (27.2)$$

Proof. The Fourier series of f is absolutely convergent, and

$$|\widehat{f}(h)| \leq \frac{\|f''\|_{L^1}}{4\pi^2 h^2}, \quad h \neq 0.$$

Summing the Fourier series over a regular q -grid therefore gives

$$\sum_{\ell=0}^{q-1} f(\ell/q) = qI_f + \text{err}_q, \quad |\text{err}_q| \leq \frac{\|f''\|_{L^1}}{12q}.$$

After removing the term at zero and summing with the weights $\mathcal{W}_k(q)$, we obtain

$$\mathsf{D}_k \int f \, d\rho_k^z = 2cI_f - 2G_k f(0) + \sum_q \mathcal{W}_k(q) \text{err}_q. \quad (27.3)$$

From (25.4),

$$\mathcal{W}_k(q) \leq \sum_{b=q+1}^k \left(\frac{k^2}{b^2} + \frac{k}{b} \right) \leq \frac{2k^2}{q}.$$

Thus the last sum in (27.3) is $O_f(k^2)$. Divide by D_k , and use

$$\frac{2G_k}{\mathsf{D}_k} = \frac{3}{\zeta(2)} \frac{\log k}{k} + O(1/k).$$

This proves (27.1). For $\phi \in C_c^2(\mathbb{R})$, the pullback $f(t) = \phi(-\frac{1}{2} \cot(\pi t))$ is periodic C^2 and is zero near $t = 0, 1$. Hence $f(0) = 0$, and the change of variables in (24.4) gives (27.2). $\square$

The coefficient in (27.1) has a direct interpretation: the regular grids would include the angle zero, but that angle corresponds to the point at infinity under (24.3). Removing it and renormalizing creates the positive first bias in compactly supported smooth statistics. A discontinuous indicator cannot be substituted into this theorem. Its endpoint may hit a root of small order, and then the correction changes.

28 Arithmetic CDF corrections and the sharp constant

We now refine the elementary bound (26.2). For $0 \leq t < 1$, write

$$\mathsf{D}_k(\mathcal{F}_k(t) - t) = 2G_k t - \mathcal{S}_k(t), \quad \mathcal{S}_k(t) = \sum_{i < j} 2g_{ij} \{q_{ij}t\}. \quad (28.1)$$

The endpoint $t = 1$ is treated separately. All limits below use right-continuous CDFs unless a left limit is displayed.

Theorem 28.1 (Pointwise arithmetic correction). *For every irrational $t \in (0, 1)$,*

$$\lim_{k \rightarrow \infty} \frac{k}{\log k} (\mathcal{F}_k(t) - t) = \frac{3}{\zeta(2)} \left(t - \frac{1}{2} \right).$$

For a reduced rational $t = r/m \in (0, 1)$,

$$\lim_{k \rightarrow \infty} \frac{k}{\log k} (\mathcal{F}_k(t) - t) = \frac{3}{\zeta(2)} \left(t - \frac{1}{2} + \frac{1}{2\psi(m)} \right).$$

The corresponding left-continuous value $\mathcal{F}_k(t-)$ has the same limit with a minus sign before $1/(2\psi(m))$.

The right/left distinction records the atom at the primitive root: subtracting the formulas yields asymptotic normalized atom mass $3 \log k / (\zeta(2)\psi(m)k)$.

Theorem 28.2 (Moving angles near rational points). *Suppose $t_k \in [0, 1)$ and $t_k \rightarrow t_0$. If t_0 is irrational, then*

$$\frac{k}{\log k} (\mathcal{F}_k(t_k) - t_k) \longrightarrow \frac{3}{\zeta(2)} (t_0 - 1/2).$$

If $t_0 = r/m \in [0, 1]$ is rational in lowest terms, assume $\delta_k = t_k - t_0 \neq 0$ has an eventually constant sign ς , and

$$\alpha = \lim_{k \rightarrow \infty} \min \left\{ 1, \frac{\log(1/|\delta_k|)}{\log k} \right\} \quad \text{exists.}$$

Then $0 \leq \alpha \leq 1$ and

$$\frac{k}{\log k} (\mathcal{F}_k(t_k) - t_k) \longrightarrow \frac{3}{\zeta(2)} \left(t_0 - \frac{1}{2} + \frac{\varsigma \alpha}{2\psi(m)} \right). \quad (28.2)$$

If $t_k = t_0 < 1$ identically, the formula holds with $\varsigma = +1$, $\alpha = 1$. Statements with eventual equality have the same meaning. For the endpoints $0 = 0/1$ and $1 = 1/1$, the permitted signs are, respectively, $+1$ and -1 .

Corollary 28.3 (Sharp Kolmogorov constant).

$$\sup_{0 \leq t \leq 1} |\mathcal{F}_k(t) - t| \sim \frac{3}{2\zeta(2)} \frac{\log k}{k} = \frac{9}{\pi^2} \frac{\log k}{k}.$$

The same constant holds after the monotone change of coordinate $y = -\frac{1}{2} \cot(\pi t)$, relative to the Cauchy distribution of scale $\frac{1}{2}$.

28.1 An exact uniform reduction

Write $H_N = \sum_{q \leq N} q^{-1}$ and $Q_N(\theta) = \sum_{q \leq N} \{q\theta\}/q$. All $O(1)$ constants in the following lemma are absolute and uniform in the real argument t .

Lemma 28.4.

$$2G_k = \frac{k^2}{\zeta(2)} \log k + O(k^2), \quad (28.3)$$

$$D_k = \frac{k^3}{3} + O(k^2 \log k), \quad (28.4)$$

$$\frac{\mathcal{S}_k(t)}{k^2} = \sum_{d=1}^{\infty} \frac{\mu(d)}{d^2} Q_k(dt) + O(1). \quad (28.5)$$

Proof. Write $i = ga$, $j = gb$ with $(a, b) = 1$. With $q = b - a$, coprimality is equivalent to $(q, b) = 1$. Consequently

$$\mathcal{S}_k(t) = \sum_{b=2}^k h_b(h_b + 1) J_b(t), \quad h_b = \lfloor k/b \rfloor, \quad J_b(t) = \sum_{\substack{1 \leq q < b \\ (q, b) = 1}} \{qt\}.$$

Since $h_b(h_b + 1) = (k/b)^2 + O(k/b)$ and $0 \leq J_b \leq b$, this gives

$$\frac{\mathcal{S}_k(t)}{k^2} = \sum_{b \leq k} \frac{J_b(t)}{b^2} + O(1).$$

Möbius inversion, including the harmless $b = 1$ term, yields

$$\sum_{b \leq k} \frac{J_b(t)}{b^2} = \sum_{d \leq k} \frac{\mu(d)}{d^2} K_{\lfloor k/d \rfloor}(dt), \quad K_N(\theta) = \sum_{n \leq N} \frac{1}{n^2} \sum_{q < n} \{q\theta\}.$$

Interchanging the finite sums and using

$$\sum_{n=q+1}^N \frac{1}{n^2} = \frac{1}{q} + O(q^{-2}) + O(N^{-1}) \quad (1 \leq q < N)$$

shows $K_N(\theta) = Q_N(\theta) + O(1)$ uniformly in θ and N . Moreover,

$$|Q_k(\theta) - Q_{\lfloor k/d \rfloor}(\theta)| \leq O(1) + \log(2d).$$

The convergent series $\sum_d (1 + \log d)/d^2$ therefore allows us to replace all cutoffs by k . Extending the outer sum to infinity changes the result by at most $H_k \sum_{d > k} d^{-2} = O(1)$, proving (28.5).

The gcd estimate is Proposition 25.2 at $m = 1$, and the degree estimate follows from (24.2). $\square$

28.2 The harmonic sawtooth lemma

This is the key step responsible for the sharp uniform constant. It uses only approximation of a Riemann-integrable periodic function by trigonometric upper and lower bounds, and a geometric-sum estimate. It does not use an unproved cancellation statement for Möbius sums.

Lemma 28.5 (Harmonic sawtooth near a limiting angle). *Let $\theta_k \rightarrow \theta_0$ be a real sequence. If θ_0 is irrational, then*

$$\frac{Q_k(\theta_k)}{\log k} \rightarrow \frac{1}{2}.$$

Suppose instead that $\theta_0 = a/b$ in lowest terms, $b \geq 1$, $\tilde{\delta}_k = \theta_k - \theta_0 \neq 0$ has eventually constant sign ς , and

$$\tilde{\alpha} = \lim_{k \rightarrow \infty} \min\{1, \log(1/|\tilde{\delta}_k|)/\log k\}$$

exists. Then

$$\frac{Q_k(\theta_k)}{\log k} \rightarrow \frac{1}{2} - \frac{\varsigma \tilde{\alpha}}{2b}. \quad (28.6)$$

For $\theta_k = a/b$ exactly, the limit is $1/2 - 1/(2b)$.

Proof. Let $\bar{B}_1(x) = \{x\} - 1/2$. For every $\varepsilon > 0$ there exist real trigonometric polynomials T^- and T^+ with

$$T^-(x) \leq \bar{B}_1(x) \leq T^+(x) \quad (x \in \mathbb{R}), \quad \left| \int_0^1 T^\pm(x) dx \right| \leq \varepsilon.$$

For completeness, first choose continuous periodic lower and upper bounds that agree with the sawtooth away from an arbitrarily small interval around its jump; then approximate these continuous functions uniformly by trigonometric polynomials and move the approximants down or up by their uniform approximation error. Their integrals can be made arbitrarily close to the sawtooth's zero integral.

Here $\|x\| = \min_{n \in \mathbb{Z}} |x - n|$ denotes distance to the nearest integer. For $h \neq 0$ and $\|h\theta\| > 0$, a geometric sum followed by partial summation gives

$$\left| \sum_{q=M'+1}^N \frac{\exp(2\pi i h q \theta)}{q} \right| \leq \frac{C}{(M'+1)\|h\theta\|}, \quad 0 \leq M' < N, \quad (28.7)$$

with an absolute constant C . In the irrational case, for the finitely many nonzero frequencies in $T^\pm$, each $\|h\theta_k\|$ is bounded away from zero for all sufficiently large k . The polynomial inequalities and (28.7) with $M' = 0$ give

$$\left| \sum_{q \leq k} \frac{\overline{B}_1(q\theta_k)}{q} \right| \leq \varepsilon H_k + O_{\varepsilon, \theta_0}(1).$$

Divide by $\log k$ and let $\varepsilon \downarrow 0$.

Now suppose $\theta_0 = a/b$ and $\tilde{\delta}_k \neq 0$. Fix $0 < \kappa < 1/(4b)$, and write

$$N_k = \min\{k, \lfloor \kappa/|\tilde{\delta}_k| \rfloor\}.$$

For all large k , $N_k \rightarrow \infty$. For $q \leq N_k$, the displacement $|q\tilde{\delta}_k|$ is too small to cross a discontinuity unless $b \mid q$. We therefore have the exact identity

$$Q_{N_k}(\theta_k) = Q_{N_k}(a/b) + \tilde{\delta}_k N_k + \mathbf{1}_{\{\zeta=-1\}} \sum_{\substack{q \leq N_k \\ b \mid q}} \frac{1}{q}.$$

Periodicity gives

$$Q_N(a/b) = \frac{b-1}{2b} \log N + O_b(1), \quad \sum_{\substack{q \leq N \\ b \mid q}} q^{-1} = \frac{1}{b} \log N + O_b(1).$$

Because $|\tilde{\delta}_k N_k| \leq \kappa$, it follows that

$$\sum_{q \leq N_k} \frac{\overline{B}_1(q\theta_k)}{q} = -\frac{\varsigma}{2b} \log N_k + O_b(1). \quad (28.8)$$

It remains to control $N_k < q \leq k$, if this interval is nonempty. For every fixed nonzero integer h , if $b \nmid h$, then $\|h\theta_k\|$ stays bounded away from zero. If $b \mid h$, then eventually $\|h\theta_k\| = |h\tilde{\delta}_k|$ and $(N_k + 1)|\tilde{\delta}_k| > \kappa$. Hence (28.7) bounds every nonconstant Fourier term over the high interval by a constant depending on h, b, κ , independently of k . Apply the polynomial bounds again to obtain

$$\left| \sum_{q=N_k+1}^k \frac{\overline{B}_1(q\theta_k)}{q} \right| \leq \varepsilon (H_k - H_{N_k}) + O_{\varepsilon, b, \kappa}(1).$$

Consequently this high sum divided by $\log k$ tends to zero, by letting $\varepsilon \downarrow 0$. Finally $\log N_k / \log k \rightarrow \tilde{\alpha}$, so (28.8) proves (28.6). The exactly rational case follows directly from the periodicity computation. $\square$

[Note of the batch-98 write: manuscript 08 writes the lower summation limit in (28.7) as M ; it is printed as M' here because M is the zero-row count of Part I.]

28.3 Proofs of the arithmetic and sharp uniform results

Proof of Theorem 28.2. If t_0 is irrational, every dt_0 is irrational. By Lemma 28.5, $Q_k(dt_k)/\log k \rightarrow 1/2$ for every fixed d . Since $0 \leq Q_k \leq H_k$ and $\sum d^{-2} < \infty$, dominated convergence in (28.5) gives

$$\frac{S_k(t_k)}{k^2 \log k} \longrightarrow \frac{1}{2\zeta(2)}.$$

If $t_0 = r/m$ is reduced, then dt_0 has reduced denominator $m/(d, m)$. The sign of $d\delta_k$ is ς , and replacing δ_k by $d\delta_k$ does not change α . Thus dominated convergence gives

$$\frac{\mathcal{S}_k(t_k)}{k^2 \log k} \longrightarrow \frac{1}{2\zeta(2)} - \frac{\varsigma\alpha}{2m} \sum_{d \geq 1} \frac{\mu(d)(d, m)}{d^2}.$$

The absolutely convergent Euler product evaluates to

$$\sum_{d \geq 1} \frac{\mu(d)(d, m)}{d^2} = \prod_{p|m} (1 - 1/p) \prod_{p \nmid m} (1 - 1/p^2) = \frac{1}{\zeta(2)} \prod_{p|m} \frac{p}{p+1} = \frac{m}{\zeta(2)\psi(m)}.$$

The same calculation with the exact rational case of Lemma 28.5 treats $\delta_k = 0$ identically. Combining these limits with (28.1), (28.3) and (28.4) proves all statements. $\square$

Proof of Theorem 28.1. The right-continuous formulas follow from Theorem 28.2. The root multiset is invariant under $t \mapsto 1 - t$, so $\mathcal{F}_k(t-) = 1 - \mathcal{F}_k(1 - t)$ for $0 < t < 1$. Apply the right-continuous formula at $1 - t$, whose reduced denominator is again m , to obtain the left formula. Alternatively it follows by counting the multiples of m in the exact fractional-part identity. $\square$

Proof of Corollary 28.3. For the lower bound, fix any irrational $t \in (0, 1/2)$ and use Theorem 28.1; then let this fixed t decrease to zero.

For the upper bound it suffices to examine an arbitrary subsequence of angles $t_k \in [0, 1]$ at which the scaled absolute discrepancies approach their limsup. Terms with $t_k = 1$ have discrepancy zero and may be discarded. After taking a subsequence, compactness gives $t_k \rightarrow t_0$. If t_0 is irrational, the absolute limit from Theorem 28.2 is strictly below $3/(2\zeta(2))$. If $t_0 = r/m \in (0, 1)$ is reduced, pass to a further subsequence on which either $t_k = t_0$ or δ_k has fixed sign and the truncated exponent has a limit $\alpha \in [0, 1]$. The absolute limit is bounded by

$$\frac{3}{\zeta(2)} \left(\left| \frac{r}{m} - \frac{1}{2} \right| + \frac{1}{2\psi(m)} \right) \leq \frac{3}{\zeta(2)} \left(\frac{1}{2} - \frac{1}{m} + \frac{1}{2m} \right) < \frac{3}{2\zeta(2)}.$$

At $t_0 = 0$, the possible bracket in (28.2) is $-1/2 + \alpha/2 \in [-1/2, 0]$; exact $t_k = 0$ gives zero. At $t_0 = 1$, it is $1/2 - \alpha/2 \in [0, 1/2]$. These exhaust all subsequences and prove the upper bound. The coordinate map to the vertical line is strictly increasing, so it preserves Kolmogorov distance. $\square$

28.4 Interpretation and useful examples

At $t = 1/2$, the right-continuous correction is $3/(2\zeta(2)\psi(2)) = 1/(2\zeta(2))$, and the left correction is its negative. For $t_k = t_0 + k^{-\alpha}$ with rational $t_0 \in (0, 1)$ and $0 < \alpha < 1$, the rational correction is α times its exact right-continuous size. Approaching from below changes its sign. If $t_k = k^{-\alpha}$ with $0 < \alpha < 1$, then

$$\frac{k}{\log k} (\mathcal{F}_k(t_k) - t_k) \longrightarrow -\frac{3(1 - \alpha)}{2\zeta(2)}.$$

If $t_k = 1/\log k$, the limit is $-3/(2\zeta(2))$, explicitly realizing the sharp constant along a moving endpoint sequence. The sharp constant therefore reflects a logarithmic range of angles approaching an endpoint, rather than a single fixed rational atom.

29 The arithmetic law at the extreme scale

The Cauchy law has unbounded support. In contrast, the largest imaginary coordinate of a finite polynomial is approximately $k/(2\pi)$. To describe this range one must both rescale the coordinate and compensate for the small proportion of zeros there.

29.1 An exact tail count

For $y > 0$, put

$$\Omega(y) = \frac{\pi}{\arctan(1/(2y))} > 2, \quad Z_k(y) = D_k \nu_k^z((y, \infty)). \quad (29.1)$$

A positive ordinate in a q -grid is $\frac{1}{2} \cot(\pi r/q)$, $1 \leq r < q/2$. The condition that it exceed y is exactly $q > \Omega(y)r$. For real $K > 0$ and $Q \geq 0$, define

$$\mathcal{P}(K, Q) = \#\{(a, q) \in \mathbb{Z}_{>0}^2 : (a, q) = 1, a + q \leq K, q > Q\}.$$

Using the representation $i = ga$, $j = g(a + q)$, we obtain the exact formula

$$Z_k(y) = 2 \sum_{g, r \geq 1} g \mathcal{P}(k/g, \Omega(y)r). \quad (29.2)$$

Every nonzero term satisfies $gr < k/\Omega(y)$, so this sum is finite. The strict inequality is important at a threshold that is itself a zero.

Lemma 29.1 (Primitive points in a truncated triangle). *Uniformly for $K \geq 1$, $Q \geq 0$,*

$$\mathcal{P}(K, Q) = \frac{(K - Q)_+^2}{2\zeta(2)} + O(K \log(2K)), \quad (29.3)$$

with an absolute implied constant.

Proof. If $Q \geq K$, both the count and the main term vanish. Otherwise let $\mathcal{P}^*(K, Q)$ be the count without coprimality. Summing the possible positive values of a for each $q > Q$, and comparing the linear function $K - v$ with its integral, gives

$$\mathcal{P}^*(K, Q) = \frac{1}{2}(K - Q)^2 + O(K + 1)$$

uniformly in Q . Möbius inversion gives

$$\mathcal{P}(K, Q) = \sum_{d \leq K} \mu(d) \mathcal{P}^*(K/d, Q/d).$$

Its main term is $\frac{1}{2}(K - Q)^2 \sum_{d \leq K} \mu(d)/d^2$, and its error is $O(\sum_{d \leq K} (K/d + 1)) = O(K \log(2K))$. Extending the main sum to infinity costs only $O(K)$. $\square$

29.2 The edge tail, its density, and its cutoff

Theorem 29.2 (Quantitative edge law). *Let Ξ be defined in (24.8). For every fixed $\varepsilon > 0$,*

$$\sup_{x \geq \varepsilon} |k \nu_k^z((kx, \infty)) - \Xi(x)| = O_\varepsilon(\log k/k). \quad (29.4)$$

The same assertion holds for the two-sided tail with limit $2\Xi(x)$.

Proof. Uniformly for $x \geq \varepsilon$,

$$\Omega(kx) = 2\pi kx + \frac{\pi}{6kx} + O_\varepsilon(k^{-3}), \quad \Omega(kx) > 2\pi kx.$$

Thus all nonzero terms in (29.2) belong to the fixed finite set $gr < 1/(2\pi\varepsilon)$. Apply Lemma 29.1; replacing $\Omega(kx)$ by $2\pi kx$ in the squared areas costs only $O_\varepsilon(1)$. We obtain

$$\begin{aligned} Z_k(kx) &= \frac{1}{\zeta(2)} \sum_{g,r \geq 1} g \left(\frac{k}{g} - 2\pi kxr \right)_+^2 + O_\varepsilon(k \log k) \\ &= \frac{k^2}{\zeta(2)} \sum_{g,r \geq 1} \frac{1}{g} (1 - 2\pi xgr)_+^2 + O_\varepsilon(k \log k) \\ &= \frac{k^2}{\zeta(2)} \sum_{n \geq 1} \sigma_{-1}(n) (1 - 2\pi xn)_+^2 + O_\varepsilon(k \log k). \end{aligned} \quad (29.5)$$

At a breakpoint $2\pi xgr = 1$, the actual term is zero: $\Omega(kx) > 2\pi kx$ and a contributing q would have to be both larger than k/g and smaller than k/g . The area term is zero as well. Hence the proof does not exclude breakpoints. Divide (29.5) by $D_k = k^3/3 + O(k^2 \log k)$, and multiply by k . Symmetry proves the two-sided statement. $\square$

On $(0, \infty)$, Ξ is decreasing, convex, continuously differentiable, and piecewise quadratic. Its negative derivative is

$$-\Xi'(x) = \frac{12\pi}{\zeta(2)} \sum_{n \geq 1} \sigma_1(n) (1 - 2\pi nx)_+, \quad \sigma_1(n) = \sum_{d|n} d. \quad (29.6)$$

Thus the locally finite measures $\nu_k^e(A) = k \nu_k^z(kA)$ on $(0, \infty)$ converge vaguely to $-\Xi'(x) dx$. This follows first on intervals bounded away from zero by subtracting the tail limits, and then for compactly supported continuous functions by step-function approximation. The limit is not a probability measure: its mass diverges toward the origin.

At $x_n = 1/(2\pi n)$, the jump in the second derivative is

$$\Xi''(x_n^-) - \Xi''(x_n^+) = \frac{24\pi^2}{\zeta(2)} n^2 \sigma_{-1}(n). \quad (29.7)$$

Divisor sums are therefore recoverable from the slope changes of the edge density. On the outermost interval,

$$\frac{1}{4\pi} \leq x \leq \frac{1}{2\pi}, \quad \Xi(x) = \frac{3}{\zeta(2)} (1 - 2\pi x)^2.$$

Proposition 29.3 (Exact extreme zeros). *For $k \geq 4$, the two zeros of largest absolute imaginary coordinate are*

$$-\frac{1}{2} \pm \frac{i}{2} \cot \frac{\pi}{k-1}, \quad (29.8)$$

each of algebraic multiplicity exactly two. In particular,

$$\max |\operatorname{Im} u| < \frac{k}{2\pi}, \quad \frac{\max |\operatorname{Im} u|}{k} \longrightarrow \frac{1}{2\pi}.$$

For $k = 3$, the only reduced zero is $-1/2$, with multiplicity two.

Proof. Every grid has $q = (j - i)/g_{ij} \leq k - 1$. The least positive angle entering a positive cotangent is $\pi/(k - 1)$, attained only at $(i, j) = (1, k)$, $r = 1$. That pair has $g_{ij} = 1$; no other grid can produce this same extreme angle. The bound follows from $\cot z < 1/z$ for $0 < z < \pi/2$. $\square$

The edge theorem describes a collection of order k^2 zeros with multiplicity, out of a total of order k^3 . It does not describe a limiting process for the finitely many largest individual roots. The latter question is finer than (29.4).

30 Matching the edge to the Cauchy tail

30.1 A divisor-sum expansion with its constant

Theorem 30.1. As $x \downarrow 0$,

$$\Xi(x) = \frac{1}{2\pi x} - \frac{3}{2\zeta(2)} \log \frac{1}{x} + \frac{3}{2\zeta(2)} \left(\frac{3}{2} - \gamma_E \right) + O\left(x \log \frac{1}{x}\right). \quad (30.1)$$

Proof. Put

$$\mathcal{H}(t) = \sum_{n \geq 1} \sigma_{-1}(n) (1 - tn)_+^2, \quad X = 1/t, \quad B_3(v) = v^3 - \frac{3}{2}v^2 + \frac{1}{2}v.$$

Regroup the divisors as $n = d\ell$. The elementary finite-sum identity

$$\sum_{\ell=1}^{\lfloor v \rfloor} (v - \ell)^2 = \frac{B_3(v) - B_3(\{v\})}{3}$$

then gives the exact equality

$$\mathcal{H}(1/X) = \frac{X}{3} \sum_{d \leq X} \frac{1}{d^2} - \frac{1}{2} \sum_{d \leq X} \frac{1}{d} + \frac{\lfloor X \rfloor}{6X} - \frac{1}{3X^2} \sum_{d \leq X} d B_3(\{X/d\}). \quad (30.2)$$

Define $\Upsilon(v) = v B_3(\{1/v\})$ for $v > 0$, with $\Upsilon(0) = 0$. The periodic extension of B_3 and its first derivative agree at integer endpoints. Thus Υ is continuous, $|\Upsilon(v)| \leq Cv$, and away from zero $|\Upsilon'(v)| \leq C(1 + 1/v)$. Splitting off $[0, 1/X]$ and bounding the remaining Riemann-sum error by the integral of $|\Upsilon'|/X$ proves

$$\frac{1}{X^2} \sum_{d \leq X} d B_3(\{X/d\}) = \mathcal{I}_3 + O(\log X/X), \quad \mathcal{I}_3 = \int_0^1 \Upsilon(v) dv. \quad (30.3)$$

Here

$$\mathcal{I}_3 = \int_1^\infty \frac{B_3(\{v\})}{v^3} dv = \frac{3}{2} \log(2\pi) - \frac{11}{4}.$$

For clarity, the constant can be evaluated without a contour argument. Two integrations by parts using the periodic Bernoulli polynomials give

$$\mathcal{I}_3 = \frac{1}{4} + 3 \int_1^\infty \frac{\{v\} - 1/2}{v} dv.$$

The last integral converges by the zero mean of its periodic numerator. Its integral through $N + 1$ is

$$N - (N + 1/2) \log(N + 1) + \log(N!),$$

which tends to $\frac{1}{2} \log(2\pi) - 1$ by Stirling's formula.

Finally,

$$\frac{X}{3} \sum_{d \leq X} d^{-2} = \frac{\zeta(2)X}{3} - \frac{1}{3} + O(X^{-1}), \quad \sum_{d \leq X} d^{-1} = \log X + \gamma_E + O(X^{-1}), \quad \frac{\lfloor X \rfloor}{6X} = \frac{1}{6} + O(X^{-1}).$$

Insert these and (30.3) into (30.2). The result is

$$\mathcal{H}(t) = \frac{\zeta(2)}{3t} - \frac{1}{2} \log \frac{1}{t} + \frac{3}{4} - \frac{\gamma_E}{2} - \frac{1}{2} \log(2\pi) + O\left(t \log \frac{1}{t}\right). \quad (30.4)$$

Since $\Xi(x) = 3\mathcal{H}(2\pi x)/\zeta(2)$, the $\log(2\pi)$ terms cancel, giving (30.1). $\square$

[Note of the batch-98 write: in the evaluation of $\mathcal{I}_3$, N is a running integer of that display only, not the diagonal index of Part I.]

The leading term matches the Cauchy tail. The next term has exactly the scale that appears in the global discrepancy theorem. The remainder in (30.1) has not been differentiated; the density formula (29.6) follows directly from the exact finite sum.

30.2 One formula covering the entire intermediate range

Theorem 30.2 (Uniform transition formula). *For $y > 0$, set $\mathfrak{m} = k/\Omega(y)$. If $\mathfrak{m} \leq 1$, both $Z_k(y)$ and $\mathcal{H}(1/\mathfrak{m})$ vanish. If $\mathfrak{m} \geq 1$, then*

$$Z_k(y) = \frac{k^2}{\zeta(2)} \mathcal{H}(1/\mathfrak{m}) + O(k\mathfrak{m} \log(2\mathfrak{m}) \log(2k)), \quad (30.5)$$

with an absolute implied constant. For sufficiently large k , its normalized form is

$$\nu_k^z((y, \infty)) = \frac{1}{k} \Xi\left(\frac{\Omega(y)}{2\pi k}\right) + O\left(\frac{\log(2\mathfrak{m}) \log(2k)}{\Omega(y)k}\right), \quad \mathfrak{m} \geq 1. \quad (30.6)$$

Proof. Apply Lemma 29.1 to (29.2) without approximating Ω . The squared areas regroup exactly into $k^2 \mathcal{H}(\Omega/k)/\zeta(2)$. Only $gr < \mathfrak{m}$ contribute. Their total error is at most

$$Ck \sum_{g \leq \mathfrak{m}} \left\lfloor \frac{\mathfrak{m}}{g} \right\rfloor \log(2k/g) \leq C' k \mathfrak{m} \log(2\mathfrak{m}) \log(2k).$$

This proves (30.5). Also $\mathcal{H}(1/\mathfrak{m}) = O(\mathfrak{m})$ for $\mathfrak{m} \geq 1$, for example by summing its divisor expression and using $\sum d^{-2} < \infty$. The normalization $D_k = k^3/3 + O(k^2 \log k)$ gives (30.6); its extra normalization error is absorbed because $\log(2\mathfrak{m})$ is bounded below by $\log 2$. $\square$

Corollary 30.3 (All mesoscopic Cauchy tails). *If $y_k \rightarrow \infty$ and $y_k = o(k)$, then*

$$\nu_k^z((y_k, \infty)) \sim \frac{1}{2\pi y_k}, \quad \nu_k^z(\{|Y| > y_k\}) \sim \frac{1}{\pi y_k}.$$

Proof. Now $\Omega(y_k) \sim 2\pi y_k$, so $\mathfrak{m}_k = k/\Omega(y_k) \rightarrow \infty$. Equations (30.4) and (30.6) imply

$$\nu_k^z((y_k, \infty)) = \frac{1}{\Omega(y_k)} \left[1 + O\left(\frac{\log(2\mathfrak{m}_k)}{\mathfrak{m}_k} + \frac{\log(2\mathfrak{m}_k) \log(2k)}{k}\right) \right].$$

Both errors tend to zero; note that $\mathfrak{m}_k < k/2$. Symmetry gives the second formula. $\square$

The exact effective argument in (30.6) is $\Omega(y)/(2\pi k)$. Replacing it by y/k is justified for growing y , but would lose the correct Cauchy coordinate for bounded y . The mesoscopic result places no extra restriction such as $y = o(k/\log k)$.

31 The transition in absolute moments

The zeros are symmetric in their imaginary coordinate, so every signed odd moment is zero. Absolute moments reveal the heavy tail. For $p > 0$, retain the notation $M_p(k) = \int |Y|^p d\nu_k^z(Y)$.

31.1 A uniform bound at every height

Before applying its multiplicity, a q -grid has at most $2q/\Omega(y)$ points with $|Y| > y$. As $\sum_{i < j} g_{ij}q_{ij} = c$, we have the useful exact bound

$$\nu_k^z(\{|Y| > y\}) \leq \min\left(1, \frac{4c}{D_k \Omega(y)}\right) \leq \min\left(1, \frac{2c}{\pi D_k y}\right). \quad (31.1)$$

The right-hand side is bounded by $\min(1, C/y)$ for an absolute constant and all $k \geq 3$. The finitely many small k 's can be included after increasing the constant, since $2c/D_k \rightarrow 1$.

31.2 Orders below and above one

Proof of (24.10). For a nonnegative random variable, the layer-cake identity gives

$$M_p(k) = p \int_0^\infty y^{p-1} \nu_k^z(\{|Y| > y\}) dy.$$

When $0 < p < 1$, (31.1) supplies an integrable majorant, $py^{p-1} \min(1, C/y)$. Weak convergence to the continuous Cauchy law gives convergence of the tails at every positive y . Dominated convergence proves moment convergence. Finally, the change of variables $z = 2y$ evaluates the limit:

$$\int_{\mathbb{R}} |y|^p \frac{2 dy}{\pi(1+4y^2)} = \frac{2^{1-p}}{\pi} \int_0^\infty \frac{z^p}{1+z^2} dz = \frac{2^{-p}}{\cos(\pi p/2)}.$$

□

Proof of (24.12). The exact cutoff in Proposition 29.3 and the layer-cake identity yield

$$k^{1-p} M_p(k) = p \int_0^{1/(2\pi)} x^{p-1} k \nu_k^z(\{|Y| > kx\}) dx.$$

By (31.1), the integrand is at most a constant times x^{p-2} . This is integrable at zero exactly when $p > 1$. The edge theorem and dominated convergence therefore give

$$\begin{aligned} \lim_{k \rightarrow \infty} k^{1-p} M_p(k) &= \frac{6p}{\zeta(2)} \sum_{n \geq 1} \sigma_{-1}(n) \int_0^{1/(2\pi n)} x^{p-1} (1 - 2\pi n x)^2 dx \\ &= \frac{12}{(2\pi)^p (p+1)(p+2)\zeta(2)} \sum_{n \geq 1} \frac{\sigma_{-1}(n)}{n^p} \\ &= \frac{12\zeta(p)\zeta(p+1)}{(2\pi)^p (p+1)(p+2)\zeta(2)}. \end{aligned}$$

Tonelli's theorem justifies the nonnegative interchange. The last Dirichlet-series identity follows directly by writing $n = d\ell$:

$$\sum_{n \geq 1} \frac{\sigma_{-1}(n)}{n^p} = \sum_{d, \ell \geq 1} \frac{1}{d^{p+1} \ell^p} = \zeta(p+1)\zeta(p).$$

□

This computation shows why cutting off a continuous Cauchy density at its largest zero would give the wrong higher-moment constants. The entire divisor-sum edge profile contributes. For instance,

$$M_2(k) \sim \frac{\zeta(3)}{4\pi^2} k. \quad (31.2)$$

31.3 The critical moment, with its constant term

Lemma 31.1. *Uniformly for integers $q \geq 1$,*

$$S_1(q) := \frac{1}{2} \sum_{\ell=1}^{q-1} \left| \cot \frac{\pi \ell}{q} \right| = \frac{q}{\pi} (\log q + \gamma_E - \log \pi) + O(1). \quad (31.3)$$

Proof. By symmetry,

$$S_1(q) = \sum_{\ell=1}^{\lfloor (q-1)/2 \rfloor} \cot(\pi \ell / q).$$

The function $h(v) = \cot(\pi v) - 1/(\pi v)$ extends smoothly across zero on $[0, 1/2]$, and

$$\int_0^{1/2} h(v) dv = \frac{1}{\pi} \log \frac{2}{\pi}.$$

Subtract $q/(\pi \ell)$ in the sum. A Riemann-sum error $O(1)$, together with the harmonic-number expansion, gives

$$S_1(q) = \frac{q}{\pi} (\log(q/2) + \gamma_E + \log(2/\pi)) + O(1).$$

The finitely many smallest values of q are absorbed in the constant. $\square$

Define the two weighted sums

$$\mathcal{S}_k^{\text{gap}} = \sum_{i < j} (j - i) \log(j - i), \quad \mathcal{S}_k^{\text{gcd}} = \sum_{i < j} (j - i) \log g_{ij}.$$

For the first, grouping by $h = j - i$ and using a Riemann sum gives

$$\mathcal{S}_k^{\text{gap}} = \sum_{h=1}^{k-1} h(k-h) \log h = c \left(\log k - \frac{5}{6} \right) + O(k^2 \log k). \quad (31.4)$$

The constant follows from $\int_0^1 v(1-v) \log v dv = -5/36$ and $\int_0^1 v(1-v) dv = 1/6$. The integrable endpoint derivative also gives the displayed Riemann-sum error.

For the second, let $\Lambda_{\text{vM}}(n)$ be the von Mangoldt function: $\Lambda_{\text{vM}}(p^a) = \log p$ for prime powers, and zero otherwise. The elementary identity $\log n = \sum_{d|n} \Lambda_{\text{vM}}(d)$ gives the exact finite formula

$$\mathcal{S}_k^{\text{gcd}} = \sum_{d \leq k} \Lambda_{\text{vM}}(d) d \binom{\lfloor k/d \rfloor + 1}{3}. \quad (31.5)$$

Since

$$d \binom{\lfloor k/d \rfloor + 1}{3} = \frac{k^3}{6d^2} + O(k^2/d),$$

we obtain

$$\mathcal{S}_k^{\text{gcd}} = -c \frac{\zeta'(2)}{\zeta(2)} + O(k^2 \log^2 k). \quad (31.6)$$

For completeness, no prime number theorem is needed for this error: $\Lambda_{\text{vM}}(d) \leq \log d$ implies

$$\sum_{d \leq k} \frac{\Lambda_{\text{vM}}(d)}{d} = O(\log^2 k), \quad \sum_{d > k} \frac{\Lambda_{\text{vM}}(d)}{d^2} = O(\log k/k).$$

Logarithmic differentiation of the absolutely convergent Euler product at 2 gives $\sum_{d \geq 1} \Lambda_{\text{vM}}(d)/d^2 = -\zeta'(2)/\zeta(2)$.

Proof of (24.11). Apply Lemma 31.1 to each grid in the exact product. Since $gq = j - i$,

$$M_1(k) = \frac{2}{\pi D_k} \{ \mathcal{S}_k^{\text{gap}} - \mathcal{S}_k^{\text{gcd}} + (\gamma_E - \log \pi) c \} + O(G_k/D_k).$$

Insert (31.4) and (31.6). Because

$$D_k = 2(c - G_k), \quad G_k = O(k^2 \log k),$$

the normalization correction multiplied by $\log k$ is $O(\log^2 k/k)$. This proves

$$M_1(k) = \frac{\log k + \gamma_E - \log \pi - 5/6 + \zeta'(2)/\zeta(2)}{\pi} + O(\log^2 k/k).$$

□

The constant after $(\log k)/\pi$ is approximately

$$\frac{\gamma_E - \log \pi - 5/6 + \zeta'(2)/\zeta(2)}{\pi} = -0.6273278444052116 \dots \quad (31.7)$$

In particular, a good numerical fit to the logarithmic coefficient alone can conceal a substantial finite constant.

31.4 An exact second moment and a shift-dependent example

The finite identity

$$\sum_{\ell=1}^{q-1} \cot^2 \frac{\pi \ell}{q} = \frac{(q-1)(q-2)}{3}$$

can be checked by logarithmically differentiating the finite sine product twice: $\sum_{\ell=0}^{q-1} \csc^2(z + \pi \ell/q) = q^2 \csc^2(qz)$. Letting $z \rightarrow 0$ and removing the common z^{-2} term gives the sum of squared cosecants, and subtracting $q-1$ gives the displayed cotangent sum. Thus

$$M_2(k) = \frac{1}{6D_k} \left\{ \sum_{i < j} \frac{(j-i)^2}{g_{ij}} - 3c + 2G_k \right\}. \quad (31.8)$$

This identity supplies an exact rational check of (31.2), independent of numerical root summation.

The positive absolute imaginary moments of the full zero measure have the exact relation

$$\int |\operatorname{Im} u|^p d\mu_{k,s}^z(u) = \frac{D_k}{\deg H_{k,s}} M_p(k). \quad (31.9)$$

For $p > 1$ and $s \gg k^2$, this gives

$$\int |\operatorname{Im} u|^p d\mu_{k,s}^z(u) \sim \frac{\mathcal{M}_p}{3} \frac{k^{p+1}}{s}.$$

Hence $\mu_{k,s}^z \implies \delta_{-1}$ can coexist with divergent p -th moments when $k^2 \ll s \ll k^{p+1}$. If $s/k^{p+1} \rightarrow \varkappa_p \in (0, \infty)$, the moment tends to $\mathcal{M}_p/(3\varkappa_p)$. This is a concrete illustration of why weak convergence and moment convergence must be kept distinct.

32 Manuscript 08's checks and numerical illustrations

Manuscript 08's program `code/07-zero-laws-reproduce.py` (delivered as `code/reproduce.py`) regenerates the data and all four figures without accessing the network. The proofs above do not rely on these computations. The purpose of the checks is to test the enumerative normalization, detect multiplicity or factor-of-two errors, and display the rate at which the asymptotic regimes become visible.

32.1 What is checked exactly

The program constructs the counts allowing zero rows, $A(n, j; u)$ of Section 2, directly from the column recurrence

$$A(n, j; u) = u \sum_{h=1}^n \binom{h+j-1}{j-1} A(n-h, j; u) \quad (n \geq 1, j \geq 1),$$

which is (2.2), uses inclusion–exclusion to obtain $T_u(n, k)$, and takes the determinant before comparing it with (24.15). Thus the determinant entries are not generated from the product being checked.

Verification	Completed checks
Complete polynomial determinant identities in $\mathbb{Q}[u]$, $1 \leq k \leq 4$, $1 \leq s \leq 3$	12
Rational specializations, including exceptional and central roots	96
Exact nonreal algebraic-root specializations	3
Degree and primitive-multiplicity identities	91
Independent root merging and exact CDF comparisons	18
Independent direct and aggregated tail counts	20
Finite cotangent identities at 70-digit precision	78
CDF maxima over complete supports, through $k = 2048$	18
Independent high-precision absolute first-moment checks	2

Table 13: Manuscript 08's checks. All listed checks passed. High-precision and tail checks are distinguished from exact polynomial and rational computations.

The empirical CDF maximum is found by checking every support angle and both its one-sided values. These values are exact rational numbers, rather than maxima sampled from a plotting grid. The second and fourth moments are also computed by exact finite cotangent identities. In edge plots, the root masses are integer counts, but comparison with transcendental thresholds uses floating-point arithmetic. Independent high-precision tail checks use thresholds away from atoms.

[Note of the batch-98 write: the program's record `data/07-zero-laws-verification_summary.txt` also lists four checks of PDF structure (trailer, cross-reference offsets and page count of each figure), which the table does not count. The shipped `data/07-zero-laws-pdf_validation.json` is a different record: it reports that the four figure PDFs were opened and rendered by PyMuPDF, a tool that is not a dependency of the program and is not shipped, so that record cannot be regenerated from the package.]

32.2 Bulk convergence and arithmetic concentrations

The computed Kolmogorov distances illustrate a useful distinction: the theorem determines the exact limiting constant, but moderate sizes do not approximate that constant especially closely. Write $\Delta_k^K = d_K(\nu_k^Z, \mathcal{C})$.

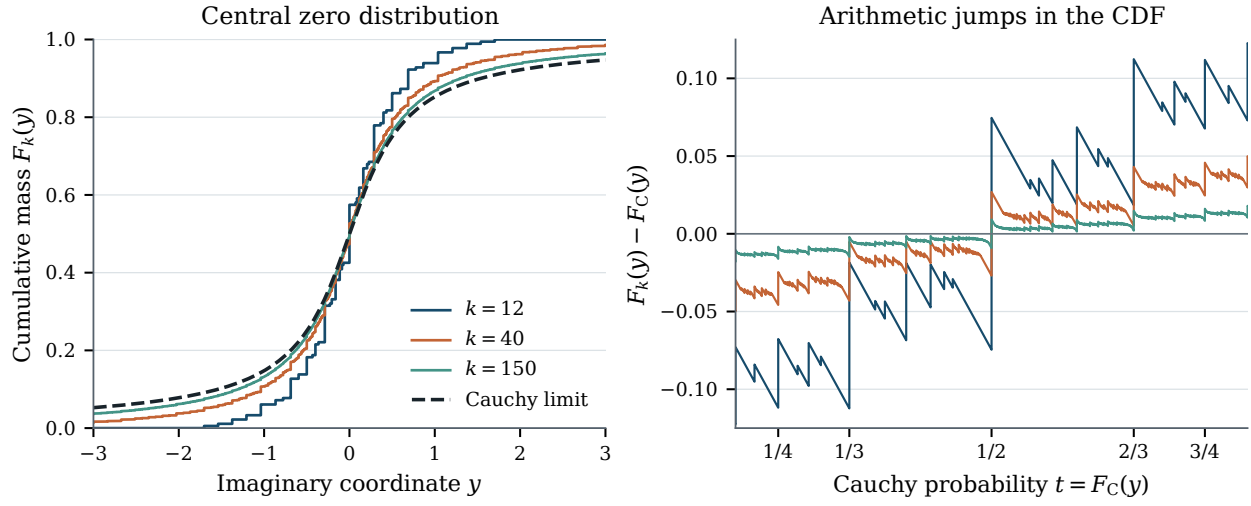


Figure 3: The reduced zero distribution for $k = 12, 40, 150$ (manuscript 08). Left: CDFs of the imaginary coordinates and the Cauchy limit. Right: discrepancy in the uniform angular coordinate $t = F_C(y)$. The arithmetic jumps are visible even when the CDFs are close on the original scale. [The figure's $F_k(y)$ is the distribution function of ν_k^z , that is $\mathcal{F}_k(F_C(y))$, and its F_C is F_C .]

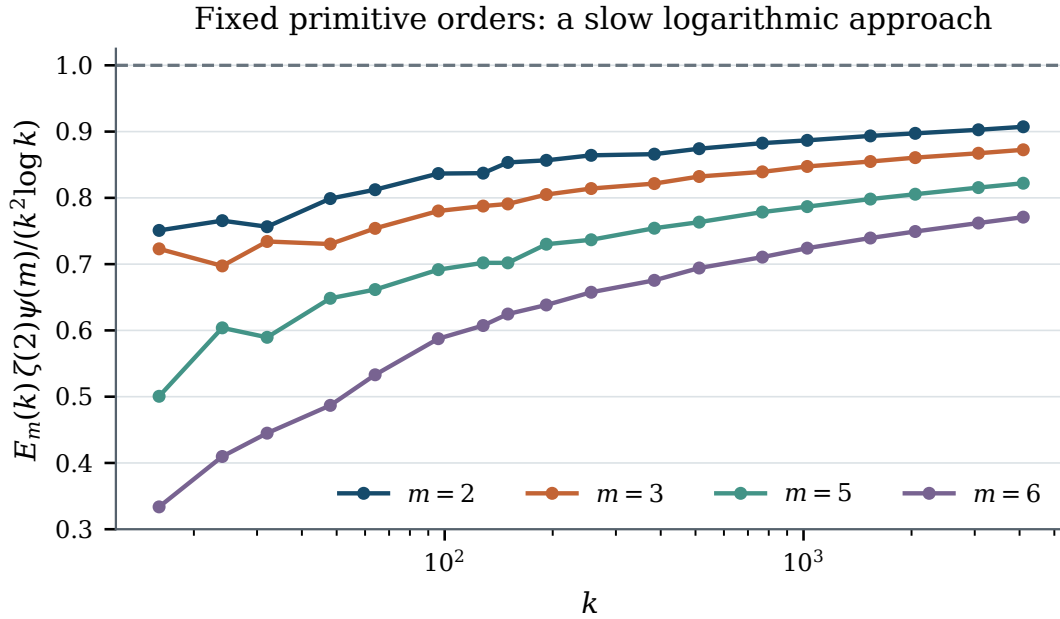


Figure 4: Exact primitive-order multiplicities divided by the leading terms $k^2 \log k / (\zeta(2) \psi(m))$, for $m = 2, 3, 5, 6$ (manuscript 08). The convergence is slow because the proved error has relative order $1/\log k$ at each fixed order. [The figure's $E_m(k)$ is $E_m(k)$.]

k	Δ_k^K	$k\Delta_k^K/\log k$	One maximizing left limit
64	0.0350281076162	0.539038690	1/7
128	0.0208961596153	0.551255513	1/7
256	0.0122991367143	0.567804913	1/7
512	0.00708644151164	0.581607767	1/11
1024	0.00406672748617	0.600785672	1/11
2048	0.00230151896685	0.618196247	1/11
Limit	$9/\pi^2 = 0.911890653\dots$		

Table 14: Complete-support exact discrepancies, displayed in decimals (manuscript 08). These data neither determine the limiting constant by themselves nor suggest that the maximum always occurs at the order-two atom.

The movement from an order-seven to an order-eleven endpoint is compatible with the moving-angle proof of the sharp constant. It is finite evidence, not a theorem predicting the denominator of every maximizing endpoint (Research question 9).

32.3 The extreme tail and the moment transition

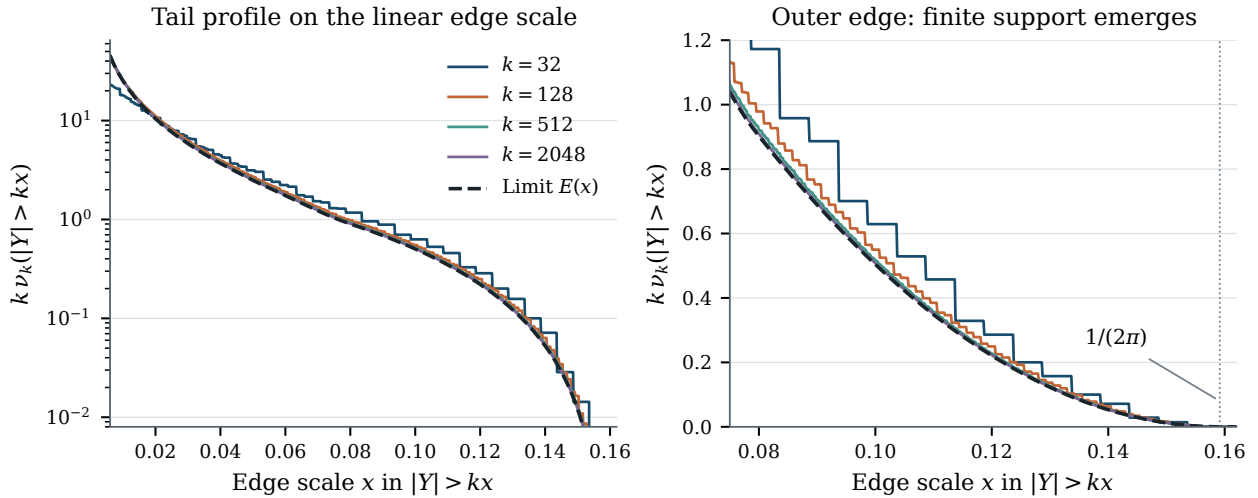


Figure 5: The two-sided extreme tail $k\nu_k^Z(\{|Y| > kx\})$ for $k = 32, 128, 512, 2048$, compared with $2\Xi(x)$ (manuscript 08). The enlarged outer region displays the finite endpoint $1/(2\pi)$. The limiting curve comes from the divisor-sum formula, not from a fitted numerical density. [The figure's ν_k is ν_k^Z , and its “Limit $E(x)$ ” is the two-sided limit $2\Xi(x)$.]

For example, at $k = 4096$ the computed values are

$$k^{-1}M_2(k) = 0.0305025370246502, \quad \frac{k^{-1}M_2(k)}{\zeta(3)/(4\pi^2)} = 1.00177611516,$$

and

$$k^{-3}M_4(k) = 0.000175765262189301, \quad \frac{k^{-3}M_4(k)}{\zeta(5)/(600\pi^2)} = 1.00377307676.$$

The constants here are consequences of (24.13); the exact finite moments provide independent numerical checks.

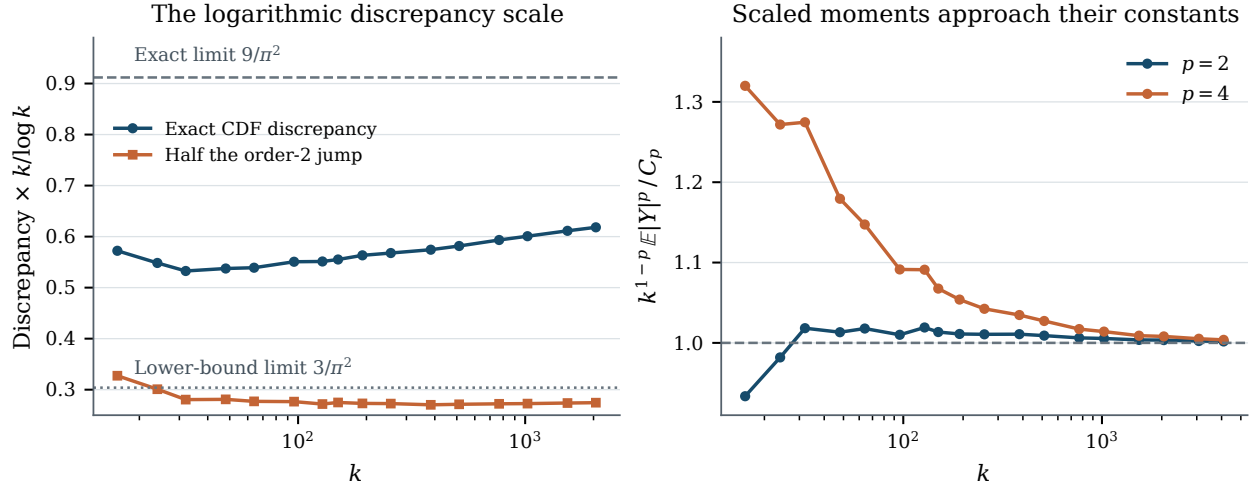


Figure 6: Left: discrepancy on the $(\log k)/k$ scale, its order-two half-jump lower bound, and the proved limit $9/\pi^2$. Right: normalized second and fourth moments. Moment convergence in these plots is substantially faster than convergence of the scaled global discrepancy (manuscript 08; an observation, not a theorem). [The figure’s $k^{1-p}\mathbb{E}|Y|^p/C_p$ is $k^{1-p}\mathcal{M}_p(k)/\mathcal{M}_p$; its lower-bound limit $3/\pi^2 = 1/(2\zeta(2))$ is that of half the central atom (25.7).]

k	$M_1(k) - (\log k)/\pi$
64	−0.505972605
256	−0.573868502
1024	−0.605429214
2048	−0.613841428
4096	−0.619330934
Limit	−0.6273278444052116...

Table 15: The critical first-moment constant (manuscript 08). These entries use extended-precision cotangent summation; two cases are independently checked at 70-digit precision. [See Remark 32.1: “extended precision” is the platform’s `long double`.]

At $k = 4096$, the residual after subtracting the theorem’s logarithm and constant, multiplied by $k/\log^2 k$, is about 0.473444. This is consistent with the error scale in (24.11); it is not an empirical determination of an optimal remainder constant.

Remark 32.1 (write; “extended precision”). [Note of the batch-98 write.] The program sums the cotangents for Table 15 and the edge-tail grid in NumPy’s `longdouble`. That type is an extended format on typical x86 Linux builds, but on Windows it is the ordinary 64-bit double; the package does not record the platform it was produced on. At the intake’s reruns on Windows, 16 of the 20 data rows of `data/07-zero-laws-first_absolute_moment.csv` and 20 of the 4,804 data rows of `data/07-zero-laws-edge_tail_grid.csv` differ from the shipped files in the last one or two of their seventeen printed digits; every other CSV file was byte-identical. The nine-digit values of Table 15 and the plotted curves are unaffected (the four PNG figures were pixel-identical), and the two 70-digit cross-checks do not depend on the type. The caption’s “extended precision” therefore depends on the platform, and the summary’s sentence “exact CSV values are reproducible” holds for the exact files, not for these two floating-point ones.

32.4 Reproduction and data

Manuscript 08’s package contains its L^AT_EX source and compiled PDF, the four vector figures and their PNG versions, the verification program, and the exact and numerical data used in its tables and plots. Its stated commands are `python3 code/reproduce.py` and a `latexmk` build of its own `.tex` file; its environment was SymPy 1.14.0, mpmath 1.3.0, NumPy and Matplotlib 3.10.8, and its finite checks and their scope are recorded in `data/07-zero-laws-verification_summary.json` and its text companion.

[Note of the batch-98 write: in this report the program, its data and its figures carry the prefix `07-zero-laws-`, the requirements file is `data/07-zero-laws-requirements.txt`, and its build script `code/07-zero-laws-build.sh` builds the delivered `.tex`, which is not shipped; the article here is a build of the merged text. The program writes unprefixes files into `data/` and `figures/` under the directory above its own, so it must be rerun on a copy with the delivered layout, as the report README describes, never in the report directory. At the intake’s rerun on such a copy (Python 3.13.5, the pinned SymPy, mpmath, NumPy 2.3.5 and Matplotlib), it exited with status 0 after about 45 s; nine CSV files were byte-identical, the text summary equal apart from line endings, the JSON summary equal apart from the recorded Python version, the four PNG figures pixel-identical and the four PDF figures identical apart from their creation dates; for the two remaining CSV files see Remark 32.1.]

33 Further questions and research

[Added 5 October 2026, batch 98.] This section collects every unproved claim of manuscript 08: its nine research questions, with its sketches; its explicit candidate for the correction of the fractional moments; its speculation about the last zeros; and its three numerical observations. Manuscript 08 states no claim that this write found wrong. Its question 8 is Part I’s Research question 5 and is printed there (see the paragraph on it below); its question 9 overlaps Section 10’s “practical order of attack” and is printed here with a note. Manuscript 08’s questions 1–7 and 9 are Research questions 9–16.

The original zero-distribution question is now answered for both normalizations, with fixed-order multiplicities, extreme tails, and a complete leading moment classification. The following questions

ask for finer information beyond those theorems. They are proposed research directions for this family, not claims that every formulation has been checked against all literature.

Research question 9 (The next term in the global discrepancy; manuscript 08). Determine a second asymptotic term for Δ_k^K beyond $(9/\pi^2)(\log k)/k$, and describe the denominators of maximizing support angles.

The sharp-constant proof uses compactness and harmonic averaging, and does not give a quantitative error in that constant. Table 14 suggests why such an error would matter in practice. A first target is a two-sided effective bound with a smaller error than $o((\log k)/k)$. A more ambitious target is an arithmetic description of the maximizing order and its transitions as k grows. The fixed rational endpoint formulas alone cannot settle a maximization over growing denominators.

[Note of the batch-98 write.] *Evidence, not proof:* in Table 14 the maximizing left limit moves from $1/7$ (at $k = 64, 128, 256$) to $1/11$ (at $k = 512, 1024, 2048$), and $k\Delta_k^K/\log k$ is still 0.618 at $k = 2048$ against the limit 0.912 . Manuscript 08 calls the movement “finite evidence, not a theorem”. The moving-endpoint example of Section 28, which realizes the sharp constant along $t_k = 1/\log k$, suggests that the maximizing angles drift to the endpoints with growing denominators; that is a heuristic, not a statement proved here.

Research question 10 (Cyclotomic orders that grow with k ; manuscript 08). For which growing ranges of $m = m_k$ can (24.6) be strengthened to a useful uniform relative asymptotic?

The present absolute error guarantees a relative equivalent for $\psi(m) = o(\log k)$, but says little when $\psi(m)$ is comparable to or larger than $\log k$. The exact expression (25.4) retains more information: when m is a fixed fraction of k , only a bounded number of unreduced denominators are multiples of m . An analysis of that transition should keep the short coprimality intervals rather than replace them by a global density.

Research question 11 (Local spacing with multiplicity weights; manuscript 08). What are the local spacing and pair-correlation laws of the distinct zeros, and how do their multiplicity weights change those laws?

The distinct angles are the reduced fractions of denominator at most $k - 1$, so their unweighted spacing belongs to the classical Farey setting. The present weights $E_m(k)$ impose a different arithmetic sampling. A natural first comparison is between a uniformly selected distinct zero and a zero selected with its algebraic multiplicity. A satisfactory result should state the coordinate used for spacing, since the cotangent map strongly stretches the ends.

Research question 12 (The last finitely many zeros; manuscript 08). Describe the successive extreme ordinates and their multiplicities at a resolution finer than Y/k .

Proposition 29.3 gives the largest ordinate exactly. The edge theorem controls order k^2 roots, counted with multiplicity, and does not resolve a fixed extreme rank. The next denominators near $k - 1$, together with their short coprimality counts, are an explicit finite starting point. This problem may exhibit dependence on arithmetic features of k that are averaged out by Ξ . [Note of the batch-98 write: that last sentence is manuscript 08’s speculation; it is not proved or refuted here.]

Research question 13 (Subleading fractional moments; manuscript 08). For $0 < p < 1$, determine the first correction to $M_p(k) \rightarrow 2^{-p}/\cos(\pi p/2)$, uniformly on useful ranges of p .

The edge expansion suggests a term of order k^{p-1} . One concrete candidate is the analytic continuation in p of the coefficient $\mathcal{M}_p$ in (24.13), using the continued zeta function. This candidate is not proved here. A proof must subtract the Cauchy tail before integrating the edge profile and control the small- x region uniformly. Merely continuing the formula established for $p > 1$ does not justify the asymptotic statement for $p < 1$.

[Note of the batch-98 write.] The candidate, stated as a claim, is: for fixed $0 < p < 1$, $\mathcal{M}_p(k) = 2^{-p} / \cos(\pi p/2) + \mathcal{M}_p k^{p-1} + o(k^{p-1})$ with $\mathcal{M}_p$ continued analytically; it is open. What is missing is the subtraction described above and a bound on the bulk error, which by Theorem 27.1 is of order $(\log k)/k$ for smooth compactly supported statistics, smaller than k^{p-1} for every $0 < p < 1$; but $|Y|^p$ is neither compactly supported nor smooth at the origin, so that theorem does not apply to it directly.

Research question 14 (An expansion through the critical moment order; manuscript 08). Can the three moment regimes be replaced by one uniform formula when $p = 1 + \lambda / \log k$ and λ remains bounded?

Such a formula should recover the logarithm and the constant in (24.11) when $\lambda \rightarrow 0$, while matching the two noncritical regimes in an overlap region. The pole of $\zeta(p)$ in $\mathcal{M}_p$ identifies the cancellation that a uniform derivation must handle. This is a more precise objective than substituting a varying p into a fixed- p theorem. [Here λ is a local parameter, not the ratio n/k of Part I.]

Research question 15 (Smaller Hankel minors and other row models; manuscript 08). Which submaximal Hankel determinants, or variants of the column enumeration, retain a vertical-line zero law?

The exact resultant product is decisive for the maximal determinant. Smaller minors contain sums of products, and their zeros need not inherit its geometry. A first project is an exact classification for small deficits in the matrix size, supported by symbolic examples and then by a structural identity. Bounded entries and alternative column weights are separate variants whose spectra must first be established. [Note of the batch-98 write: this continues the last sentence of Part I's Research question 5, which asks which minors other than the maximal ones have a definite sign, and the variants of Part I's Research question 8 and of Part II's paragraph on weighted rows and restricted alphabets (Section 20).]

Manuscript 08's question 8: a combinatorial model for the positive product. Manuscript 08's eighth question asks whether the coefficients of (24.15) can be explained directly by counted objects or a sign-reversing involution. It says itself that the question “was already present in the repository report”: it is Part I's Research question 5, “A combinatorial explanation of determinant positivity” (Section 10), and is not repeated here. Manuscript 08's addition to it, which arithmetic factors a model would have to explain, is printed there in a dated note. The zero analysis does not answer it.

Research question 16 (Effective bounds and formal certificates; manuscript 08). Develop explicit constants and a proof-producing implementation for the algebraic and arithmetic parts of the theory.

The determinant, cyclotomic multiplicities, exact degree identities, and full-support CDF comparisons are finite objects. They offer a practical first formalization stage. The next stage is the primitive-lattice error with an explicit constant, followed by the CDF bound (26.2). The sharp constant requires the harmonic approximation argument; the numerical verification suite does not

currently formalize that analytic step. Effective constants would also permit a principled switch between exact data and asymptotic evaluation. [Note of the batch-98 write: the first stage extends the first target named by both earlier manuscripts, the reduced denominator, the resultants and the maximal Hankel product (Section 10, “A practical order of attack”, and Section 20). Nothing of Part III is formalized, and this paragraph is a proposal, not a plan of the repository.]

Observations that are not theorems. Manuscript 08 records three numerical observations, kept here as such: (i) the drift of the maximizing endpoint from $1/7$ to $1/11$ (Table 14; Research question 9); (ii) the residual 0.473444 of the first moment at $k = 4096$ after scaling by $k/\log^2 k$, “not an empirical determination of an optimal remainder constant” (Section 32); whether $O(\log^2 k/k)$ in (24.11) is the true order is open; and (iii) “moment convergence in these plots is substantially faster than convergence of the scaled global discrepancy” (Figure 6), a finite comparison with no rate proved for either side beyond the theorems above.

A Provenance and editorial notes

[Added 3 October 2026, batch 85B.]

Sources. The report was built from two manuscripts of batch 85 of ProveIt’s incoming-reports intake, written independently on the same subject.

- *Manuscript 06* (Part I, the base): the archive `Matrix_Compositions_Uniform_Asymptotics.zip` (21 files), main file `matrix_compositions.tex` (2,078 lines, 100 labels), a 31-page PDF titled *Matrix Compositions: Exact Spectra, Hankel Products, and Uniform Asymptotics*, dated October 3, 2026 (Pacific time). Its author line and PDF metadata read “OpenAI ChatGPT”, with “Research prepared for Vladimir Reshetnikov”. Its pin is the search-index commit `ce37e13f4aa16819c87c3ccc611362d15758f2ac` (2026-10-03, 17:05) [7]. It arrived in commit `345284ac8`.
- *Manuscript 01* (Part II): the archive `OEIS_Matrix_Compositions_Research.zip` (26 files), main file `matrix_compositions.tex` (1,561 lines, 98 labels), a 22-page PDF titled *Matrix compositions: finite spectra, uniform asymptotics, and two Poisson limits*, dated 3 October 2026, author line “Research report prepared for Vladimir Reshetnikov”. Its pin is the tree revision `d68b65ea064084fb121df9bb431b21d086f7be81`; it also records the search-index commit `ce37e13f4` [9]. It arrived in commit `317c1ce2e`, with the other archives of batch 85; the placement of batch 85A (`713149ded`) held it back so that it could be merged with manuscript 06.

Both were placed together by commit `fa2f3e419`, which deleted both archives from `docs/incoming`; they survive in their arrival commits. Neither manuscript is an edition of the other: they share about 0.7 % and 0.5 % of their word 8-grams (bibliography entries and one phrase), and their pins, code and notation differ.

Why manuscript 06 is the base. On every theorem the two share, manuscript 06 has the weaker hypotheses or the more general statement: recurrence minimality for every column weight $u > 0$ (manuscript 01: $u = 1$); a pole bound uniform in k , also weighted (manuscript 01: a factor $k - 1$); the coverage expansion of the whole generating function in complex w (manuscript 01: $\mathbb{P}(M = 0)$ only); and the correct attribution of the recurrence’s existence to [4, Proposition 25]. It also has more new mathematics (the Hankel determinant product, the sharp total-variation rate,

the joint Gaussian–Poisson law, the column central limit theorem, the prescribed-column threshold). Manuscript 01 is more general only in the kernel lemma, and that lemma only serves compact rays, where both theorems coincide.

Where the merge had to choose.

- Shared results are printed once, in manuscript 06’s form, with a bracketed note naming manuscript 01’s version (Table 5). Manuscript 01’s proofs that are the same argument are summarized in those notes, not reprinted; its genuinely different devices are recorded there (Bonferroni truncation for the coverage probability, the tail bound $|\mathbf{W}_n(w)| \leq e^{|w|}$, the direct uniqueness argument for the maximum on the circle) or printed in Part II where a new result needs them.
- Manuscript 01’s k -dependent pole bound is printed as Remark 4.2, because it is sharper for $k \leq 4$.
- Manuscript 01’s kernel theorem is printed in full in Part II as the general form of Lemma 7.2, and Remark 7.3 says so where the lemma stands. Manuscript 06’s lemma is kept unchanged.
- Manuscript 06’s moments of D in Corollary 7.5 are kept, with a pointer to the stronger Theorem 16.2.
- Manuscript 01’s closed form of β_1 is printed as a note after Corollary 7.4, being the same expression; its fifty-digit constants and β_3 are in Part II.
- Manuscript 01’s overstated claim about the order conjecture is narrowed (Section 11.2), its draft OEIS text is printed with the correction (Section 22), and its example (2.8) is credited to [4, Remark 26].
- Manuscript 01’s research question on joint column laws is re-scoped as partly answered by manuscript 06; manuscript 06’s Research question 2 is re-scoped as answered for the event $M = 0$ by Theorem 13.1.
- Manuscript 01’s inversion section is printed with a note that it is an instance of the transseries volume’s apparatus; manuscript 01’s source manifest (its Appendix C) is replaced by this appendix and the report README, and its rebuild instructions by the README.
- Manuscript 01’s three tables (Tables 6, 7 and 8) are typeset, unchanged, from its shipped table bodies, the files `data/01-two-poisson-X.tex` with X one of `defect_table`, `error_table` and `coupon_table`. Its fourth body, `diagonal_table` (the same diagonal errors at more indices and digits), was not used by its article and is not used here.

What the write changed in manuscript 06’s text.

- Every label gained the prefix `mxc:` (100 labels; every reference was updated); nothing was renamed otherwise and nothing was removed.
- Added: the second author and date lines of the title, the PDF metadata, the editorial note after the abstract, the part heading, Section 1.5 with Table 2, Remarks 2.3, 4.2 and 7.3, bracketed notes marked “[Added 3 October 2026, batch 85B]” in Sections 1, 2, 5, 7, 9 and 10, Part II, this appendix, and five bibliography entries [9, 10, 11, 12, 8].
- The figure path is now `figures/06-hankel-coupon_window.pdf`.

- No delivered sentence of manuscript 06 was changed or removed. Every non-claim is kept: the abstract’s last sentence; the scope table’s caption; the bounded audit; “The theorem controls the unrestricted count $A(n, k)$ uniformly” and what it does not say; the moving-mean remark; the compactness paragraph after Corollary 7.5; the prescribed-column caveat; the Hankel caveat on total positivity; the numerics caveats; and the caveat on the research questions.

Manuscript 01’s text is restated in manuscript 06’s letters (Table 2), with some proofs condensed; none of its statements was strengthened, and every one of its non-claims is kept (Section 11, the remarks after Theorems 13.1 and 16.2, Remark 15.1, Sections 18, 19, 21 and 22).

Overloaded letters in manuscript 06. Manuscript 06 reuses some letters; none is renamed, and in each case the section determines the meaning. D is $\binom{k+1}{2}$ in Section 2, the constant $D_u = 16b/L_u^2$ in Section 4, the function $D(\eta)$ in Section 5 and $n - Q$ in Section 7.6. d is $\binom{k+1}{2}$ and c is $\binom{k+1}{3}$ in Section 3, beside the growth constants d_k, c_k . b is $u/(1+u)$ in Sections 2–4, also u in Section 3, the variance κ_2 in Section 7, and b_j are factorial moments. $P_j(t)$ in Section 3 and $P_j(w)$ in Section 7.4 are different polynomials; $E_n(w)$ is a product, $E_j(\lambda)$ a saddle coefficient and $E_n = B\rho^{n-1}$ an error budget; r is $r_k = 1 - a_k$, the saddle, and a summation index; v is $Lr/2$ in Section 7 but a tilting parameter in the proof of Theorem 6.2; and m indexes d_m and counts columns.

Checks by the intake. The placement reran both packages on copies: manuscript 06’s suite passed with its outputs byte-identical or equal apart from line endings and timings and its figure pixel-identical; manuscript 01’s suite passed likewise, and its checksum ledger matched 25/25. Manuscript 06’s diagonal constants and manuscript 01’s agree in every printed digit, and their coverage probabilities at $k = 20, n = 60$ agree. The intake also verified Proposition 25 and Remark 26 in the published paper, the impulse of Remark 2.4 at $k = 1, 2$, the instances $H_{3,s}(u)$, $H_{2,1}(1) = 8$ and $H_{3,1}(1) = 62208$, the constants of Theorem 4.1 and Remark 4.2, the moments in Corollary 7.5, the increment expansion (2.11), the cumulant recursion at $j = 3$, and P_3 numerically. These checks do not replace an independent review of the proofs.

Part III (batch 98). [Added 5 October 2026, batch 98.] Part III is built from manuscript 08 of batch 98, the archive `a261781_hankel_zero_laws.zip` (30 files, a 29-page PDF, main file of 2,041 lines with 110 labels), which arrived in commit 2172df76a and was placed in commit 0fba5167f. Its pin is d18416ec7 [15]; this report’s text did not change between the pin and the write. Its article, PDF and delivery README are not shipped; it has no checksum ledger. Its other 27 files are shipped byte-identical with the prefix `07-zero-laws-`, and they use the delivery names; its `requirements.txt` (shipped as `data/07-zero-laws-requirements.txt`) is byte for byte a generic file of an unrelated report in the Fabius tree. Manuscript 08 is a continuation written against this report, not an edition of either earlier manuscript: 1.25 % of its word 8-grams occur in this report, and 0.50 % of this report’s in it.

Where the batch-98 write had to choose.

- Manuscript 08’s Section 2 re-derives Theorem 3.1 by the same argument; it is printed as a pointer (Remark 24.4), which keeps its two new small cases $H_{1,s}, H_{2,s}$ and its remarks on the convention $s \geq 1$.
- Its question 8 is Part I’s Research question 5; it is printed once, there, with manuscript 08’s addition in a dated note, and a paragraph of Section 33 points to it. Its question 9 is printed with a note naming the formalization targets of Parts I and II. Its questions 1–7 and 9 are

Research questions 9–16; the question counter of Part I continues, so no existing number changes.

- Manuscript 08’s letters collide with Parts I and II in about thirty places (among them ν_k , M , M_p , D_k , d_k , c_k , Δ_k , γ , Λ , L , ρ_k , τ and σ). Part III renames them, with sans-serif type for its colliding Latin objects and a superscript z on its zero measures, adopts Section 3’s unindexed $c = \binom{k+1}{3}$ and writes $\binom{k+1}{2}$ out; every rename is in Table 10. Two internal overloads of the manuscript are resolved by the same table: H_N (harmonic numbers) beside $H_{k,s}$, and σ (a sign) beside the divisor functions $\sigma_{\pm 1}$; its lower summation limit M and running integer N are marked in notes. Its labels are printed as `mx:hz:` followed by the delivered name, including the `ar:` labels of its Section 6.
- Its figures are delivered files and keep its letters; the captions translate them. Its tables are retyped from its source with the renamed letters and unchanged numbers.
- Its Section 12 (source audit, attribution and package contents) is folded into Section 23, Section 32 and this appendix; its bibliography entries for A261781, A261784 and Munarini–Poneti–Rinaldi are this report’s [2, 3, 4], and three entries were added [13, 14, 15].
- Following the intake’s standing rule on unproved and wrong claims, every unproved claim of manuscript 08 stands in Section 33: its nine questions, its candidate for the correction of the fractional moments (Research question 13, stated there as a claim), its speculation that the last zeros depend on arithmetic features of k (Research question 12), and its three numerical observations. No claim of manuscript 08 was found wrong. Two imprecisions are recorded in remarks of the write: “extended precision” and “exact CSV values are reproducible” depend on the platform (Remark 32.1), and the uniformity in m of Proposition 25.2 is empty for $m \geq k$ (Remark 25.3).

What the batch-98 write changed. It added to the preamble eight macros for Part III’s notation, a third title, author and date line and the PDF metadata; the batch-98 editorial note; bracketed notes marked “[Added 5 October 2026, batch 98]” in Sections 1 and 1.5, at the end of Section 3, in Research questions 5 and 7 and in “A practical order of attack” of Section 10; Part III (Sections 23–33); these four paragraphs; and three bibliography entries. All its labels carry the prefix `mx:hz:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of manuscripts 06 or 01 was changed or removed. Manuscript 08 is printed in full apart from its Section 2 and its Section 12, with renamed letters, notes of the write and light rewording where a sentence named its own package layout; none of its statements was strengthened, and every one of its non-claims is kept (Section 23).

Checks by the intake for Part III. The placement and the write each reran manuscript 08’s program on a copy with the delivered layout (Section 32): exit status 0, all counted checks passed, outputs as described there. The placement read every proof of Sections 25–31 and found no error; it re-derived by hand, among others, the Euler product for ψ , the grid and weight bounds of Theorem 27.1, the exact low-range identity of Lemma 28.5, the case analysis of Corollary 28.3, the jump formula (29.7), the Bernoulli identity of Theorem 30.1, the weights of Theorem 24.3, $\mathcal{M}_2 = \zeta(3)/(4\pi^2)$, $\mathcal{M}_4 = \zeta(5)/(600\pi^2)$ and the identity (31.8). An independent mpmath computation of its own (not shipped) found $\mathcal{I}_3$ equal to $\frac{3}{2} \log(2\pi) - \frac{11}{4}$ to 10^{-15} ; at $k = 500$ and 1500 it found $D_k = 2(c - G_k)$ exactly, $M_1(k) - (\log k)/\pi = -0.5923$ and -0.6107 against the limit -0.62733 , $(k/\log k)(\mathcal{F}_k(t) - t)$ at $t = \sqrt{2} - 1$ equal to -0.127 and -0.130 against -0.156 , and $k\nu_k^z(kx, \infty)$ at

$x = 0.1$ equal to 0.2570 and 0.2545 against $\Xi(0.1) = 0.2520$: slow approaches consistent with the error terms proved. The write checked $H_{1,s}$, $H_{2,s}$, the bound of Remark 25.3 and the translation of every renamed formula against the delivered source. These checks do not replace an independent review of the proofs.

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